

Governance Conflict and Exit en Masse

A Behavioral Foundation for Theories of State Development*

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August 29, 2026

ABSTRACT. Theories of state development assume that fragmentation lets subjects turn to alternative authorities, but direct evidence of such behavior is scarce. Tokugawa Japan offers a test: nearly 300 domains gave villages unequal access to neighboring rulers and the central authority (the shogunate). Exit included collective flight, often temporary, and appeals through a neighboring lord or to the shogunate for concessions. I use difference-in-differences around provisionally identified domain conversions that expanded neighboring villages' access to the shogunate. Neighboring villages governed by other lords recorded 3.9 percentage points more exit but no detectable change in confrontation. Across places and years, easier access to another authority predicted exit but not confrontation. Among resisting villages, proximity to another domain's capital predicted exit rather than confrontation. Domains whose villages were closer to other authorities reported lower tax rates when neighboring domains could absorb migrants. The evidence links fragmentation to exit through access to another authority.

Total Words (Main Text): 13,913

*I thank Scott Abramson, Casey Petroff, Hein Goemans, Sergio Montero, Matteo Bertoli, Amy Catalinac, Nathan Feldman, Mark Fey, Francisco Garfias, Jenny Guardado, Yukari Iwanami, Daiki Kishishita, Jefferson Leal, Alexander Lee, Bingru Li, Chiaki Moriguchi, Emily Sellars, Yu Xia, and all participants of JPOSS (Japanese Politics Online Seminar Series), the University of Rochester Graduate Research Seminar and Brownbag Seminar, and JSQPS 2024 Summer Meeting for their helpful comments and suggestions.

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1. INTRODUCTION

Many influential theories of state development rest on a common premise: political fragmentation gives subjects access to alternative authorities, and that access disciplines extraction and shapes institutions. This premise carries the central causal weight of leading explanations for the Great Divergence (Cox 2017; Fernández-Villaverde et al. 2023; Jones 2003; Mokyr 2017); for accounts linking Europe’s polycentric structure to constrained taxation (Abramson 2017; Dincecco and Wang 2018; Sng and Moriguchi 2014; Stasavage 2016); for fiscal-federalism arguments that competing jurisdictions tame Leviathan (Qian and Weingast 1997; Rodden 2003; Tiebout 1956; Weingast 1995); and for recent theories positing that the modern territorial state system itself arose as a cartel against subject mobility (Acharya and Lee 2018; 2023). In each case, the mechanism is the same: fragmentation offers subjects credible exit options, and the threat or use of exit forces rulers to compete or cooperate. Yet the behavioral foundation of this chain has yet to be directly demonstrated: whether access to authority outside the home jurisdiction shaped how subjects resisted. I test this behavioral link.

Direct evidence supporting this premise is scarce for two main reasons. First, data on exit are limited (Bailey 2014, 76): unlike open rebellion, which leaves abundant documentary traces, exit is often subtle and covert. As a result, most existing research infers exit indirectly rather than observing it directly at the community level. Second, it is difficult to find usable variation in access to alternative authority. Macro-level studies that adopt this premise typically compare entire regions, such as Europe and China, whose polities differ across many dimensions, providing little within-case variation suitable for testing a behavioral mechanism at the level where subjects actually make choices. This paper addresses both challenges.

Tokugawa Japan (1615–1868) provides a clean laboratory for this mechanism. Nearly 300 feudal domains formed a patchwork of units autonomous in local administration but not sovereign, each with its own legal code, tax system, and border controls, under a central authority, the Tokugawa shogunate. The shogunate retained supra-domain arbitration, attainder power, direct peasant-appeal authority, and power to assume direct rule (Moore 1993; Ravina 1999; White 1995). Three features make the setting exceptional.

First, the post-1615 peace shuts off the war-finance channel that [Rosenthal and Wong \(2011\)](#) identify as overwhelming the exit-discipline mechanism in early modern Europe, isolating the mechanism itself ([Moore 1993](#)). Second, intense inter-domain competition for agricultural labor made mass migration a credible peasant strategy, and villages varied widely in their geographic exposure to rival rulers. Third, newly digitized records of peasant resistance compiled by [Aoki \(1975\)](#) document direct confrontation and exit through alternative authority at the village level across the entire period. I use *exit through alternative authority* for collective flight into a neighboring domain (*chōsan*), a petition routed through a neighboring lord, or an appeal over the home lord’s head to the shogunate (*osso*). These routes created different forms of leverage, but each moved villagers or their claim beyond the home lord’s exclusive control.^[3]

The empirical analysis begins with two behavioral tests and then examines a downstream fiscal trace. First, I use difference-in-differences around provisionally identified domain conversions to direct shogunal rule (*tenryo*), which installed a shogunal intendant at the former capital and expanded surrounding villages’ access to the shogunate. After direct rule began, nearby villages governed by other lords recorded 3.9 percentage points more exit but no detectable change in direct confrontation, and event studies show no differential pre-trends. Because episode destinations are unobserved, I interpret this pattern as evidence consistent with improved access to outside authority, not as a definitive stand-alone causal estimate. A grid-year panel tracking changes as nearby capitals were established, abolished, or relocated finds the same pattern: easier access to another active capital predicts more exit but no more direct confrontation.

Second, among villages that resisted, greater geographic exposure was associated with choosing exit over direct confrontation. I measure each village’s exposure as the ratio of its distance to the home-domain capital to its distance to the nearest active

^[3]Tokugawa mass exit operated as active resistance rather than permanent relocation: peasants presented demands to home-domain officials while seeking permission from neighboring rulers to cross the border, with some documented cases recording return once concessions were granted. [Moore \(1993, p. 255\)](#), drawing on Borton’s catalog, recorded 106 such mass desertions across the Tokugawa period as evidence of “the solidarity of the Japanese village”; my catalog substantially extends this count and covers the full archipelago. The same active form appears in medieval Europe ([Bailey 2014](#)), precolonial Southeast Asia ([Adas 1981](#); [Scott 1985](#)), and colonial India ([Gandhi and Parel 1997, 95](#)), suggesting the Tokugawa case is one window onto a broader phenomenon.

non-home capital. This ratio is a spatial proxy for access to alternative authority, not a direct measure of competition, bargaining, or another ruler’s willingness to receive migrants. Across all recorded resistance, this exit-*share* result holds robustly under home-neighbor dyad fixed effects. Decomposing the ratio into its two absolute distances shows that foreign-capital proximity is the larger and more robust component. Home-capital remoteness is positive but less precise under spatial inference.

As a downstream fiscal trace, a late-period domain cross-section associates greater geographic exposure with lower reported tax rates, especially where neighboring domains had greater capacity to absorb labor.

The paper makes three core contributions. First, it provides direct, micro-level evidence for a central behavioral premise running through four traditions in political economy, each resting on the same link between access to outside authority and exit: the Great Divergence argument that ruler competition for mobile subjects checked extraction and fueled European growth (Jones 2003; Mokyr 2017); state-formation accounts in which exit forced ruler bargains and built representation (Dincecco and Wang 2018; Stasavage 2016); fiscal-federalism claims that competition among jurisdictions tames Leviathan through taxpayer and labor mobility (Rodden 2003; Tiebout 1956; Weingast 1995); and the cartel theory of the territorial state, in which modern borders are erected specifically to suppress subject exit (Acharya and Lee 2018; 2023). The evidence links better access to outside authority with communities’ use of exit, supplying behavioral evidence for the first link in this shared causal chain.

Second, the paper brings exit from the background to the center of the study of state development. In Hirschman’s (1970) terms, the dominant analytical traditions are built around voice: revolution and rebellion drive regime change (Acemoglu and Robinson 2001; Skocpol et al. 1982), collective mobilization makes states (Hintze 1975; Tilly 1978), and the threat of revolt disciplines extraction (Besley and Persson 2009; Fearon 2011). When exit appears, it shapes incentives (Besley and Persson 2009; Fearon 2011) or long-run sorting (Jones 2003) without being directly observed. Historians of everyday resistance have long documented otherwise: exit often predominated over revolt (Adas 1981; Bailey

2014; Camp 2004; Diouf 2014; Kolchin 1990; Scott 1976; 1985; Sreenivasan 2004). This paper documents exit directly as a strategy of resistance alternative to open rebellion.

Third, the paper provides community-level evidence on recourse to authority outside the home jurisdiction. It distinguishes horizontal exit through neighboring jurisdictions from hierarchical appeal to the shogunate, while complementing work on flight beyond state reach (Carneiro 1970; Herbst 2000; Pruett 2024; Scott 2009).^[4] Within studies of horizontal exit, the Tiebout tradition interprets mobility primarily as individual preference-sorting rather than collective resistance (Peterson 1992; Tiebout 1956). Other strands conceive of exit as a spur to voice (Hirschman 1993; Pfaff and Kim 2003; Schoppa 2022), analyze government-driven emigration (Lueders 2025; Michel, Miller, and Peters 2023), or infer horizontal mobility indirectly—from the persistence of indigenous identity near jurisdictional havens (Guardado and Franco-Vivanco 2025), from legal efforts to restrict peasant movement (Matranga and Natkhov 2025), or from tax-base reactions to cross-border competition (Kleven et al. 2020)—but do not directly capture community-level exit as resistance. The Tokugawa case pairs direct observation of mass community exit and external appeal with a fragmented but hierarchically ordered institutional setting.

2. FRAGMENTATION, EXIT, AND STATE DEVELOPMENT

Theories of state development constrain rulers through war and exit, but direct tests have focused on war. Financing armies required rulers to extract men, money, and supplies from holders of capital, land, and labor. The resulting bargains produced tax bureaus, representative assemblies, and eventually citizenship (Tilly 1985; 1992). Research now specifies when war built states rather than destroyed them (Dincecco and Wang 2018; Spruyt 2017). Exit operates differently: divided authority gives subjects an outside option, so excessive extraction can cost rulers their tax base. This mechanism carries the Great Divergence, fiscal-federalism, and territorial-cartel arguments, yet direct evidence for it remains scarce.

^[4]Moore (1993, p. 230) characterized the late Tokugawa order as marked by “vertical and horizontal fissures... perhaps equally important,” foregrounding the horizontal axis without operationalizing it; this paper supplies the community-level measurement.

Tilly’s account contains the exit mechanism but confines it to elites and regional trajectories. In Europe’s capital-intensive zones, merchants and financiers could “escape with their capital to another business site if the ruler’s demands became unbearable,” forcing rulers in Genoa, Amsterdam, and the Hanseatic cities to offer favorable terms (Tilly 1992, 53). In coercion-intensive zones, rulers and landlords instead fixed peasants in place. Russia’s Assembly Code of 1649 and Prussian serfdom bound cultivators to the land because departure would have undermined extraction (Tilly 1992, 22–23, 141). Tilly thus treats mobility as a difference between regional trajectories and peasant exit as an option rulers foreclosed. He does not treat horizontal exit as a strategy within a state system or include it in his typology of popular collective action (Tilly 1992, 187).

Europe cannot separate war from exit because the same fragmentation that gave subjects somewhere to go gave rulers military rivals. Rosenthal and Wong (2011) note that exit can moderate taxes “particularly when the international scene is peaceful,” but endemic warfare overwhelmed this restraint (pp. 169, 184). Cross-regional comparisons face the reverse problem. Europe and China differ in law, kinship, religion, bureaucratic recruitment, and geography, while Europe’s fragmentation arose partly from medieval ecclesiastical competition (Grzymala-Busse 2023). Their contrast cannot isolate fragmentation itself.

Tokugawa Japan separates the two mechanisms. Peace after 1615 closed the war-finance channel for two and a half centuries, leaving competition over subjects as the main form of inter-domain rivalry. Fragmentation varied within a common legal, linguistic, and religious order: nearly 300 domains differed in size, wealth, and villages’ proximity to rival rulers. The setting holds constant the differences that confound regional comparisons and removes the warfare that complicates Europe. It also places the competing units beneath a hegemon, a structure absent from those theories.

2.1. What Political Fragmentation Means Across Cases

Fragmentation varies along three dimensions that shape exit. The first is sovereignty: early modern European polities recognized no common superior, whereas Tokugawa domains held territory at the shogunate’s pleasure. The second is cultural and legal

distance, which determines what subjects surrender by moving. A peasant leaving Bavaria for Bohemia changed language, law, and often confession. One leaving Tsuyama for Tottori crossed neither a linguistic nor a religious divide and remained within the shared shogunal framework. The third is the density of alternatives, which varies across villages in this paper.

Each dimension enters a different link in the exit mechanism. Sovereignty determines whether an outside ruler disciplines extraction only by accepting subjects or can also appeal to a common superior. Cultural and legal distance determines what villagers lose by placing themselves under another ruler. Density and accessibility determine how far they must travel and how many authorities they can plausibly approach. As those costs fall, exit becomes more credible and the home ruler has stronger reasons to moderate extraction before losing cultivators or facing outside intervention.

These dimensions do not move together, and the comparison cases combine them differently. Europe paired sovereign units with high cultural and legal distance between them. Qing China instead placed local jurisdictions within a single sovereign hierarchy, while movement generally entailed less cultural and legal change. Tokugawa Japan combined subordinate units and low cultural distance with dense alternatives, as roughly 300 separately governed domains occupied an archipelago smaller than France. Treating these configurations as one variable called fragmentation obscures how much subjects had to give up, and which authority they could invoke, when attempting exit.

The Tokugawa configuration permits close observation of exit but not its ultimate institutional consequences. Low cultural and legal distance makes exit depend more on the geography of authority than on assimilation costs, while a common legal order holds constant differences that separate Europe from China. Across villages and years, rival centers vary in number and proximity, while the shogunal hierarchy and shared order remain fixed. But domains subject to abolition by a hegemon could not develop the sovereign institutions, standing armies, and fiscal bureaucracies that state-building theories seek to explain. The setting therefore illuminates the behavioral link in those theories, not their institutional outcome.

2.2. Hegemonically Ordered Fragmentation

The shogunate did more than govern one territory among many. It arbitrated disputes between lords, heard peasant petitions over their lords' heads, converted domains to direct rule, and stripped lords of their territories. These powers created two distinct mechanisms of discipline. A village that invoked a rival lord engaged in horizontal competition and threatened its ruler's tax base; a village that invoked the shogunate engaged in vertical oversight and threatened the ruler's tenure.

That oversight applied on both sides of the domain boundary. Bakufu rules permitted petitions against its own intendants because appeals conveyed information about crop failure and abusive local officials. In privately ruled Aizu, several thousand villagers crossed into neighboring domains in 1642, publicly petitioned the bakufu, and threatened permanent resettlement; their lord canceled arrears partly to forestall intervention, and the unrest contributed to—though was not the formal charge for—his later attainder (Vlastos 1986, 35–38, 42–43). These examples establish review within shogunal holdings and oversight of daimyō rule, although they do not by themselves show that every end-run appeal received adjudication.

Repeated appeals may also have reinforced the shogunate's position as the final receiver and investigator of grievances. The bakufu's 1633 rules permitted appeal to higher authority when petitioners obtained no satisfaction at the district level, and the resulting information helped the center monitor crop failures and misconduct by local officials. Direct appeals subsequently created precedents for accepted, if formally unlawful, political action and provided a recurring mechanism for resolving tax disputes (Vlastos 1986, 42–43, 55–59). In 1720, the shogunate formalized bottom-up monitoring through petition boxes whose submissions went directly to the shogun, were investigated, and remained part of governance through the end of the period (Sng and Moriguchi 2014, 8). End-run pressure therefore did more than exploit an existing hierarchy: repeated appeals may also have reinforced the shogunate's arbitrating role.

The two mechanisms worked together in neighbor-routed and escalated exit. Crossing a border placed villagers beyond their lord's immediate reach but gave them no standing to compel him. Petitioning Edo directly remained a separate vertical route, but it required

a costly journey, risked a hostile reception, and often ended in the representatives' execution.^[5] For villagers who first sought refuge, a neighboring lord combined shelter with an institutional route to Edo by presenting their grievance as a dispute between lords. He supplied protection and standing for the appeal, while the shogunate supplied the sanction that could compel the home lord.

In that sequence, horizontal protection came first. After villagers left, the neighboring lord sheltered them from retaliation, received their complaint, and approached the home lord as an institutional equal. This mediation could produce a negotiated return. If the home lord refused, the neighbor could move the dispute upward, where shogunal authority made the unresolved claim enforceable through its power over domain tenure. Shelter thus allowed villagers to survive the initial departure, and hierarchical sanction gave the resulting grievance force against the home lord.

Hegemonic ordering changed rather than eliminated competition. Lords competed for subjects as fiscal-federalism and Great Divergence theories suppose, but discipline ran through a superior authority that could confiscate their domains. Under this hegemonically ordered fragmentation, local rulers were autonomous but not sovereign. Tokugawa Japan is therefore a demanding case: cross-domain migration was prohibited, and domains could not leave the system. Evidence that exit shaped resistance and extraction under these constraints cannot be attributed to unusually free movement.

2.3. Exit and Direct Confrontation

This institutional structure shapes not only whether exit is feasible but also how exit relates to collective confrontation. The compact formal model gives each village one contemporaneous action among compliance, exit, and direct confrontation. At that decision point, emigration opportunities can weaken mobilization by raising a migrant's opportunity cost of participation and lowering beliefs that collective action will reach its threshold (Sellars 2019). Greater exposure can therefore shift some villages toward exit and away from confrontation in the benchmark. This event-level assumption does

^[5]Unmediated petition to Edo, associated with Sakura Sōgorō, entered later legend because it was exceptional and fatal (Walthall 1997). Appeals were overwhelmingly addressed to Edo (Section 3.2), but a sheltering neighbor provided a survivable route.

not make the tactics substitutes across an entire protest episode. The empirical designs estimate whether easier access to outside authority was associated with recorded exit and with exit's share of recorded resistance. They do not identify whether exit at one stage reduced confrontation at another; in the occurrence designs, confrontation generally does not decline.

The distinction is between a recorded action and a broader protest sequence. A single action is coded as exit or direct confrontation, but complementarity was also present in the Tokugawa context: villages could use both within the same dispute. Aoki's 1853 Morioka records, for example, include an uprising and separate end-run appeals (Aoki 1975). Such combinations show how exit could facilitate collective action rather than simply displace it.

The relationship depends on what resistance requires. Exit is more likely to substitute for confrontation when departure removes participants whose physical presence is indispensable. It can instead complement confrontation when people outside the jurisdiction provide money, information, organization, shelter, or political pressure (Clark, Golder, and Golder 2017; Hirschman 1993; Karadja and Prawitz n.d.; Pfaff and Kim 2003). Both channels were possible in Tokugawa Japan: departure could reduce immediate participation in a march, while protection by another lord or appeal to the shogunate could increase pressure on the home ruler. The model formalizes the first channel; the historical discussion now explicitly recognizes the second.

2.4. The Mechanism and Three Predictions

Villages facing an extractive ruler may first petition through ordinary channels. When that petition does not change the demand, the compact benchmark gives each village one contemporaneous choice among compliance, exit to a neighboring jurisdiction, and collective confrontation. Exit costs fall with proximity to a rival capital, whereas confrontation becomes more effective as more villages from the home domain participate. Rulers set taxes in anticipation of these responses. Because a ruler can concede after villagers leave, an exit episode can end in negotiated return rather than permanent migration.

The mechanism begins when the ruler sets extraction in anticipation of village responses. After observing the demand and failing to obtain relief through an ordinary petition, villagers choose compliance, confrontation, or exit. Those who leave may receive shelter and mediation from a neighboring lord, followed by shogunal intervention if mediation fails. The home lord can then concede and induce their return. Villages can thus react immediately to a change in access, whereas rulers adjust standing extraction later in response to geographic exposure.

The two strategies depend on different conditions. Exit returns vary with the neighboring domain's yield and tax rate and the cost of reaching it. Confrontation returns depend on the home tax rate and participation by other home villages. A more attractive neighboring domain therefore changes the exit margin without changing confrontation returns.

The first prediction concerns the immediate aftermath of conversion. Installing a shogunal intendant nearby may improve access to an outside authority before surrounding lords can reset their standing tax policies. The compact model does not contain a conversion indicator; its fixed-tax comparative static only supplies the conditional logic that a more attractive outside option raises exit without mechanically changing confrontation returns. The conversion design tests that reduced-form implication rather than identifying a structural change in the neighbor's yield or tax rate. The zero-confrontation result is likewise conditional: comparison-based grievances or outside resources could change confrontation in a richer setting. The next two predictions concern persistent geographic exposure. They ask how villages sort between tactics under standing geography and how rulers later set taxes in response.

PREDICTION 1 (Outside-Option Conversion): If conversion improves the neighboring outside option before tax rates adjust, nearby villages' exit frequency rises without a mechanical change in direct confrontation.

In the compact formalization, standing geographic exposure affects both resistance margins through a compositional change. Where more villages lie near a rival capital, more face low exit costs and become willing to leave. Because each village takes one

action at a time in the model, this expands the pool available for exit and reduces the contemporaneous pool available for confrontation. That event-level abstraction does not rule out complementary tactics across a broader protest sequence. The prediction tested below is that geographic exposure raises both exit's share among resisting villages and the overall rate of exit.

PREDICTION 2 (Geographic Exposure and Exit): Greater geographic exposure to a rival capital raises both exit's share among resisting villages and its rate.

Geographic exposure gives the ruler a retention incentive: keeping villages that can credibly leave requires lower taxes. The formal single-action benchmark also permits a countervailing force because a smaller contemporaneous pool for confrontation can permit higher taxes. The retention force dominates only when the neighboring jurisdiction is attractive enough that villages would use the outside option.

The retention force lowers taxes because extraction must compete with a credible neighboring option. In the formal benchmark, confrontation relief can pull in the opposite direction when an exit choice removes a participant from simultaneous collective action at home. Neighbor attractiveness determines the relative strength of these forces. An attractive neighbor makes departure feasible and strengthens retention; an unattractive neighbor weakens that constraint, so the negative tax gradient should attenuate and could, within the benchmark, turn positive.

PREDICTION 3 (Governance Conflict and Extraction): Rulers more exposed to governance conflict levy lower taxes when the neighboring jurisdiction makes exit credible; this negative tax gradient attenuates when the outside option is weak.

Each prediction maps onto an empirical design below. The supplementary information provides a compact formalization of the mechanism and proofs of the corresponding propositions ([Supporting Information A](#)).^[6]

^[6]The mechanism differs from nearby alternatives in two respects. Tiebout fiscal competition treats mobility as costless sorting ([Tiebout 1956](#)), whereas exit here is costly and unevenly distributed. Exit-

3. EMPIRICAL EVIDENCE

3.1. Setting: Tokugawa Japan

During the Tokugawa period (1603–1868), nearly 300 domains governed Japan. [Figure 1\(a\)](#) maps the 264 capitals active in 1868, and [Figure 1\(b\)](#) tracks the number active as domains were created, merged, transferred, or absorbed into shogunal land. These domains originated in territories controlled by local warriors during the Warring States period and, earlier, in private estates known as *shoen*. Although subordinate to the shogunate, each domain maintained its own government and military forces and set its own tax rates ([Ravina 1999, 1000](#)).

Capital-centered fiscal administration links domain fragmentation to village choices. Domains shared an administrative structure centered on the castle town ([Hall 1955](#)), while the village contract system (*murauke*) made villages responsible to the daimyō for tax collection ([Saxonhouse 1995; Walthall 1997, 89; White 1995](#)). Domain capitals housed the officials who received petitions, decided on shelter and mediation, and kept records that could reach Edo ([Hall 1955; Ravina 1999](#)). A village therefore needed to reach an administrative center, not merely cross a border. A neighboring domain governed from a distant castle town offered less access than one governed from nearby. Consistent with this distinction, border distance predicts neither exit nor tax rates after controlling for distance to a rival capital ([Supporting Information D.3.2, Supporting Information E.3.2](#)).

The village contract made extraction a collective obligation administered through identifiable offices. When a tax dispute arose, the village addressed the same institutional system that collected its dues: home officials received its demands, an outside capital could decide shelter or mediation, and Edo could receive the resulting record. This connection between fiscal obligation and administrative venue makes resistance observable. The relevant choice was not simply whether individual households moved. It was whether the village kept its dispute within the home domain or placed it before an outside

voice models treat a single organization ([Gehlbach 2006; Hirschman 1970](#)), whereas two rulers make each one's restraint depend on the other. The closest ruler-side precursor is [Acemoglu and Wolitzky \(2011\)](#).

authority. The national catalog records that choice in forms that can be classified as exit or direct confrontation.

The long peace further connects the setting to the argument. Domains fought no major wars against one another until the shogunate’s collapse in the 1860s.^[7] Domain rivalry therefore operated through competition for subjects and the “everyday forms of resistance” it enabled (Scott 1985), rather than through the warfare central to conventional state-building accounts (Hintze 1975; Tilly 1978; Weber 1919). Peace did not eliminate rivalry. It shifted rivalry toward competition for cultivators, shelter of fugitives by neighboring domains, and shogunal resolution of disputes between jurisdictions.

This separation makes the Tokugawa case useful for studying exit apart from war finance. In Europe, fragmentation simultaneously expanded subjects’ outside options and rulers’ exposure to military rivals, so mobility and wartime extraction operated together. Here, access to alternative rulers varied within a domain system in which inter-domain war remained absent for most of the period. A rise in exit near outside authority can therefore be studied without treating military mobilization as the source of the change. The period also left a rich documentary record of resistance. The national catalog introduced below distinguishes exit from direct confrontation, making it possible to observe the behavioral choice that the theory seeks to explain.

3.2. A Typology of Resistance: Direct Confrontation and Exit

Aoki’s (1975) catalog distinguishes coercive appeals (*gōso*), destruction (*uchikowashi*), rebellion (*hōki*), desertion (*chōsan*), and end-run appeals (*osso*).^[8] I group the first three

^[7]The Shimabara Rebellion of 1637–38 was an anti-Christian and anti-Matsukura uprising suppressed by a shogunal coalition, not unrestricted inter-*daimyō* war. The Chōshū War (1864–1866) similarly pitted the shogunate against Chōshū, and the Boshin War (1868–1869) accompanied regime collapse. Institutional peace held between Shimabara and the 1860s. The tax data cover the period’s end, but tax policies remained stable even in the final years (Smith 1958).

^[8]Later studies follow this classification. White (1995) analyzes protest repertoires and their evolution; Vlastos (1986) treats end-run appeals as a distinct form that dominated the mid-Tokugawa period; Walthall (1997) translates primary documents from all five categories.

^[9]“Direct confrontation” is more accurate than “fight” because coercive appeals were ritualized and often only implicitly violent.

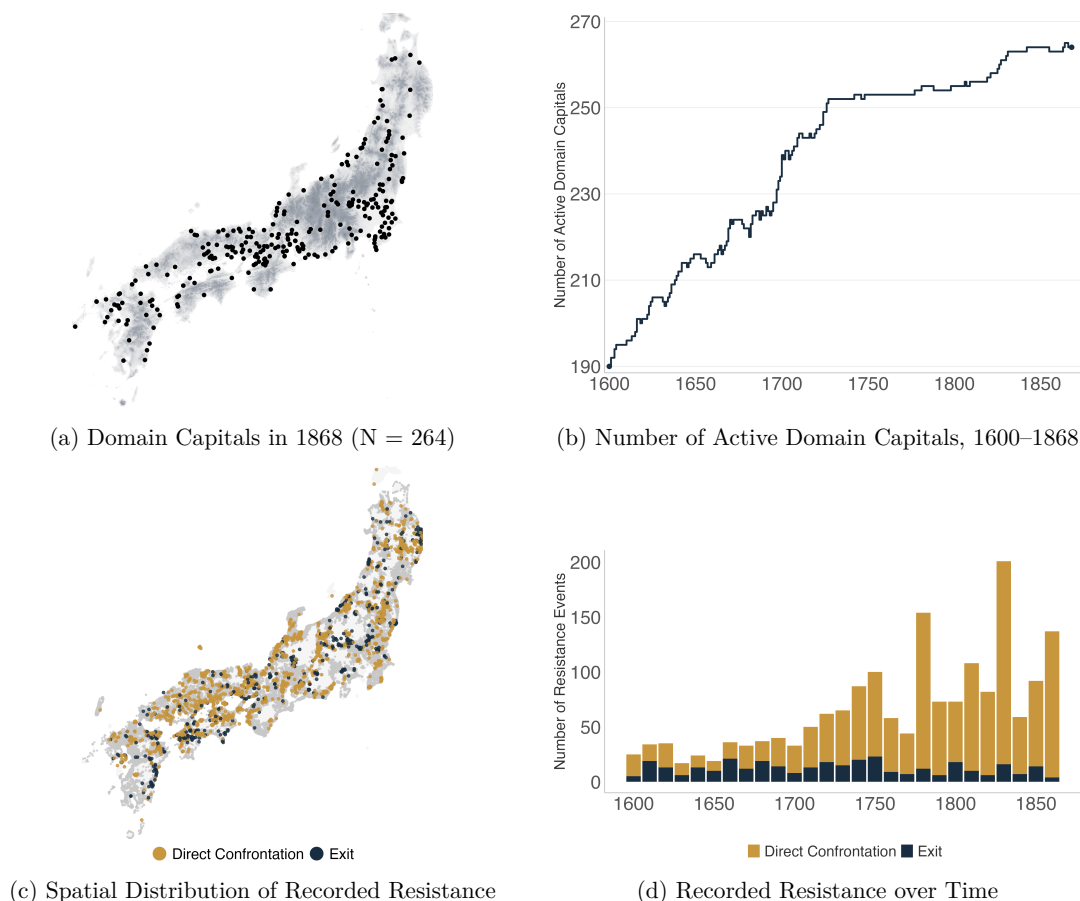


Figure 1: The Tokugawa Domain System and the Record of Peasant Resistance

Note: Tokugawa Japan was divided into roughly 300 domains, each governed from a castle town that served as its capital and housed the officials who received petitions. (a) Locations of the 264 capitals active in 1868, plotted over terrain; the archipelago runs about 1,900 km southwest to northeast. (b) The number of active capitals each year, 1600–1868, which changed as the shogunate created, merged, transferred, and abolished domains. (c) Villages recorded as resisting, colored by whether they exited (left the domain or petitioned an outside authority) or confronted their lord directly; gray points are villages present through the end of the period with no recorded resistance. (d) Recorded episodes per decade.

as *direct confrontation* and the last two as *exit*.^[9] Figure 1(c) maps these incidents, and Figure 1(d) plots their frequency over time.

I call the common category *exit through alternative authority*. Desertion and end-run appeals both moved a dispute outside the home domain and enlisted another authority, so physical movement alone is not the defining feature. Across three routes, villagers first removed either themselves or their claim from the home lord’s exclusive control. An outside authority then received the villagers, recognized their petition, or protected their

representatives. That intervention transmitted leverage back to the home lord through lost revenue, inter-domain mediation, or shogunal sanction.

In *physical relocation* (*chōsan*), villagers collectively crossed into a neighboring domain. The receiving lord then decided whether to shelter them or return them to the home domain. Shelter removed the villagers and the tax obligations assessed to their community from the home lord's immediate control, threatening his revenue directly. The receiving lord's protection made departure sustainable while the two sides bargained.

A second route involved a *neighbor-routed appeal*. Its leverage came from intervention rather than fiscal loss alone. Villagers placed themselves under a rival lord's protection and enlisted him in the dispute. The receiving lord could shelter them, transmit their demands, and mediate with the home lord. His involvement publicized the home lord's failure to govern and converted a village grievance into a dispute between jurisdictions.

Villagers could also make a *direct appeal to the shogunate* (*osso*), placing the claim, rather than necessarily the entire village, outside the home domain. Representatives bypassed the home lord and addressed the shogunate, invoking hierarchical oversight rather than competition for subjects. The shogunate could thereby discipline the home lord without requiring a rival lord to compete for the villagers. All three routes thus reached beyond the home domain, but they differed in who moved, who received the claim, and how the resulting pressure reached the home lord.

These routes imply different relationships with geographic exposure. Physical relocation and neighbor-routed appeal require a reachable rival ruler willing to receive villagers. Direct appeal to Edo does not: even a village far from rival capitals could send representatives to the realm's capital. The pooled exit measure therefore combines routes that exposure should and should not predict. To isolate horizontal exit by action rather than inferred destination, [Supporting Information C.5](#), [Supporting Information D.2.8](#), and [Supporting Information D.2.9](#) repeat each principal behavioral analysis for desertion alone. The findings persist under this restriction.

For clarity, six institutional terms recur in the analysis. A *daimyō* held a domain in return for service to the shogunate, which could reduce, relocate, or abolish his holding. *Tenryō* was land governed directly by the shogunate and administered by a shogunal

intendant (*daikan*) from a district office (*daikansho*). *Chōsan* was a village’s collective departure, whereas *osso* was a petition that bypassed the home lord for an authority above or outside him. These distinctions determine who could protect a village and who could punish its lord.

The three forms of direct confrontation differed in intensity (Aoki 1975). Coercive appeals were mass marches on domain offices or castles to present demands backed by an implicit threat of violence. Crowds of thousands wore straw raincoats and carried bamboo spears as symbols of collective resolve (Walthall 1997, 131–132). Destruction targeted the property of officials, merchants, or wealthy villagers associated with oppressive policies. Organized groups “smashed” (*uchikowashi*) houses and storerooms (Walthall 1997, 89–130). Rebellion was armed uprising against domain authority, the rarest form at roughly 1% of recorded incidents (White 1995, 142). All three sometimes won concessions but often met severe repression.

The categories describe recorded actions, not a mutually exclusive repertoire across an entire protest sequence. As the Morioka example in Section 2.3 shows, the analysis retains combined tactics as distinct event records rather than forcing the broader sequence into one category.

Episodes are coded by the villagers’ action, not the lords’ subsequent response. When a neighboring lord negotiated for a sheltered village, the episode remains exit even though the dispute shifted to the two lords. Armed conflict between domains would instead be an inter-domain dispute, but no sampled episode took that form.^[10]

3.2.1. Why Non-Compliance Was Consequential

Exit publicized governance failure more readily than direct confrontation. The Buke Shohatto (Laws of Military Houses) required domains to be “well-governed,” and failure gave the shogunate grounds to confiscate, reduce, or transfer a lord’s territory (Steele, Paik, and Tanaka 2017, 3–5). Direct confrontation usually remained internal, allowing the daimyō to contain information or control its reporting. Exit necessarily crossed jurisdictions and placed villagers under a rival lord or shogunal intendant, who had

^[10]Inter-domain warfare was effectively absent after 1615, and the shogunate adjudicated disputes between lords.

independent reasons to report the episode. A fleeing village thus exposed its lord to rivals and the shogunate, making attainer risk harder to suppress (Steele, Paik, and Tanaka 2017). This publicity gave daimyō strong incentives to moderate extraction where villages could reach outside authority. Mass flight was accordingly “what the warrior class feared most” (Kuromasa 1927, 69) and was prohibited by strict decree.

3.2.2. Landholding, Tax Liability, and the Cost of Exit

The fiscal mechanism applies most directly to registered landholding households. Seventeenth-century cadastral surveys in shogunal territory matched parcels to cultivating stem families, assessed output and tax, and protected household claims conditional on payment (Nakabayashi et al. 2020, 158, 175). Peasants could use, buy, sell, rent, and inherit land; partible inheritance was rare, and the main family generally inherited all or most land (Kumon 2024; 2026, 158, 160; Nakabayashi 2025). A durable cultivation claim and recurring fiscal duty were thus two sides of the same institution.

Tenure nevertheless varied regionally. Less productive areas retained joint village ownership, covering about 30% of assessed national output, under which villages assigned and sometimes reallocated parcels (Brown 2011; Nakabayashi et al. 2020, 175–176). Kaga combined corporate repartition with inheritable and transferable household use rights (Brown 1993, 94–95, 107–108). Land access was therefore place-specific—attached either to a registered household or to membership in a landholding village. Prolonged departure could weaken practical access, and permanent resettlement abandoned a valuable asset, but the sources do not show that border crossing automatically extinguished title. The 1873 Land Tax Reform standardized and transformed these prior claims rather than creating them from nothing (Nakabayashi et al. 2020, 176–177).

The costs on each side played different strategic roles. Outside shelter and possible absorption enabled villagers to remain away if the home lord refused to concede, making continued loss of cultivators and tax payments a feasible sanction rather than an empty threat. Villagers nevertheless lost income, incurred shelter costs, and risked weakening access to household- or village-held land while away. Once the lord granted sufficient relief, returning allowed villagers to recover those benefits while restoring his tax base. Continued departure is therefore the sanction that sustains the threat; concession followed

by return is the benchmark path assumed in the formalization, not a bargaining outcome derived there. Kaga officials followed this pattern by waiving arrears, offering rice, promising to correct abuses, and calling cultivators home (Brown 1993, 138). Nobeoka likewise granted the departing villagers' demands before they returned (Steele, Paik, and Tanaka 2017, 356).

Aoki prints peasant uprisings separately from urban disturbances, *murakata sōdō*, and province-wide petitions. The project comparison ends with the uprising chronology; all 294 unambiguous event-page matches, including 115 in the analytic sample, fall there, so excluding the separately printed categories removes no observation (Table A.6). Household tenure within included entries is less certain. Removing all nineteen incidents mentioning tenants, landlords, or *komae* smallholders leaves the preferred village-choice estimate positive and statistically significant (Table A.7). I therefore interpret the estimates as behavior by recorded village groups, not a landholder-only effect.

3.2.3. Exit as Deliberate Protest

Desertion and end-run appeals both depended on outside authority, and villagers sometimes combined them by petitioning while others sheltered abroad (Hosaka 2002; Miyazaki 1995). Receiving authorities supplied shelter, guards, mediation, or recognition. Continued shelter deprived the home lord of cultivators, publicized the grievance, and preserved possible absorption; return after concessions was therefore a bargaining outcome, not evidence that departure lacked force (Miyazaki 1995; Vlastos 1986). Aoki records return after seven sampled exits or attempts, but sparse duration and resolution details support only the claim that return occurred, not a representative rate (Table A.8).

Receiving lords had two reasons not to collude in automatic return. Absorbing cultivators increased their revenue, while surrendering them benefited a rival (Miyazaki 1995). Reporting misgovernment could also prompt shogunal sanctions, whereas quiet return yielded no comparable gain. This asymmetry is the obstacle a ruler cartel must overcome (Acharya and Lee 2018).

Systematic destinations are unavailable. A project-derived field codes a destination or addressee for 176 of 328 unique exit events, but Aoki prints no destination column and the coding rule is undocumented (Table A.9). Its *foreign* and *edo* values mix physical

locations, home authorities, internal destinations, Osaka, Kyoto, and home lords' Edo residences. I therefore draw no institutional claim from them. Distance to a foreign capital clearly governs desertion, but only some end-run appeals passed through a neighbor. Desertion-only estimates isolate horizontal exit and reproduce the main findings ([Supporting Information C.5](#); [Supporting Information D.2.8](#); [Supporting Information D.2.9](#)).

3.2.4. How Exit Worked: Illustrative Cases

Three *hyakushō ikki* episodes show how neighboring rulers made exit effective through shelter, mediation, and possible escalation to the shogunate.

In Nobeoka (1690), 1,418 peasants fled oppressive taxation and floods to Takanabe. Its officials sheltered them, mediated, guarded representatives, and helped escalate stalled negotiations to Edo. After ten months, Nobeoka granted all demands, including tax relief, without punishment, and the villagers returned ([Steele, Paik, and Tanaka 2017, 356](#)). The shogunate transferred the Nobeoka lord to a much smaller domain the next year ([Ishikawa 1981](#)).

In Yamashirodani (1837), border villages left Tokushima for Imabari. The Imabari lord sent guards, mediated, and threatened to absorb them if Tokushima rejected their petition ([Okamura 2009](#)). Absorption would have increased Imabari's revenue ([Miyazaki 1995](#)).

In Matsuyama (1741), famine-stricken villages fled to neighboring Ōzu and requested migration. Their lord accepted most demands, punished his chief minister rather than revolt leaders, and the villagers returned ([Nakagawa 1980](#)).

The 1798 Mimasaka case attributed to Kurushima's chapter shows the vertical route. Village federations coordinated across four administrative divisions of shogunal land, including a tract administered in custody by Tatsuno. After the finance commissioner returned their *ishidai* appeal, representatives petitioned a senior councillor; investigation

^[11]The Kurushima monograph was unavailable for direct inspection. I verified the chapter and page range (pp. 185–198) through the National Diet Library and Ibaraki Prefectural Library catalogs and cross-checked the episode against the University of Tokyo dissertation abstract and the official *Tsuyama City History*, vol. 4, pp. 279, 281–289.

produced withdrawal of the challenged rule and a modified formula (Kurushima 2002).

[11] The case concerns shogunal land, not ordinary daimyō territory.

Customary legitimacy and ritual also shaped action (Vlastos 1986; Walthall 1997; White 1995). The strategic analysis is complementary: norms shaped the available repertoire, while relative costs shaped the selected tactic.

3.3. Exposure and Exit Occurrence

Two designs examine whether recorded exit occurs more often under greater geographic exposure: a conversion-based difference-in-differences design and an observational grid-year panel.

3.3.1. Conversion-Based Test: Tenryō Conversion

This design associates conversion with 3.9 percentage points more recorded exit among nearby villages governed by surrounding lords and no detectable average change in their direct confrontation. I use the Tokugawa shogunate’s abolition of domains and conversion of their territories into direct shogunal land (*tenryō*), an institutional change that introduced a powerful nearby authority.

Conversion was a general instrument of shogunal lordship, not a policy directed at neighboring villagers. Daimyō governed fiefs as invested vassals: the shogunate could revoke, reduce, or transfer their holdings, but could not tax the inhabitants while the territory remained daimyō land. When a ruling house failed to produce an heir or a lord lost his tenure through misconduct or political sanction, direct administration shifted the fiscal rights to the shogunate, installed its own officials, and preserved its discretion to re-enfeoff the territory later. This explains why the same disposition followed heterogeneous internal triggers. Attainder and transfer also fell disproportionately on smaller *fudai* houses whose tenure the shogunate could alter most readily, rather than on domains selected because neighboring peasants had become unusually mobilized (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989; Ravina 1999, 17–21).

A tenryō conversion opened a credible appeals route to the shogunate. A shogunal intendant (*daikan*) installed at the former domain capital allowed nearby villagers to petition locally rather than travel to Edo. Conversion also set a recent precedent for

absorbing adjacent territory, while contiguity with shogunal land lowered the administrative cost of further attainer. The Buke Shohatto’s “well-governed” mandate ([Section 3.2](#)) had already underwritten the seizure of the Arima daimyō after the Nobeoka revolt (1690). Conversion thus made exit-driven petitions cheaper to file and more likely to expose surrounding lords to seizure.

These conversions test [Prediction 1](#) before home rulers adjust their tax rates. A nearby shogunal appeals route lowers the effective cost of exit and should raise exit, but not direct confrontation, in villages of neighboring domains. Because internal disorder sometimes triggered conversion, the null on nearby confrontation also tests whether unrest simply spilled across domain borders.

This selection process does not itself violate the difference-in-differences design. The shogunate selected a focal daimyō’s tenure for conversion, whereas the outcomes exclude that domain and count resistance only in villages belonging to surrounding lords. Succession failure, punishment, or fiscal reorganization inside the converted domain threatens identification only if it reflects a regional shock that independently changes subsequent resistance in those foreign villages. Focal-domain fixed effects absorb persistent differences around capitals, year fixed effects absorb shocks common to all domains, and the pre-conversion hazard estimates show that lagged nearby exit and confrontation do not jointly predict conversion timing ($p = 0.344$; [Supporting Information C.13](#)). Excluding conversions connected to documented unrest also leaves the exit estimate positive and confrontation null ([Table A.19](#)). These checks do not require every conversion to have the same cause; they test whether the causes generate the specific cross-border outcome trend that would confound the design.

[Aoki \(1975\)](#) catalogs 7,664 incidents of peasant resistance from local histories, official documents, and diaries.^[12] Each observation records the date, home domain, nearest neighboring domain,^[13] and resistance mode, which I classify as exit or direct confronta-

^[12][Hosaka \(2002\)](#) characterizes [Aoki \(1975\)](#) as the culmination of postwar Japanese research on peasant uprisings and “one of the largest contributions in Japanese historical studies after WW2.”

^[13]I used [Kodama and Kitajima \(1989\)](#) to identify the neighboring domains existing at the time of each resistance event. The nearest foreign domain is identified using time-varying domain existence data: for each event, only domains active at that date are potential alternative authorities.

tion under [Section 3.2](#). I exclude 574 peaceful petitions and inter-village resource or boundary disputes.

The catalog identifies 2,790 sites. I georeference 2,571 of them using several sources and exclude those whose locations remain imprecise.^[14] The resulting data contain 910 resistance episodes involving 2,571 villages and 2,741 tactic choices ([Table A.5, Supporting Information B](#)). Because large mass exits were more likely than small flights to enter the historical record, unrecorded minor cases are unlikely to account for the main results. Capital locations and the creation and abolition dates of all 294 domains come from [Kimura, Fujino, and Murakami \(2015\)](#) and [Kodama and Kitajima \(1989\)](#) ([Table A.10, Supporting Information B](#)); I assign branches (shi-han) to their main domains (hon-pan).

I construct an annual domain-year panel for 1615–1868. For each domain-year, I count exit and confrontation events in villages of other domains within $r \in \{30\text{km}, 40\text{km}, 50\text{km}\}$ of the capital, the zone exposed to the new exit opportunity. [Figure 2](#) illustrates the design.

Consider a domain abolished in year t and converted to tenryo. The outcomes do not come from villages inside that converted territory. They come from villages that remain governed by surrounding lords but lie within radius r of the converted capital. For the treated domain, the design compares resistance in those foreign villages during the five years before conversion with resistance during the five years after a shogunal intendant arrives. It then compares that before-and-after change with the contemporaneous change in foreign villages surrounding domains that never converted. The converted capital supplies the new outside authority; villages in neighboring domains supply the measured exit and confrontation outcomes. Domain fixed effects absorb persistent differences around focal capitals, and year fixed effects absorb shocks common to all domains. The coefficient estimates whether resistance around a converted capital changed more at conversion than resistance around never-converted capitals observed in those same years.

I exclude four abolitions that transferred territory directly to another domain rather than to tenryo. The 55 treated domains come from 79 dissolutions across all 294 domains

^[14]These sources include [Honda, Natsume, and Nemoto \(2022\)](#), the geocoding API provided by the Ministry of Land, Infrastructure, Transport and Tourism of Japan, and the Historical Location Name Comprehensive Database provided by the Center for Open Data in the Humanities.

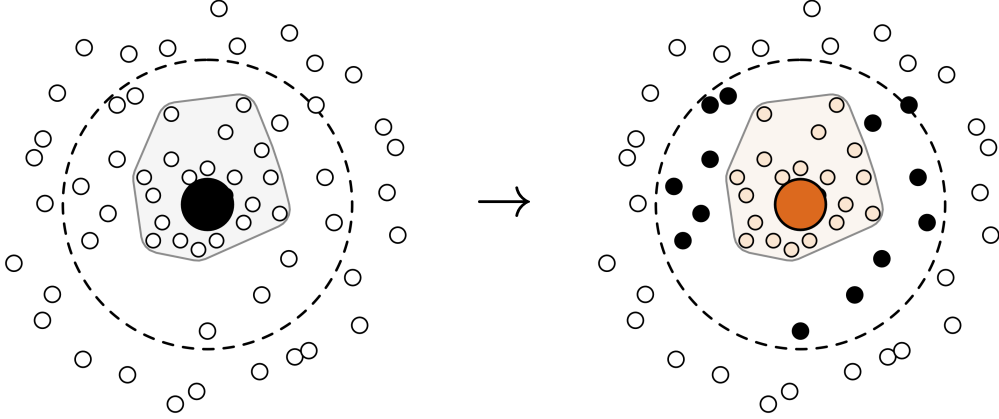


Figure 2: DiD Design: Tenryō Conversion

Note: Stylized illustration. The large central circle is the focal capital, the gray polygon is its territory, small circles are villages, and the dashed circle is the analysis radius r . Left: before conversion, villages of surrounding lords within r lack a nearby shogunal authority. Right: abolition converts the focal capital and territory to direct shogunal rule (*tenryō*, orange); black circles identify surrounding foreign villages inside r , while pale circles inside the gray territory are excluded from the outcomes and open circles fall outside the treated foreign-village set. No village-to-territory arrows are shown because episode destinations are unobserved. The DiD compares changes in exit and confrontation in surrounding foreign villages with changes around never-converted domains.

and are listed by year and stated reason in [Supporting Information C.1](#) (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989). I restrict converted domains to five years before and after their first conversion, retaining strict pre-conversion years and post-conversion years under direct shogunal rule. Never-converted controls enter only for the same calendar years. I estimate a domain-year linear probability difference-in-differences model with domain and year fixed effects:

$$\text{NonCompliance}_{j,t} = \beta_1 (\text{Converted}_j \times \text{Post}_{j,t}) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

Here, $\text{NonCompliance}_{j,t} \in \{\text{Exit}_{j,t}, \text{Confront}_{j,t}\}$ indicates any nearby exit or confrontation within r kilometers of domain j 's capital in year t . Converted_j identifies ever-converted domains, and $\text{Post}_{j,t}$ identifies years after first conversion under direct shogunal rule.

Destination misclassification motivates the first checks. The design cannot observe the destination of every exit. Sources usually record where villagers came from but rarely

where or how they traveled, so proximity identifies converted territory as a plausible outside option rather than a confirmed destination. [Figure 2](#) accordingly depicts exposure within the analysis radius without village-to-territory arrows. At the tighter 40km and 30km radii, the exit coefficients remain positive at 0.0262 ($p = 0.025$) and 0.0163 ($p = 0.038$), respectively, while nearby confrontation remains near zero at every radius. The exit estimate also remains positive when the outcome is restricted to desertion ([Supporting Information C.5](#)). A general problem of geographically mismeasured unrest would not readily produce all three patterns.

Conversion endogeneity motivates a separate exclusion exercise. A page-level local-source and Aoki chronology audit identifies three cases clearly connected to peasant unrest or petitioning—Saijō (1665), Shimabara (1668), and Hōjō (1712)—and two mixed cases, Numata (1681) and Ōmi Komuro (1788). Removing the three clear cases increases the exit estimate from 0.0385 to 0.0421 (two-way cluster $p = 0.0067$); removing all five yields 0.0397 ($p = 0.0130$). A maximal screen also removes five conversions preceded within ten years by an Aoki event in the same named domain or province, even though the histories do not identify those events as causes; the exit estimate is 0.0443 ($p = 0.0117$). The confrontation placebo remains near zero in all three samples ([Table A.19](#)). These screens show that the result does not depend on the observed suspect cases, but silence in a domain history is not affirmative proof that no unrest occurred.

Local source volumes permit a separate treatment-status audit of all 55 rows. Twenty establish direct shogunal administration at the recorded date, twenty-one are partial reductions, family partitions, transfers, or otherwise noncomparable events, ten leave the intervening administrator unstated, and four use the wrong date. Correcting Takayanagi to 1739, Kōfu to 1724, and Sagara to 1787 yields a strict 23-case screen; Numazu is excluded because no transition occurred in 1724. The exit estimate remains positive at 0.0385 (cluster $p = 0.0903$; Conley $p = 0.0338$). An ambiguity-inclusive corrected screen of 33 cases yields 0.0310 (cluster $p = 0.0682$; Conley $p = 0.0340$). After the three clear unrest cases are also removed, the corresponding estimates are 0.0447 and 0.0366 ([Table A.18](#)). The strict subset also produces a positive confrontation estimate, whereas confrontation remains statistically indistinguishable from zero in the ambiguity-inclusive

Outcome:	Any Nearby Exit			Any Nearby Confrontation		
Sample Radius:	30km	40km	50km	30km	40km	50km
<i>Variables</i>						
Converted $\times$ Post	0.0163	0.0262	0.0385	0.0084	0.0033	0.0076
Cluster SE	(0.0078)**	(0.0116)**	(0.0147)***	(0.0093)	(0.0092)	(0.0122)
Conley SE (70km)	(0.0084)*	(0.0107)**	(0.0131)***	(0.0104)	(0.0108)	(0.0136)
<i>Controls</i>						
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num. Obs.	33,877	33,877	33,877	33,877	33,877	33,877
Tenryo-Converted/All Domains	55/294	55/294	55/294	55/294	55/294	55/294
R2	0.038	0.052	0.062	0.078	0.101	0.136
Mean Dep. Var.	0.010	0.018	0.026	0.020	0.031	0.044

Table 1: Domain-to-Tenryo Conversion and Nearby Exit

Note: LPM estimates from an annual domain-year panel. “Nearby” denotes resistance in villages of *other* domains within the specified radius. Converted domains are restricted to `rel_year` in $[-5, +5]$ and matched to never-converted controls in the same calendar years. SEs in parentheses are (i) two-way clustered by domain and year and (ii) Conley (70km) on capital coordinates; stars use the corresponding estimator’s p -value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

screen. The audit therefore preserves the positive exit association but shows that the conversion design’s exit-specific contrast depends on the broader defensible treatment envelope; the grid and village-choice designs carry the tactic-specific inference.

Table 1 reports the main result. At 50km, tenryo conversion is associated with a 0.0385 increase in nearby exit probability ($p = 0.010$), or 3.9 percentage points. This difference is roughly 59% of the control group’s pre-conversion mean of 0.065. By contrast, the confrontation estimate is only 0.0076 ($p = 0.532$). The baseline treatment universe therefore shows more exit without a detectable parallel change in direct confrontation.

The conversion-timing checks begin with the assumption that converted and never-converted domains would have followed parallel resistance trends absent conversion. I assess this assumption with an event study:

$$\text{NonCompliance}_{j,t} = \sum_{s \neq -1} \beta_s (\text{Converted}_j \times \mathbf{1}[t - t_j^* = s]) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

where Converted_j denotes tenryo-converted domains, t_j^* is the first conversion year, and $s = -1$ is the reference period. In Figure 3, the pre-treatment coefficients are individually

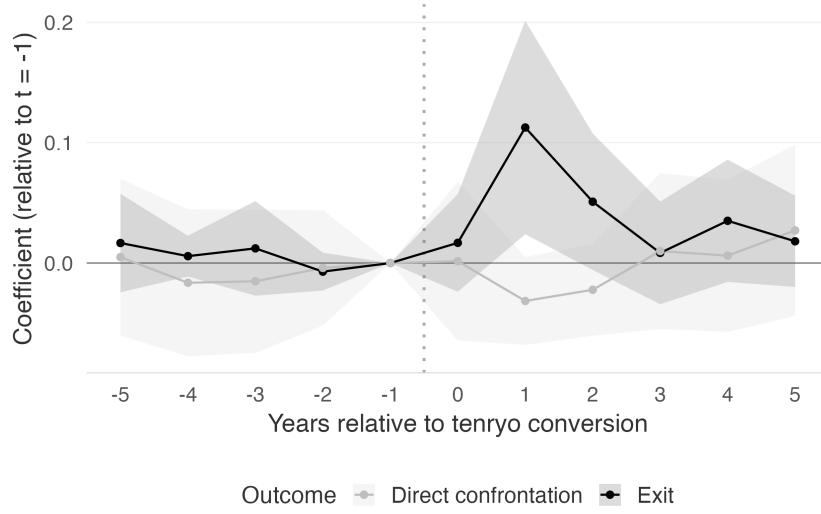


Figure 3: Tenryo-Conversion Event-Study Coefficients

Note: Yearly event-study coefficients for nearby exit (black) and direct confrontation (light gray), with $s = -1$ as the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between $s = -1$ and $s = 0$. “Nearby” denotes resistance in villages of *other* domains within 50km of the focal capital. Standard errors are two-way clustered by domain and year.

small and jointly insignificant ($F = 1.26$, $p = 0.283$; confrontation: $F = 0.18$, $p = 0.947$). After conversion, exit rises at $t = 1$ ($\hat{\beta}_1 = 0.113$, $p < 0.05$) and remains elevated, while confrontation stays flat.

Three diagnostics bound the design’s identifying assumptions. The [Rambachan and Roth \(2023\)](#) relative-magnitudes breakdown is $\overline{M}^* = 0.20$; under larger departures the estimate remains positive but confidence intervals include zero ([Supporting Information C.9](#), [Supporting Information C.10](#)). Lagged nearby exit and confrontation do not jointly predict conversion timing ($p = 0.344$; [Supporting Information C.13](#)). Clean treated-versus-never-treated comparisons receive 94.5% of the Goodman-Bacon weight, and the Sun–Abraham estimator yields an exit ATT of 0.0413 while confrontation remains null ([Supporting Information C.7](#), [Supporting Information C.6](#)). Exclusions for nearby conversions and potentially exposed controls preserve a positive exit coefficient ([Supporting Information C.11](#)).

Spatial, functional-form, and randomization checks lead to the same qualitative pattern. Conley bandwidth sweeps, count models, and two permutation schemes produce a positive exit estimate and a near-zero confrontation estimate ([Supporting](#)

Information C.14, Supporting Information C.15, Supporting Information C.4, Supporting Information C.12). The association is concentrated in the pre-1750 cohort, which contains 49 of the 55 conversions, when attainer remained especially credible (Ravina 1999; White 1995); the later cohorts are too small to support comparable inference (Supporting Information C.8).

3.3.2. General Test: Grid-Year Panel

The conversion design estimates the response to a specific improvement in access to shogunal authority. The grid-year panel broadens the comparison: greater geographic exposure is associated with more recorded exit and no detectable increase in confrontation as nearby capitals are established, abolished, or relocated. This design tests the rate margin of Prediction 2.

I measure location-level exposure with a *distance ratio*:

$$\tilde{\rho}_{i,t} := \frac{d_{i,t}^1}{d_{i,t}^2}$$

where $d_{i,t}^1$ and $d_{i,t}^2$ are straight-line distances from location i to the nearest and second-nearest active capitals in year t . The ratio approaches one when a new capital appears closer than the prior second-nearest or when the nearest capital disappears or moves farther away. Values near zero indicate one dominant capital; values near one indicate two equally accessible rulers. Section 3.4.1 defines the village-level analog.

A village-year panel would be preferable, but domain affiliation is observed only at the end of the Tokugawa period. The governing domain of most non-resisting villages is therefore unknown in earlier years. I address this limitation by assigning villages to fixed 5km hexagons and computing $\tilde{\rho}_{i,t}$ from the two nearest active capitals without using domain affiliation. The 46,086-village late-Tokugawa *Kyūdaka-Kyūryō Torishirabe Chō* catalog defines populated cells (Section 3.4). The resulting panel contains 2,225,802 cell-years across 8,763 cells, with annual counts of exit and confrontation in each hexagon.

The estimating equation is

$$\text{NonCompliance}_{i,t} = \beta_3 \tilde{\rho}_{i,t} + \lambda_i + \eta_t + \varepsilon_{i,t}$$

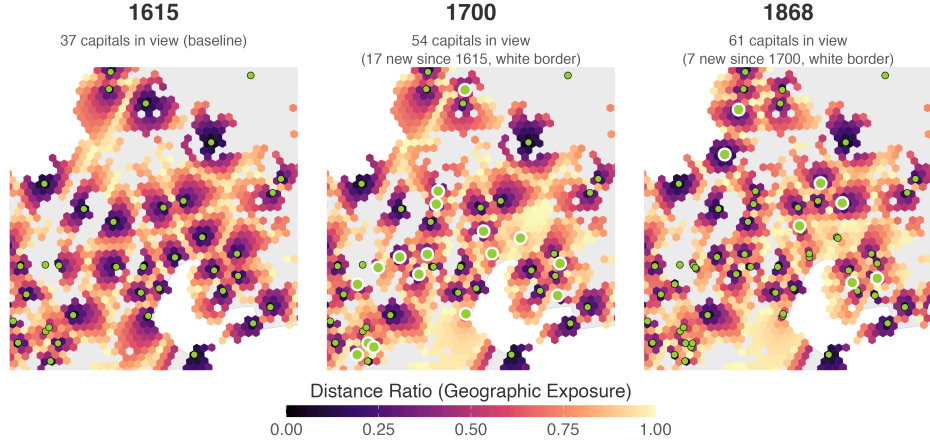


Figure 4: Time-Varying Distance Ratio across 5km Hexagonal Grid: Kinki Region

Note: Each panel maps $\tilde{\rho}_{i,t}$ (nearest-capital distance / second-nearest-capital distance) in a $\approx 200\text{km} \times 200\text{km}$ window of central Japan (lat 34.4–36.2°N, lon 135.5–137.6° E), covering Kyoto, Osaka, Nagoya, and Lake Biwa. This region has the most cells whose ratios changed across the three years. Yellowgreen circles mark active capitals; white borders mark capitals added since the previous snapshot. Yellow indicates greater geographic exposure and purple a dominant nearest capital. Changes reflect domain creation and dissolution.

with hexagon and year fixed effects, where $\text{NonCompliance}_{i,t}$ is the exit or confrontation count. Identification comes from capital dissolutions, transfers, and reestablishments that reorder the nearest authorities. Figure 4 maps $\tilde{\rho}_{i,t}$ in three years: broad spatial patterns persist, while domain creation and dissolution change exposure locally. Hexagon and year fixed effects absorb time-invariant geography and common shocks.

Table 2 shows that the distance ratio is associated with exit but not confrontation, whereas remoteness from any authority predicts neither. In columns (1)–(3), a one-unit increase in the ratio is associated with a 0.035-percentage-point increase in cell-year exit probability. The estimate is significant at 5% with both cell-and-year clustering ($p = 0.048$) and Conley inference ($p = 0.037$). The smaller positive estimate for confrontation is not statistically significant. Because the baseline exit probability is only 0.022%, a one-standard-deviation increase in the ratio ($\text{SD} = 0.26$) corresponds to roughly 40% of that mean.

Columns (4)–(6) distinguish geographic exposure from simple remoteness by using distance to the closest active capital $d_{i,t}^1$ and discarding the second-nearest. The exit coefficient is small and marginal with clustered standard errors ($p = 0.092$) but null with

Outcome:	Distance ratio ($\tilde{\rho}_{i,t}$)			Closest-capital dist. ($d_{i,t}^1$, km)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any confr.	Any confr. (strict)	Any exit	Any confr.	Any confr. (strict)
Coefficient	0.00035	0.00026	-0.00012	0.00000	0.00000	0.00000
Cluster SE	(0.00017)**	(0.00019)	(0.00011)	(0.00000)*	(0.00000)	(0.00000)
Conley SE (10km)	(0.00017)**	(0.00023)	(0.00012)	(0.00000)	(0.00000)	(0.00000)
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802
R ²	0.006	0.007	0.006	0.006	0.007	0.006
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00022	0.00064	0.00026

Table 2: Geographic Exposure Predicts Recorded Exit

Note: LPMs on any-event indicators at the 5km hexagon-year level for all 8,763 cells. Columns (1)–(3) use the annual distance ratio $\tilde{\rho}_{i,t} = \frac{d_{i,t}^1}{d_{i,t}^2}$ from the two closest active capitals. Columns (4)–(6) use distance to the closest active capital $d_{i,t}^1$ (km), which measures remoteness without the second-nearest capital. “Confrontation (strict)” includes only destruction and uprising. All models include grid and year FE. SEs are (i) two-way clustered by cell and year and (ii) Conley (10km). See [Supporting Information D.2.11](#) for ever-active cells. Signif. Codes: .***: 0.01, **: 0.05, *: 0.1.

Conley standard errors ($p = 0.162$); confrontation is also null. The contrast between the two sets of columns points to the relative geometry of competing rulers, rather than distance from the nearest ruler, as the relevant condition.

Across alternative samples and measures, the exit coefficient remains positive, although inference is weaker in some specifications. This is true among the 1,466 ever-active cells ([Supporting Information D.2.11](#)), in Poisson and Negative Binomial fixed-effects models ([Supporting Information D.2.10](#)), for desertion alone ([Supporting Information D.2.9](#)), with terrain-weighted least-cost-path distances ([Supporting Information D.2.12](#)), and when daikansho enter the choice set ([Supporting Information D.2.14](#)).

3.4. Exposure Shifts the Form of Non-Compliance Toward Exit

The preceding tests ask whether exit occurrence is higher under greater exposure. This section conditions on recorded non-compliance and asks which tactic a resisting village chose: exit or direct confrontation. It estimates composition within the resisting sample, not whether exposure makes a village resist. Separating these margins matters because

more exit events could reflect more disputes rather than substitution between tactics. I test the share margin of [Prediction 2](#): whether exposure predicts a shift toward exit once a village resists.

The village and tax analyses use 46,086 villages compiled primarily from the *Kyū-daka–Kyūryō Torishirabe Chō* (Ledgers of Former Assessments and Territories, hereafter LFAT), a Meiji land census that records each village’s home domain at the end of the Tokugawa period. Village locations come from [Honda, Natsume, and Nemoto \(2022\)](#). When several domains taxed one village, I assign the village to the domain holding the largest share of its assessed productivity (*kokudaka*, a rice-denominated index of aggregate agricultural output rather than literal rice harvest alone). The data cover virtually all arable land in Japan except high-altitude, non-cultivable regions ([Figure 1\(a\)–\(b\)](#); [Figure 6](#)).

3.4.1. Measuring Exposure to Alternative Authority

I measure each village’s exposure with ρ_i , the distance to its own lord’s capital divided by the distance to the nearest other lord’s capital. Unlike the grid-level $\tilde{\rho}_{i,t}$ in [Section 3.3.2](#), which uses the two nearest active capitals, ρ_i anchors on the village’s known home domain. Proximity to an outside authority lowered exit costs by facilitating the intervention on which mass exit depended ([Section 3.2](#)): guards or mediators could protect fleeing peasants, villagers could more easily reach neighboring capitals or shogunal offices, and authorities could more readily transfer allegiance. The measure is:

$$\rho_i := \frac{d_i^{\text{HC}}}{d_i^{\text{FC}}}$$

Read the ratio as relative access. A village 10km from its home capital and 20km from the nearest foreign capital has $\rho_i = 0.5$; reversing those distances gives $\rho_i = 2$. Higher values place outside authority closer relative to home authority. Because the ratio rises when the home capital is farther away or the foreign capital is closer, Models 6 and 7 later separate the two distances. Here, d_i^{HC} is the straight-line distance from village i to its home-domain capital and d_i^{FC} is the distance to the nearest active non-home capital at the time of resistance. The data do not reconstruct village-specific historical roads, passes, checkpoints, guards, or realized travel routes, so the ratio measures geographic

access to administrative centers rather than each village’s exact travel cost.^[15] I calculate both distances from the georeferenced resistance data and the time-varying capital panel in [Section 3.3.1](#). Active capitals include existing daimyō seats and the former seats of domains converted to tenryo, where a shogunal intendant (*daikan*) replaced the daimyō ([Vlastos 1986, 49](#)). The latter remain in the choice set throughout the tenryo period.^[16]

3.4.2. Empirical Strategy: Home-Neighbor Dyad Fixed Effects

The analysis now returns from exit occurrence to the choice between exit and confrontation. Each observation is a village involved in recorded resistance in [Aoki \(1975\)](#), and the outcome indicates which of the two strategies it used. [Figure 5](#) illustrates how the design compares otherwise similar villages facing the same pair of rulers.

Home-neighbor dyad fixed effects isolate within-pair variation in the distance ratio. Each dyad is an ordered pair of home and nearest-rival domains (for example, A–B and A–C in [Figure 5](#)). Dyad fixed effects absorb time-invariant home- and neighbor-domain characteristics, including persistent differences in coercive capacity, ruler ideology, agricultural endowments, and administrative quality, as well as stable pair-specific cross-border ties, transit corridors, and border controls. They remove persistent features shared by villages with the same home lord and nearest rival; they do not make those villages identical or absorb village geography. Within an A–B dyad, the estimate compares A villages whose nearest rival is B but whose relative position between the two capitals differs, as shown within each dashed circle in [Figure 5](#). The controls account for the measured local geography listed below, and decade fixed effects absorb period-wide shocks.

The main specification is:

$$\text{Exit}_i = \beta_2 \ln(\rho_i) + X_i' \gamma_2 + \underbrace{\varphi_{j,k}}_{\text{home-neighbor dyad FE}} + \underbrace{\omega_t}_{\text{decade FE}} + \varepsilon_i$$

^[15]The water-safe village subsample in [Supporting Information D.2.4](#) excludes cross-island straight lines, and the grid-level analysis in [Supporting Information D.2.12](#) replaces great-circle distance with a terrain-weighted least-cost path.

^[16]In some cases, the shogunate may have administered converted territory from a nearby *daikansho* (a shogunal intendant’s district office) rather than from the former castle town. To address this, [Supporting Information D.2.13](#), [Supporting Information D.2.14](#), and [Supporting Information D.3.1](#) replicate all main analyses with the nearest capitals redefined to include *daikansho* locations.

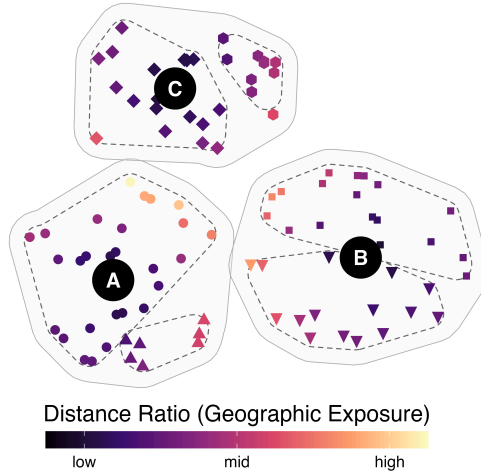


Figure 5: How Home-Neighbor Dyad Fixed Effects Isolate Within-Pair Variation

Note: A schematic, not data. Three domains (A, B, C) are drawn with their capitals as black circles and their villages as shapes shaded by the distance ratio ρ_i , the village's distance to its own lord's capital divided by its distance to the nearest other lord's capital. Dark shading marks villages sitting close to their own capital and far from any rival, light shading the reverse. A village's home domain and its nearest rival together define a dyad, shown by the dashed circles. Including a fixed effect for each dyad discards comparisons across dyads and retains only comparisons among villages facing the same pair of lords, so the estimate cannot be driven by differences between regions or between particular lords.

Here, i indexes a village-event, while j and k index its home and nearest-rival domains and t its decade. Exit_i equals one for exit and zero for direct confrontation. The logged ρ_i uses the capitals active at event i .^[17] X_i contains latitude, longitude, their interaction, logged elevation and ruggedness, logged distance to the nearer of Edo or Kyoto, and logged wet-rice suitability.^[18] The dyad fixed effects $\varphi_{j,k}$ absorb time-invariant features of each ordered home-rival pair, while ω_t absorbs decade-specific shocks.

3.4.3. Geographic Exposure Predicts Exit Share

Table 3 shows that exposure predicts exit rather than confrontation conditional on resistance. The coefficients are positive throughout and statistically significant in the preferred controlled specifications. In Model 5, with home-neighbor dyad fixed effects

^[17]The logged ratio constrains the coefficients on the two logged distances to equal magnitude with opposite signs. Linear hypothesis tests find no evidence against this restriction (Supporting Information D.3.5).

^[18]Wet rice suitability is measured as potential output under irrigation and medium input, based on the Caloric Suitability Index developed by Galor and Özak (2016).

Outcome:	Exit _{<i>i</i>}						
	Distance Ratio					Abs. Distances	
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Variables</i>							
Distance ratio ($\ln(\rho_i)$)	0.0458	0.0519	0.0201	0.0212	0.0244		
Cluster SE	(0.0154)***	(0.0141)***	(0.0071)***	(0.0072)***	(0.0068)***		
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0078)***	(0.0078)***	(0.0085)***		
$\ln(d_i^{\text{HC}})$						0.0160	0.0203
Cluster SE						(0.0104)	(0.0083)**
Conley SE (10km)						(0.0106)	(0.0113)*
$\ln(d_i^{\text{FC}})$						−0.0306	−0.0309
Cluster SE						(0.0129)**	(0.0135)**
Conley SE (10km)						(0.0138)**	(0.0148)**
<i>Controls</i>							
Decade FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No	Yes	No
Home-Nbr Dyad FE	No	No	No	No	Yes	No	Yes
Geography	No	No	No	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>							
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.225	0.621	0.626	0.678	0.627	0.678
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246	0.246	0.246

Table 3: Exposure to Alternative Authority Predicts Exit (Exit versus Direct Confrontation)

Note: OLS estimates of each village’s exit-versus-confrontation choice conditional on resisting. Models (1)–(5) use the log distance ratio; Models (6)–(7) replace it with the two absolute distances (d_i^{HC} : home capital; d_i^{FC} : nearest foreign capital). Full results in [Table A.32](#). SEs in parentheses: (i) clustered on home, nearest-neighbor, decade; (ii) Conley (10 km) on village coordinates; stars reflect that estimator’s p -value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

and full controls, a one-standard-deviation increase in the log distance ratio (≈ 1.58) raises exit probability by 3.9 percentage points, or 16% of the 24.6% baseline rate. The less restrictive home- and neighbor-FE specification (Model 4) yields a statistically indistinguishable 3.4-point estimate.

Models 6 and 7 separate the two distances because the ratio combines two distinct accounts. Greater distance from the home capital could increase exit through weaker administration at the periphery; shorter distance to the foreign capital instead increases access to an outside authority. In the preferred dyadic specification (Model 7), the home-capital coefficient is 0.0203 (cluster $p = 0.022$; Conley $p = 0.073$), while the foreign-capital coefficient is -0.0309 (cluster $p = 0.031$; Conley $p = 0.037$). The two coefficients are statistically consistent with the ratio’s equal-and-opposite restriction ($p = 0.924$; [Supporting](#)

Information D.3.5). A one-standard-deviation decrease in $\ln(d_i^{\text{FC}})$ (≈ 0.92) corresponds to 2.8 percentage points more exit, or 11% of the baseline rate. Foreign-capital proximity is therefore the larger and more robust component, although the home-capital estimate does not permit ruling out weaker peripheral administration.

Measurement and sample composition motivate one set of robustness checks. The preferred exit coefficient remains positive with alternative productivity and political-distance controls (Supporting Information D.2.1), after excluding repeat-resistance villages (Supporting Information D.2.2), and when dropping cross-island dyads whose straight-line routes cross sea straits (Supporting Information D.2.4). It also remains positive when adding shogunal intendant offices (*daikansho*) to the measured choice set (Supporting Information D.2.13).

Restricting exit to desertion retains 58–70% of the pooled point estimates across all three behavioral designs. The 50km conversion estimate is 0.0266 (cluster $p = 0.031$; Conley $p = 0.027$), the preferred village-choice coefficient is 0.0141 (cluster $p = 0.051$; Conley $p = 0.041$), and the grid-year coefficient is 0.000241 (cluster $p = 0.082$; Conley $p = 0.048$). These estimates isolate horizontal exit by observed action without inferring unrecorded destinations (Supporting Information C.5; Supporting Information D.2.8; Supporting Information D.2.9).

Alternative explanations motivate the other checks. Controlling for distance to the nearest domain border changes the distance-ratio coefficient little, while border distance itself is not associated with exit choice (Supporting Information D.3.2). A sequential choice decomposition merges the resisting sample with the comprehensive village ledger. It finds no detectable association between exposure and resistance rather than compliance in the main specification and yields an exit-versus-confrontation estimate similar to Model 5. A symmetric time-averaged exposure measure produces a small negative first-step estimate while preserving the positive composition estimate (Supporting Information D.3.3). The evidence therefore distinguishes resistance incidence from tactic composition rather than treating compliance as absent.

3.5. Exposure and Constrained Taxation

If viable exit disciplines rulers, exposed domains should tax less to retain their population (Prediction 3). The economic sequence begins with competition over scarce labor. A neighboring ruler able to employ additional cultivators has reason to receive them, making the village's exit option credible. The home ruler then anticipates losing both cultivators and the tax base they support and moderates extraction. When neighbors lack the capacity to absorb labor, geographic access alone creates a weaker threat. Systematic tax data survive only as a late-period cross-section, so this section seeks a suggestive fiscal trace rather than a causal estimate: a pooled relationship between exposure and taxation cannot distinguish exit from state-capacity decay. Neighbor characteristics provide a sharper test. The exit mechanism predicts a negative tax gradient only where neighboring domains are attractive enough to absorb migrants. Generic peripheral state-capacity decay instead predicts lower taxation where home administration is weak, regardless of neighboring labor demand, and therefore predicts no such asymmetry. I measure domain exposure by averaging the village distance ratio within each domain. Figure 6 maps the log ratio (panel a) and reported effective tax rates (panel b).

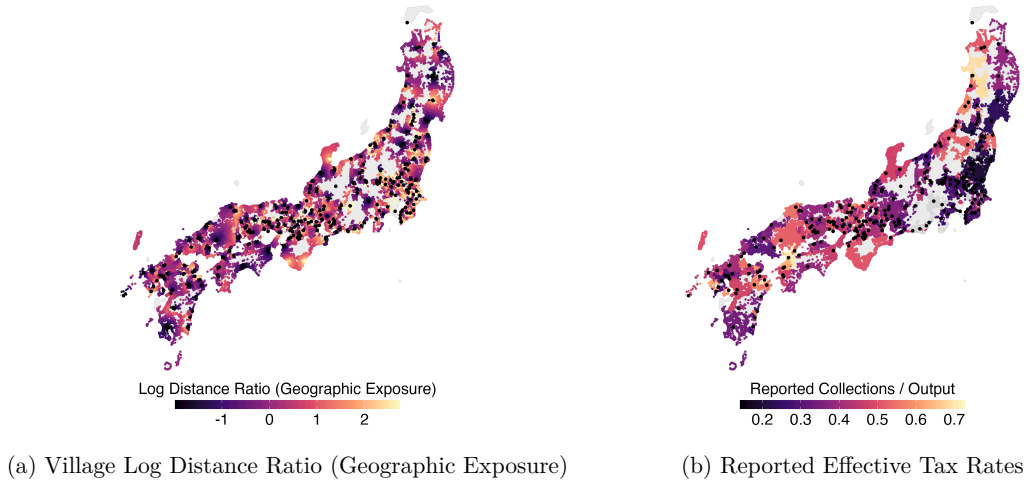


Figure 6: Geographic Exposure and Tax Rates Across Villages

Note: Panel (a) plots village locations shaded by the log distance ratio $\ln(\rho_i) = \ln(d_i^{\text{HC}}/d_i^{\text{FC}})$, where HC is the home-domain capital and FC is the nearest foreign-domain capital; higher values indicate greater geographic exposure. Panel (b) shades those villages by their domain's reported collections divided by reported output, the effective tax-rate measure used in the analysis. Gray points denote villages in domains without tax data, and black dots mark domain capitals.

3.5.1. Empirical Strategy

Capital locations could be endogenous if rulers placed capitals or exchanged border villages to deter exit. Most capitals, however, were inherited from the Sengoku period (Hall 1955) and fixed by the 1615 one-castle-per-domain edict (*ikkoku ichijō rei*); later relocations required shogunal approval and were rare. The shogunate also placed fudai vassals strategically and kept tozama lords on the periphery (Ravina 1999). I therefore control for each daimyō’s alliance status.^[19] This also absorbs political ties that could affect both shogunal approval of relocations or village transfers and extractive capacity. Any deliberate adjustment to deter exit would reduce exposure and thus attenuate the negative tax relationship.

The home-capital distance could also proxy for domain scale, peripheral state capacity (Sng and Moriguchi 2014), or village development. Controls therefore include domain population,^[20] agricultural productivity, territory size, the *gōshi system* (lower-ranking samurai serving as village officials, which raised monitoring costs), and the share of villages named “shinden” (new rice field). Tax-delivery costs are unlikely to drive the result because villages used nearby storage facilities rather than domain capitals.^[21] Tokugawa institutions also limited local agents’ authority (Section 3.1). Residual confounding by scale or coercive capacity should attenuate the estimate because larger domains could field larger armies and enforce higher rates.

The outcome is the domain effective tax rate from Kodama and Kitajima (1989) (pp. 425–431): *shunodaka* (reported collections in rice equivalents) divided by *uchidaka* (the domain’s internal record of actual production, distinct from its public *kokudaka* assessment). This ratio measures the share of reported output collected.^[22] The Meiji

^[19]Daimyō are classified according to their relationship with the shogunate: shinpan (relatives), fudai (hereditary vassals who fought for the shogunate at Sekigahara in 1600), and tozama (outside lords who became allies only after the battle).

^[20]Population is measured at the end of the Tokugawa period, so variables that use it should be read as late-period proxies rather than time-invariant characteristics.

^[21]Fukui Prefecture (1994: 335-346) documents this practice in Fukui Domain.

^[22]The measure is reported collections relative to reported output, not the ruler’s chosen extraction rate. It absorbs accounting conventions, regional assessment traditions, incentives when reporting to the new Meiji government, and late-period fiscal disturbances. I therefore treat the tax results as associational and emphasize neighbor heterogeneity rather than the pooled coefficient.

government compiled these data in 1869–1870 from each domain’s average production and collections over the preceding five years. Averaging reduces harvest noise, while widespread *jōmen-hō* (fixed-rate taxation) kept many domains’ rates administratively stable in the final Tokugawa decades (Iinuma 1974, 20–24; Smith 1958).

Because *shunodaka* and *uchidaka* cover 1865–69, they measure a late-period steady state rather than annual change. This window remains informative: *jōmen-hō* fixed nominal rates across decades (Smith 1958), limiting year-to-year variation, while Prediction 3 concerns long-run extraction rather than fiscal shocks. Mid-Tokugawa tax scatters in Smith (1958) are qualitatively consistent with the late-period cross-section, but comprehensive mid-period domain data are unavailable. Tax rates also varied within domains, yet systematic village-level data do not survive; the analysis therefore uses domain averages. That aggregation matches the formal model’s common domain-tax restriction but cannot test discriminatory village-level schedules, which lie outside the benchmark.^[23]

I exclude non-daimyō territories, including shogunal *goryō* and shrine- or temple-administered *jisharyō*, because they lack comprehensive fiscal data and had limited tax autonomy (Murakami 1996). ρ_j averages the village distance ratio from Section 3.4.1 within each domain using end-period capitals.

3.5.2. The Tax–Exposure Link Binds Where Neighbors Attract Labor

The pooled benchmark regresses domain tax rates on exposure and controls:

$$\tau_j = \alpha_4 + \beta_4 \ln(\rho_j) + X_j' \gamma_4 + \varepsilon_j$$

Here τ_j is domain j ’s tax rate, ρ_j is its average village distance ratio, and X_j contains three control groups. Core controls are logged domain size and wet-rice suitability,^[24] the share of new-farmland villages, shogunal alliance, and the *gōshi* system. Geographic controls are logged elevation and ruggedness and the monthly shares of rainy and snowy

^[23]Public goods could in principle substitute for tax relief, making tax rates an incomplete measure of oppression. Yet public works required *corvée* labor (*bueki*) from villages and thus added to their burdens, while petitions and resistance records center on excessive taxes and unfair assessments rather than missing infrastructure (Aoki 1975).

^[24]The wet-rice measure is an agro-climatic potential-yield surface (Galor and Özak 2016) rather than an observed annual output series; I use it only as a proxy for labor-absorption capacity.

days.^[25] Neighbor controls are weighted means of adjacent domains' territory, population, rice productivity, new farmlands, family ties to the focal domain, and shogunal alliance; weights count the focal villages for which each domain is the nearest neighbor.

Table 4 reports a negative cross-sectional association between exposure and reported tax rates. Across Models 1–4, a one-standard-deviation increase in the log distance ratio ($SD = 1.34$) corresponds to a 1.5–3.2 percentage-point decrease. The association remains negative with the full set of controls and in a pure spatial-lag model, although it is only marginal in the fully controlled spatial-lag model (Anselin and Rey 2014) (Models 5–6). Its direction matches Steele, Paik, and Tanaka (2017), who find that collective desertions, but not the broader category of insurrection, predict lower taxes at the domain-period level.

Several checks assess whether model choice or omitted variables account for the pooled association. The sign remains negative with alternative covariates, including kokudaka, samurai share, and population relative to wet-rice suitability (Supporting Information E.2.1). Applying the sensitivity analysis of Diegert, Masten, and Poirier (2022) shows that, when exposure- and outcome-side confounding are bounded equally, both SD ratios must reach about 0.554 before the identified set admits a nonnegative coefficient (Supporting Information E.2.2). Double/debiased machine learning, following Chernozhukov et al. (2018), tempers the result: ridge nuisance models retain negative estimates, but flexible stacked and random-forest specifications produce estimates near zero with wide confidence intervals (Supporting Information E.2.3). The sign also remains negative when daikansho enter the choice set (Supporting Information D.3.1). Thus the pooled OLS association is stable across conventional specifications but is not uniformly corroborated by flexible DML; the neighbor-heterogeneity test below asks whether the OLS pattern reflects exit rather than state-capacity decay.

To test the exit channel, I interact exposure with $\hat{y}_{-j}$, a discrete measure of neighboring domains' outside-option quality:

^[25] Weather variables represent proportion of rainy/snowy days per month, imputed as weighted averages using daily weather data from 40 locations, with distance to domain capital as weight. Data from [Historical Weather Database](#) by Professor Minoru Yoshimura.

Outcome:	τ_j					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Distance ratio ($\ln(\rho_j)$)	-0.0241	-0.0203	-0.0110	-0.0112	-0.0140	-0.0094
HC SE	(0.0053)***	(0.0058)***	(0.0055)**	(0.0055)**	(0.0047)***	(0.0051)*
Conley SE (70km)	(0.0066)***	(0.0067)***	(0.0056)**	(0.0055)**	(0.0058)**	(0.0050)*
<i>Controls</i>						
Core	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbors	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	232	230	230	230	230	230
R2	0.074	0.141	0.370	0.389	0.312 ^[a]	0.439 ^[a]
Mean Dep. Var.	0.392	0.394	0.394	0.394	0.394	0.394

Table 4: Pooled Benchmark: Domain-Level Geographic Exposure and Tax Rates

Note: OLS estimates (ML in Models 5–6) of reported effective tax rates (collections divided by reported output) on the log domain-average distance ratio. Full results in Table A.52. SEs in parentheses: (i) HC1 (ML SEs in 5–6); (ii) Conley (70 km) on capital coordinates, computed via two-step filtered-residual approximation (Anselin and Rey 2014, pp. 34–36); stars reflect that estimator’s p -value. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. ^[a] Nagelkerke pseudo-R2

$$\tau_j = \alpha_5 + \beta_5 [\ln(\rho_j) \times \hat{y}_{-j}] + \bar{\beta}_5 \ln(\rho_j) + \tilde{\beta}_5 \hat{y}_{-j} + X'_j \gamma_5 + \varepsilon_j$$

I proxy outside-option quality with neighboring domains’ demand for labor. Labor-scarce rulers attracted workers through land redistribution, tools and rations, irrigation, and other public goods (Hyugano 1957; Miyazaki 1995), with stronger incentives under greater scarcity. Persistent shortages in Kantō led domains to adopt peasant recruitment policies (*iri-byakushō seisaku*) offering tax exemptions, food and seed grants, and other settlement inducements (Gorai 1950).

The moderator operationalizes this labor-absorption capacity. $\hat{y}_{-j}$ equals “Low” when neighbors’ weighted average labor-to-productivity ratio is above its median and “High” otherwise.^[26] I weight each neighbor by the number of domain j villages for which its capital is nearest and divide its population by potential wet-rice yield from Galor and

^[26] Hainmueller, Mummolo, and Xu (2019) show that continuous multiplicative-interaction models rest on a linear-interaction assumption that frequently fails in practice, and recommend replacing the continuous moderator with a discrete one precisely for settings like this, in which the moderating relationship is inherently nonlinear.

Özak (2016).^[27] High $\hat{y}_{-j}$ thus denotes neighbors with greater capacity and incentives to absorb migrants. The interaction absorbs neighbors' average rice yield and territory size, which I drop from X_j .

The interaction probes the retention mechanism. Exposure should induce the home ruler to moderate extraction most strongly when a labor-scarce neighbor can credibly absorb departing cultivators. The compact formalization also contains a possible *confrontation-relief force*: because it assigns each village one contemporaneous action, selecting exit removes that village from direct confrontation at that moment. This benchmark force could attenuate the tax response when the outside option is weak, but the interaction does not identify it or imply that exit and confrontation could not complement one another across a broader sequence. The observable prediction is that the negative tax gradient should be strongest where absorption is credible.

Figure 7 shows the predicted asymmetry in every specification. The negative tax gradient is concentrated among domains with high $\hat{y}_{-j}$, whose neighbors had low labor-to-potential-yield ratios and strong demand for migrants. With geographic controls (Models 3 and 4), the association is statistically significant only for this group; estimates for domains facing labor-abundant neighbors are small and statistically insignificant. The pattern supports the retention branch of Prediction 3 and its prediction that the negative gradient attenuates when the outside option is weak. It does not establish that exposure raises taxes among domains facing unattractive neighbors, the stronger implication that would follow if confrontation relief fully dominated. The interaction sharpens the comparison but does not eliminate domain- or region-level confounding, so the asymmetry remains associational rather than a causal estimate of ruler moderation. The same asymmetry appears when population density proxies for outside-option quality (Supporting Information E.3.3) and when the neighbor set comprises all domains within 100 km. A Hainmueller-Mummolo-Xu kernel-smoothed plot using the continuous moderator yields a negative association through the middle of the support but is flat rather than monotonically sharpening (Supporting Information E.3.4).

^[27]I use the potential yield measure under irrigated conditions with medium inputs.

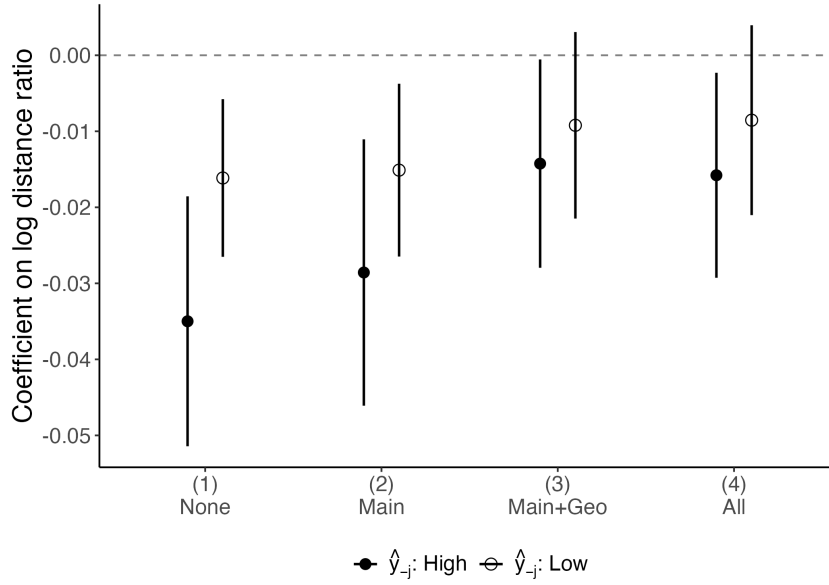


Figure 7: The Tax-Exposure Link Strengthens with Outside-Option Quality

Note: Group-specific OLS slopes on $\ln(\rho_j)$. Filled points marked High denote below-median neighbor labor-to-potential-yield ratios, hence greater capacity and incentive to absorb migrants; open points marked Low denote above-median ratios. Lines are 95% heteroskedasticity-consistent confidence intervals. Controls are cumulative: (1) none; (2) domain; (3) +geography; (4) +neighbor. Full table in [Table A.53](#).

3.5.3. Additional Analyses

Neither simple remoteness nor border proximity explains the tax pattern. Adding average logged home-capital distance, the predictor implied by state-capacity decay ([Sng and Moriguchi 2014](#)), leaves the exposure coefficient negative ([Supporting Information E.2.4](#)); replacing the ratio with remoteness produces no significant negative association once geographic controls enter ([Supporting Information E.3.1](#)). Distance to the nearest border also does not predict taxes ([Supporting Information E.3.2](#)),^[28] matching the village result in [Section 3.4](#). The relevant exposure is to rival authority.

4. CONCLUSION

The four designs recover successive links between outside access, exit, tactic choice, and extraction. Tenryō conversion and the grid-year panel associate improved outside access with more recorded exit. Among resisting villages, proximity to a rival capital supplied the

^[28]The boundary data were constructed following the procedure described in ([Honda, Natsume, and Nemoto 2022](#)).

more robust component of the shift toward exit. Exposed domains reported lower taxes when neighbors offered attractive outside options, an associational pattern consistent with ruler moderation. Across distinct units and sources of variation, the evidence depicts exit as bargaining sustained by shelter, mediation, and recognition. Reachable alternative authority made that strategy viable and placed rulers' tax bases at risk.

The evidence is strongest on village behavior; taxation supplies downstream evidence. It supports a premise shared by Great Divergence, fiscal-federalism, and territorial-cartel theories: credible alternatives can discipline rulers. Tokugawa's hegemonic order makes this a demanding case because migration was prohibited and domains held territory at the shogunate's pleasure. A neighboring lord could protect and mediate, while the shogunate could sanction the home lord. The analysis establishes this behavioral link within a subordinate domain system; it does not show that the responses accumulated into sovereign states.

The same strategic structure suggests two applications beyond the case. In civil wars, civilians may shift support and resources between governments and insurgent authorities; rival protection can constrain predation only where another armed actor can govern and protect those who move (Kalyvas 2006). In competition for foreign investment, firms rather than villages are mobile, and governments face discipline when another jurisdiction offers a credible productive location (Acharya and Lee 2023; Kleven et al. 2020). The Tokugawa designs identify neither relationship. They show why both analogies require an alternative authority that mobile actors can reach and reasonably expect to receive them.

DATA AVAILABILITY STATEMENT

Replication data and code will be deposited in the International Organization Dataverse (<https://dataverse.harvard.edu/dataverse/IOJ>) upon conditional acceptance. The release will include the documented intermediate geographic inputs, scripts, and codebook needed to reproduce the reported tables and figures.

ETHICAL STATEMENT

This research uses publicly available historical records and archival data on Tokugawa-era Japan and does not involve human subjects. No IRB review was required.

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A. FORMAL ANALYSIS

This section retains a narrow formal benchmark for the three predictions in the main text. Under the stated conditions, village responses and noncooperative ruler choices are unique. The tax result is conditional. Exposure lowers taxation only when the neighboring domain is sufficiently attractive. Otherwise, the smaller confrontation pool can permit higher taxation. Rulers use a common domain tax and cannot collude, while post-exit bargaining is reduced to an outside-option payoff. The model therefore does not establish results for discriminatory tax schedules, ruler cartels, or bargaining failure. The paper’s empirical contribution does not depend on those extensions.

A.1. Setup

Consider two neighboring domains $j \in \{1, 2\}$. All primitives are common knowledge. Each ruler governs a continuum of villages of measure one and simultaneously chooses a common domain-level tax component $\tau_j \in [0, 1]$. A single rate matches the domain-average empirical estimand but excludes schedules keyed to observed village exit costs. Rulers cannot make side payments. The ruler equilibrium is Nash, and an enforceable tax cartel lies outside the model.

The timing is: (i) rulers set taxes; (ii) villages observe both rates and may petition their home ruler; (iii) conditional on ordinary petitioning not changing the rate, each village $i_j \in [0, 1]$ chooses $a_{i_j} \in \{\text{comply, fight, exit}\}$; and (iv) outside shelter or mediation may follow exit, after which the home ruler can concede and induce return. Here, *fight* is the model’s compact label for direct confrontation. Let μ_j^a denote the share choosing action a , and let each village in domain j produce yield $y_j > 0$.

A compliant village receives $y_{j(1-\tau_j)} + \varepsilon_{i_j}$, where $\varepsilon_{i_j} \sim \text{Uniform}[-\delta, \delta]$ and $\delta > 0$. A fight succeeds with probability μ_j^{fight} , yields y_j if successful and zero otherwise, and costs $k > 0$. Exit yields the neighboring domain’s net-of-tax income less village-specific exit cost: $y_{-j}(1 - \tau_{-j}) - c_{i_j}$, where $c_{i_j} \in \{c_L, c_H\}$ and $c_H > c_L > 0$. The share $\theta_j \in [0, 1]$ with cost c_L measures geographic exposure to a rival capital; the remaining $1 - \theta_j$ face cost c_H .

Exit is a temporary bargaining strategy. A departing village bears c_{i_j} , removes labor and payments from the home tax base, and activates outside shelter or mediation. The model assumes that a post-exit concession matches the outside option and induces return. It does

not derive the earlier petition failure, concession size, exit duration, or division of bargaining surplus.

Village utility is

$$u_{i_j}(a_{i_j}; \tau_j, \tau_{-j}) = \begin{cases} y_j(1 - \tau_j) + \varepsilon_{i_j} & \text{if } a_{i_j} = \text{comply} \\ y_j \mu_j^{\text{fight}} - k & \text{if } a_{i_j} = \text{fight} \\ y_{-j}(1 - \tau_{-j}) - c_{i_j} & \text{if } a_{i_j} = \text{exit} \end{cases}.$$

Ruler j collects taxes only from compliant villages and maximizes

$$u_{j(\tau_j, \tau_{-j})} = [1 - \mu_j^{\text{fight}} - \mu_j^{\text{exit}}] y_j \tau_j = \mu_j^{\text{comply}} y_j \tau_j.$$

Before stating any proposition, the following assumption partitions resistance strategies: high-exit-cost villages choose between compliance and fight, whereas low-exit-cost villages choose between compliance and exit.

ASSUMPTION 1: For each $j \in \{1, 2\}$,

$$\frac{k - c_L}{\delta - c_L} > \frac{y_j(1 - \theta_j)}{2\delta} \quad \text{and} \quad c_H > y_{-j} + \delta.$$

The results maintain the following regularity and interiority conditions: $\delta > c_L$; $2\delta > y_j(1 - \theta_j)$; $0 < \mu_j^{\text{fight}}, \mu_j^{\text{exit}} < 1$; $0 < \tau_j^* < 1$; $0 < p_j < 1$ and $0 < p_j + q_j < 1$; and $P_j > 0$, for both domains. These conditions ensure positive denominators, interior village cutoffs, and interior ruler best responses.

A.2. Equilibrium Results

PROPOSITION 1 (Resistance Composition under Neighboring-Option Changes):

Under [Assumption 1](#) and interior village cutoffs, for fixed tax rates $(\tau_j, \tau_{-j}) \in (0, 1)^2$:

- (i) $\partial \mu_j^{\text{exit}} / \partial y_{-j} > 0$ and $\partial \mu_j^{\text{fight}} / \partial y_{-j} = 0$; (ii) $\partial \mu_j^{\text{exit}} / \partial \tau_{-j} < 0$ and $\partial \mu_j^{\text{fight}} / \partial \tau_{-j} = 0$.

Only the exit payoff contains neighboring-domain quantities. The proposition is therefore a fixed-tax outside-option comparative static that motivates, but does not structurally reproduce, the conversion-based test: conversion is not a primitive in the model, and the empirical design does not identify a change in y_{-j} or τ_{-j} . Its zero fight derivatives are not a general claim: comparison-based grievances or outside resources, shelter, and political pressure could

instead raise confrontation. The empirical null concerns average incidence and does not rule out those channels in particular episodes.

PROPOSITION 2 (Geographic Exposure and Exit): Under [Assumption 1](#) and the stated regularity conditions: (a) whenever $y_{-j} > \kappa_j^*$, the equilibrium exit share among non-compliant villages is strictly increasing in θ_j ; and (b) the equilibrium exit rate is strictly increasing in θ_j .

Greater exposure expands the low-cost exit pool and contracts the high-cost fight pool. The ruler's tax response offsets only part of the resulting increase in exit because further tax reductions eventually sacrifice more revenue per compliant village than they recover through retention.

PROPOSITION 3 (Governance Conflict and Equilibrium Tax Rate): Under [Assumption 1](#) and the stated regularity conditions, for each $j \in \{1, 2\}$ there exists a unique threshold $\kappa_j^* > 0$ such that $\partial \tau_j^* / \partial \theta_j < 0$ if and only if $y_{-j} > \kappa_j^*$.

Within the model's single-action benchmark, exposure tightens the exit constraint but also shrinks the contemporaneous pool available for direct confrontation. An attractive neighbor makes the retention force dominate, so the ruler lowers taxes; below the threshold, confrontation relief can instead permit higher taxes. This event-level abstraction does not rule out the complementary use of exit and confrontation across a multi-stage protest sequence.

A.3. Equilibrium and Proofs

Under [Assumption 1](#), the resistance shares are

$$\mu_j^{\text{fight}(\tau_j)} = \frac{(1 - \theta_j) [\delta - k - y_{j(1-\tau_j)}]}{2\delta - y_{j(1-\theta_j)}},$$

$$\mu_j^{\text{exit}(\tau_j, \tau_{-j})} = \frac{\theta_j}{2\delta} [y_{-j}(1 - \tau_{-j}) - c_L - y_{j(1-\tau_j)} + \delta].$$

For the fight share, a high-cost village fights when $\varepsilon_{i_j} \leq y_j \mu_j^{\text{fight}} - k - y_{j(1-\tau_j)}$. With the uniform distribution clipped at zero and one, the induced map on $[0, 1 - \theta_j]$ has slope zero on clipped portions and $(1 - \theta_j) \frac{y_j}{2\delta} < 1$ on its interior. It is a contraction with a unique fixed point. Interiority places that point at the expression above, so all-fight and no-fight corners

are not additional equilibria. The exit expression follows from the low-cost village's exit-versus-compliance cutoff.

A.3.1. Proof of Proposition 1

PROOF: Fix j and hold (τ_j, τ_{-j}) fixed. Differentiating the resistance-share expressions above gives

$$\partial \frac{\mu_j^{\text{exit}}}{\partial} y_{-j} = \frac{\theta_j(1-\tau_{-j})}{2\delta} > 0, \quad \partial \frac{\mu_j^{\text{exit}}}{\partial} \tau_{-j} = -\theta_j \frac{y_{-j}}{2\delta} < 0.$$

The fight expression contains neither y_{-j} nor τ_{-j} , so both corresponding derivatives are zero.

A.3.2. Equilibrium Tax Rates

Define

$$\begin{aligned} A_j &:= \frac{(1-\theta_j)y_j}{2\delta - y_j(1-\theta_j)} + \frac{\theta_j y_j}{2\delta}, \\ B_j &:= \frac{(1-\theta_j)(\delta - y_j - k)}{2\delta - y_j(1-\theta_j)} + \frac{\theta_j}{2\delta} [y_{-j} - c_L - y_j + \delta], \\ C_j &:= \frac{\theta_j y_{-j}}{2\delta}. \end{aligned}$$

Then total resistance is $B_j + A_j \tau_j - C_j \tau_{-j}$, and ruler utility is

$$u_j(\tau_j, \tau_{-j}) = y_j \tau_j [1 - B_j + C_j \tau_{-j} - A_j \tau_j].$$

Because $\partial^2 \frac{u_j}{\partial} \tau_j^2 = -2y_j A_j < 0$, the unique best response is

$$\tau_j = p_j + q_j \tau_{-j}, \quad p_j = \frac{1 - B_j}{2A_j}, \quad q_j = \frac{C_j}{2A_j}.$$

Moreover, $A_j \geq \theta_j \frac{y_j}{2\delta}$ implies $q_j \leq \frac{y_{-j}}{2y_j}$ and hence $q_1 q_2 \leq \frac{1}{4} < 1$. The two best responses therefore intersect once and yield

$$\tau_j^* = \frac{p_j + q_j p_{-j}}{1 - q_j q_{-j}}.$$

A.3.3. Proof of Proposition 2

PROOF: At fixed tax rates, $\partial \frac{\mu_j^{\text{exit}}}{\partial} \theta_j = \frac{\mu_j^{\text{exit}}}{\theta_j} > 0$. Writing $D_j = 2\delta - y_j(1-\theta_j) > 0$ and $N_j = \delta - k - y_j(1-\tau_j) > 0$, differentiation gives $\partial \frac{\mu_j^{\text{fight}}}{\partial} \theta_j = -2\delta \frac{N_j}{D_j^2} < 0$.

First suppose $y_{-j} \leq \kappa_j^*$. Proposition 3 gives $d \frac{\tau_j^*}{d} \theta_j \geq 0$, and the other ruler's affine best response gives $d \frac{\tau_{-j}^*}{d} \theta_j = q_{-j} d \frac{\tau_j^*}{d} \theta_j \geq 0$. Let $W_j = y_j - y_{-j} q_{-j}$. Since $A_{-j} \geq \theta_{-j} \frac{y_{-j}}{2\delta}$, $W_j \geq \frac{y_j}{2} > 0$. For $E_j = y_{-j}(1 - \tau_{-j}^*) - c_L - y_j(1 - \tau_j^*) + \delta$, therefore, $d \frac{E_j}{d} \theta_j = W_j d \frac{\tau_j^*}{d} \theta_j \geq 0$, and

$$d \frac{\mu_j^{*,\text{exit}}}{d} \theta_j = \frac{E_j}{2\delta} + \theta_j \frac{W_j}{2\delta} \cdot d \frac{\tau_j^*}{d} \theta_j > 0.$$

Now suppose $y_{-j} > \kappa_j^*$, so $d \frac{\tau_j^*}{d} \theta_j < 0$. At the optimum, compliance equals $A_j \tau_j^*$. Because $A'_j < 0$,

$$\frac{d(\mu_j^{\text{fight}} + \mu_j^{\text{exit}})}{d} \theta_j = -A'_j \tau_j^* - A_j d \frac{\tau_j^*}{d} \theta_j > 0.$$

Meanwhile,

$$d \frac{\mu_j^{\text{fight}}}{d} \theta_j = \frac{[-2\delta N_j + (1 - \theta_j) y_j D_j d \frac{\tau_j^*}{d} \theta_j]}{D_j^2} < 0.$$

Thus exit increases because total resistance increases while fight decreases, proving part (b). For part (a), the quotient-rule numerator for the exit share is $\left(d \frac{\mu_j^{*,\text{exit}}}{d} \theta_j\right) \mu_j^{*,\text{fight}} - \mu_j^{*,\text{exit}} \left(d \frac{\mu_j^{*,\text{fight}}}{d} \theta_j\right) > 0$.

A.3.4. Proof of Proposition 3

PROOF: Let $\Omega_j := y_{-j}$. Define the Ω_j -free part of B_j and its associated intercept as

$$\tilde{B}_j := \frac{(1 - \theta_j)(\delta - y_j - k)}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j}{2\delta}(\delta - c_L - y_j), \quad \tilde{p}_j := \frac{1 - \tilde{B}_j}{2A_j}.$$

Then $B_j = \tilde{B}_j + \theta_j \frac{\Omega_j}{2\delta}$ and $p_j = \tilde{p}_j - q_j$. Since

$$A'_j = \frac{y_j}{2\delta} - \frac{2\delta y_j}{[2\delta - y_j(1 - \theta_j)]^2} \leq 0,$$

$\partial \frac{q_j}{\partial} \theta_j = \Omega_j K_j > 0$, where $K_j = \frac{A_j - \theta_j A'_j}{4\delta A_j^2} > 0$. Let $P_j := \partial \frac{\tilde{p}_j}{\partial} \theta_j > 0$ and $R_j := \theta_j \frac{q_{-j}}{4\delta A_j} \geq 0$. The sign of $\partial \frac{\tau_j^*}{\partial} \theta_j$ is the sign of

$$\Xi_j(\Omega_j) = (1 - R_j \Omega_j) P_j + \Omega_j K_j [p_{-j} + q_{-j} \tilde{p}_j - 1].$$

This expression is affine in Ω_j with a strictly negative slope under best-response interiority, and $\Xi_{j(0)} = P_j > 0$. It therefore crosses zero exactly once, at

$$\kappa_j^* := \frac{P_j}{[R_j P_j - K_j(p_{-j} + q_{-j} \tilde{p}_j - 1)]} > 0.$$

Hence $\partial \frac{\tau_j^*}{\partial} \theta_j < 0$ exactly when $y_{-j} > \kappa_j^*$.

A.4. Non-Vacuousness

The maintained conditions hold on a non-empty open set. One example is $\delta = 10$, $y_1 = y_2 = 18$, $k = \frac{7}{2}$, $c_L = \frac{1}{2}$, $c_H = 29$, and $\theta_1 = \theta_2 = \frac{5}{6}$. At these values, $A_j = \frac{63}{68}$, $B_j = \frac{77}{272}$, $C_j = \frac{3}{4}$, $p_j = \frac{65}{168}$, $q_j = \frac{17}{42}$, and $\tau_j^* = \frac{13}{20}$. The equilibrium resistance shares are $\mu_j^{\text{fight}} = \frac{1}{510}$ and $\mu_j^{\text{exit}} = \frac{19}{48}$, while $P_j = \frac{2}{21} > 0$ and $\kappa_j^* \approx 8.49 < y_{-j} = 18$. Every maintained inequality is strict, so the admissible parameter set contains an open neighborhood of this vector.

B. DESCRIPTIVE STATISTICS

B.1. Village Resistance Data

	Unique	Missing %	Mean	SD	Min	Median	Max	Histogram	Source
Basic									
Home Ruler	168	0	—	—	—	—	—	—	Aoki (1975)
Nbr Ruler	224	0	—	—	—	—	—	—	Aoki (1975)
Year	218	0	1,777.3	59.0	1,615.0	1,782.0	1,867.0		Aoki (1975)
Decade	26	0	1,772.7	59.2	1,610.0	1,780.0	1,860.0		Aoki (1975)
Outcome									
Exit	2	0	0.2	0.4	0.0	0.0	1.0		Aoki (1975)
Direct Confrontation	2	0	0.8	0.4	0.0	1.0	1.0		Aoki (1975)
Predictor									
ρ_i	2,529	0	7.1	61.3	0.0	1.0	2,792.0		—
$\ln(\rho_i)$	2,529	0	0.2	1.5	-5.6	0.0	7.9		—
Dist. to Home Cap	2,517	0	45.8	85.5	0.0	21.2	1,096.0		—
Dist. to Nbr Cap	2,508	0	24.0	29.7	0.0	16.4	494.5		—
ρ_i^{LCP}	168	0	7.2	28.1	0.2	1.7	601.7		—
Controls									
Latitude	2,458	0	35.7	2.0	27.7	35.4	42.0		Honda, Natsume, and Nemoto (2022) and other sources
Longitude	2,445	0	136.5	3.3	128.9	136.6	142.0		Honda, Natsume, and Nemoto (2022) and other sources
Dist. to Edo	2,484	0	413.6	235.6	29.4	413.2	1,356.8		—
Dist. to Edo or Kyoto	2,484	0	234.0	152.4	0.0	191.1	1,041.7		—
Elevation	103	0	19.9	20.9	0.0	12.0	108.0		Geospatial Information Authority of Japan
Ruggedness	146	0	3.7	3.8	0.0	2.5	22.6		—
Wet-Rice Suitability	716	0	18,549.9	3,417.3	0.0	19,003.2	22,570.0		Galor and Özak (2016)
Avg. Caloric Suitability	1,133	0	2,221.3	242.7	941.1	2,297.5	2,531.2		Galor and Özak (2016)

Table A.5: Summary Statistics of the Village Resistance Data ($N = 2,741$)

B.2. Source-Category and Actor Audit

Source section	Printed pages	PDF pages	OCR start	OCR end	Used in audit
Peasant uprisings	7–364	9–366	51	23323	Yes
Urban disturbances	365–410	367–412	23324	25841	No
Village conflicts (murakata sodo)	411–620	413–622	25842	36731	No
Province-wide petitions	621–636	623–638	36732	37490	No

Table A.6: Source-Section Boundary in Aoki’s Chronology

Note: Aoki prints separate chronologies. The project comparison script reads OCR lines 51–23,324 (lines[50:23324] in Python’s zero-based indexing); the last included line is the heading of the next section, so substantive comparison content ends at line 23,323. The separately printed *murakata sōdō* chronology begins at PDF p. 413 (printed p. 411; OCR line 25,842) and is outside that window. Page boundaries were checked against the PDF’s form-feed markers and title pages. This audit establishes exclusion of the separate source category, not the household tenure status of every actor named within included peasant-uprising entries. Source: (Aoki 1975).

Sample	Estimate	Cluster p	Conley p	Events	Rows
Baseline	0.0244	0.0013	0.0041	909	2739
Exclude tenure-keyword events	0.0218	0.0008	0.0113	890	2672

Table A.7: Village-Choice Sensitivity to Tenure-Related Narratives

Note: The restricted sample removes entire resistance events whose demand narrative contains *kosaku* (tenant), *jinushi* (landlord), or *komae* (smallholder). This excludes nineteen events and sixty-seven village rows, only one of which is coded as exit. Both rows use the preferred home-neighbor dyad and decade fixed effects with the main geographic controls. Cluster inference is three-way by home ruler, neighboring ruler, and decade; Conley inference uses a 10 km cutoff. The screen is a conservative content proxy, not a household-level tenure classification.

B.3. Recorded Return after Exit

ID	Year	Home	Village	Recorded end	Aoki p.
809	1759	松山	久妙寺	Returned after reconciliation	123
2822	1788	宇和島	惣川村西	Required to return; later punishment	166
801	1800	松山	波万	Persuaded and returned that day	181
2823	1813	宇和島	岩木	Domain persuasion; return and imprisonment	196
2922	1818	吉田	俵津蒲	Returned on the twenty-fifth	204
2924	1820	吉田	奥野川	Stopped or persuaded; return and repression	207
2824	1827	宇和島	富野川	Persuaded to return; later exile or relocation	218

Table A.8: Exit Events Explicitly Recording Return

Note: These seven analytic-sample narratives explicitly record return after flight or attempted flight. They establish that some recorded exits were temporary, but they do not identify a representative return rate because Aoki did not code duration or final disposition systematically. Only ID 809 states that return followed reconciliation; the other entries record persuasion, an order to return, administrative transfer, or subsequent punishment. Source: (Aoki 1975).

B.4. Audit of Destination Codes

Tactic	Foreign-coded	Edo-coded	Kyoto-coded	Uncoded	Total
Desertion	53	1	0	126	180
End-run appeal	7	112	1	20	140
Unclassified exit	2	0	0	6	8
Total	62	113	1	152	328

Table A.9: Coverage of Derived Destination Codes

Note: Counts cover the 1615–1868 sample and collapse repeated village records to one event using `revolt_id`. Aoki’s printed chronology has no destination column; these are undocumented derived labels in the constructed data, not validated institutional categories. A spot audit finds **foreign** codes naming home authorities or internal locations and **edo** codes naming Osaka, Kyoto, or a lord’s Edo residence. The table therefore reports the labels literally and does not equate them with neighboring domains, daikansho, or the shogunate. At the village-record level, 471 of 765 sample records have a nonmissing derived code. Source chronology: (Aoki 1975).

B.5. Domain Data

	Unique	NA %	Mean	SD	Min	Median	Max	Hist.	Source
Outcome									
Exits	23	0	2.2	5.5	0.0	0.0	37.0		Aoki (1975)
Direct Confrontations	39	0	7.0	14.4	0.0	1.0	164.0		Aoki (1975)
Resistance Events	45	0	9.2	17.7	0.0	2.0	201.0		Aoki (1975)
Exit Rate	56	40	0.3	0.4	0.0	0.1	1.0		Aoki (1975)
Tax Rate (τ_j)	256	7	0.4	0.1	0.1	0.4	0.7		Kodama and Kijima (1989)
Predictor									
ρ_j	241	12	7.3	15.3	0.1	2.2	138.5		—
$\ln(\rho_j)$	241	12	1.0	1.3	-2.4	0.8	4.9		—
Dist. to Home Cap	241	12	49.0	66.7	2.5	22.4	442.5		—
Dist. to Nbr Cap	242	12	16.1	10.8	3.9	12.5	80.2		—
ρ_j^{LCP}	242	12	239.0	1,988.8	0.1	2.7	24,956.3		—
Controls: Main									
Wet-Rice Suitability	242	12	3,011,700.3	4,937,787.1	48,040.8	1,232,259.1	41,226,133.2		Galor and Özak (2016)
Territory Size	258	6	811.0	1,785.9	9.5	257.1	16,262.8		Honda, Natsume, and Nemoto (2022) and other sources
Population	255	7	87,781.0	159,087.3	2,466.0	30,833.0	1,065,910.0		LFAT
Pop. Density	249	9	166.5	125.1	11.9	150.2	1,285.0		—
# New Land	32	12	7.8	27.9	0.0	1.0	232.0		—
Shogunate Kin (0/1)	2	0	0.6	0.5	0.0	1.0	1.0		Kimura, Fujino, and Murakami (2015)
Goushi System (0/1)	2	0	0.1	0.3	0.0	0.0	1.0		Kimura, Fujino, and Murakami (2015)

Table A.10: Summary Statistics of the Domain Data ($N = 275$)

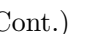
	Unique	Missing %	Mean	SD	Min	Median	Max	Histogram	Source
Controls: Geography									
Latitude	265	1	35.5	1.7	31.6	35.3	41.4		Kimura, Fujino, and Murakami (2015)
Longitude	265	1	136.5	3.3	128.8	136.6	141.5		Kimura, Fujino, and Murakami (2015)
Mean Elevation	242	12	16.1	13.6	2.6	12.0	88.4		—
Mean TRI	240	12	3.2	1.8	0.2	2.9	8.9		—
Dist. to Edo or Kyoto	242	12	188.5	148.9	25.2	135.0	671.3		—
% Rainy Days	251	3	0.1	0.0	0.1	0.1	0.2		Yoshimura HWDB ^[29]
% Snowy Days	251	3	0.0	0.0	0.0	0.0	0.1		Yoshimura HWDB
Controls: Neighbor									
Prop. Same Fam.	65	12	0.1	0.2	0.0	0.0	1.0		Kimura, Fujino, and Murakami (2015)
% New Farmland	78	12	32.9	69.2	0.0	7.5	421.0		—
Avg. Wet-Rice Suitability	242	12	2,135,287.7	3,139,273.3	288,682.9	1,350,670.1	41,057,071.8		Galor and Özak (2016)
Avg. Territory Size	242	12	433.8	602.5	12.4	278.0	6,926.3		Honda, Natsume, and Nemoto (2022) and other sources
Avg. Population	239	13	56,178.5	84,147.0	6,264.6	31,723.2	1,061,486.4		LFAT
Avg. Pop. Density	239	13	175.5	107.1	39.9	161.0	1,028.7		—
Avg. Pop. Density (Prod.)	239	13	0.0	0.0	0.0	0.0	0.2		—
Controls: Alternative									
Samurai Pop.	230	16	70,945.3	134,463.8	614.0	21,519.5	896,808.0		Kimura, Fujino, and Murakami (2015)
# Villages	170	6	196.5	304.5	1.0	82.5	2,243.0		LFAT
Dist. to Edo	242	12	370.8	244.7	42.2	309.4	1,039.5		—

Table A.11: Summary Statistics of the Domain Data (Cont.)

^[29] Historical Weather Database made available online by Minoru Yoshimura: <http://tk2-202-10627.vs.sakura.ne.jp/>

C. EVENT STUDY ROBUSTNESS

C.1. Tenryo-Conversion Universe

The treatment set is constructed from the event seed `analysis/data/domain/domain_events.csv`, itself built from (Kimura, Fujino, and Murakami 2015) and (Kodama and Kitajima 1989). Across the 1615–1868 panel window, these sources record 79 domain dissolutions. Of these, 75 (95%) were followed by direct shogunal administration (tenryō); the remaining four passed directly to another daimyō and are excluded from the treatment group per the rule stated in Section 3.3.1. The 75 dissolve-to-tenryō events correspond to 72 unique domains, three of which experienced a second dissolve-to-tenryō event in a later spell. Restricting to domains active in the panel and retaining each domain’s first qualifying dissolution drops four domains whose earliest in-panel dissolution was a daimyō-to-daimyō transfer (their later tenryō conversion occurs outside the matched-control window). These filters yield the 55 treated domains listed in Table A.12.

The list spans the full sample window, with 49 cases concentrated before 1750 and 6 after. The event seed supplied a nonblank reason for 40 of 55 cases, none with a volume or page citation. The local audit in Supporting Information C.2 now supplies a page-level disposition for all 55, fills the fifteen blank formal reasons, and corrects Kaibara. It confirms Nagamine, Nariwa, and Tamanawa as conversions, while identifying noncomparable events, ambiguous intervening administrations, and four date errors. The table is therefore both a typology and an audit trail for the baseline coding; the sensitivity analysis does not presume that every row is cleanly comparable. The cohort split in Supporting Information C.8 confirms that the baseline exit response loads on the pre-1750 sub-sample that holds 49 of the 55 provisional cases.

No.	Year	Domain	Province	Provisional Reason
1	1617	Itabashi (板橋藩)	Shimotsuke	transfer
2	1618	Nagamine (長峰藩)	Echigo	transfer
3	1619	Musashi Komuro (武蔵小室藩)	Musashi	attainder (Ina house)
4	1632	Ōmori (大森藩)	Ōmi	kokudaka halved
5	1639	Fukōzu (深溝藩)	Mikawa	residence moved
6	1642	Nariwa (成羽藩)	Bitchū	tenryō after transfer
7	1645	Fuchū (府中藩)	Hitachi	no heir
8	1650	Kaibara (柏原藩)	Tanba	family dispute (oie sōdō)
9	1653	Toyooka (豊岡藩)	Tajima	no heir
10	1656	Funai (府内藩)	Bungo	no heir
11	1657	Marugame (丸亀藩)	Sanuki	no heir
12	1658	Kakegawa (掛川藩)	Tōtōmi	no heir
13	1658	Yashima (矢島藩)	Dewa	early-period abolition
14	1660	Sakura (佐倉藩)	Shimōsa	misconduct
15	1665	Mibu (壬生藩)	Shimotsuke	no heir
16	1665	Saijō (西条藩)	Iyo	misconduct (negligence)
17	1666	Miyazu (宮津藩)	Tango	family dispute (oie sōdō)
18	1668	Shimabara (島原藩)	Hizen	misgovernment
19	1670	Harima Shingū (播磨新宮藩)	Harima	no heir
20	1671	Ichinoseki (一関藩)	Mutsu	Date affair
21	1676	Asō (麻生藩)	Hitachi	no heir
22	1677	Omigawa (小見川藩)	Shimōsa	no heir (sub-daimyō)
23	1678	Yamazaki (山崎藩)	Harima	no heir
24	1679	Kururi (久留里藩)	Kazusa	insanity
25	1679	Niwase (庭瀬藩)	Bitchū	no heir
26	1681	Numata (沼田藩)	Kōzuke	misgovernment (illicit construction)
27	1681	Takada (高田藩)	Echigo	Echigo sōdō
28	1683	Tatebayashi (館林藩)	Kōzuke	political (Tsunayoshi enthronement)

Table A.12: The 55 Tenryō-Conversion Events Used as Treatment in Table 1 (Part 1 of 2)

Note: Cases are drawn from (Kimura, Fujino, and Murakami 2015) and (Kodama and Kitajima 1989) via the event seed in `analysis/data/domain/domain_events.csv`. The universe of abolitions recorded in the file is 79; removing the 4 cases coded as direct transfers to another daimyō leaves 75 provisional dissolve-to-tenryō events. Restricting to domains active in 1615–1868 and retaining the first qualifying dissolution per domain yields the 55 rows shown. The reason field is an audit aid, not a completed typology: only 40 source rows have a nonblank reason, none has a volume or page citation, and web verification uncovered corrections and several reductions, partitions, or temporary transfers that require source-volume review. Pre-1750 cohort: 49 cases; post-1750 cohort: 6 cases.

No.	Year	Domain	Province	Provisional Reason
29	1684	Sanuki (佐貫藩)	Kazusa	misconduct
30	1687	Fukumoto (福本藩)	Harima	partition-abolition
31	1687	Karasuyama (烏山藩)	Shimotsuke	violation of bakufu law
32	1687	Sōmi (沢海藩)	Echigo	territory confiscated
33	1689	Fukami (深見藩)	Sagami	attainder (Genroku 2)
34	1689	Kitami (喜多見藩)	Musashi	attainder (Genroku 2)
35	1692	Hachiman (八幡藩)	Mino	no heir
36	1692	Takayama (高山藩)	Hida	political (Kanamori transfer)
37	1693	Matsuyama-Bitchū (備中松山藩)	Bitchū	no heir
38	1698	Fukuyama (福山藩)	Bingo	no heir
39	1698	Sano (佐野藩)	Shimotsuke	political (Hotta transfer)
40	1698	Yoshii (吉井藩)	Kōzuke	political (Hotta transfer)
41	1701	Akō (赤穂藩)	Harima	Asano incident (violence)
42	1702	Nagashima (長島藩)	Ise	violence
43	1703	Tamanawa (玉縄藩)	Sagami	transfer to Ōtaki
44	1704	Kōfu (甲府藩)	Kai	political (Ienobu enthronement)
45	1704	Yamura (谷村藩)	Kai	political (linked to Kōfu)
46	1712	Hōjō (北条藩)	Awa	misgovernment (Man'goku uprising)
47	1724	Numazu (沼津藩)	Suruga	political (Mizuno transfer)
48	1725	Matsumoto (松本藩)	Shinano	insanity (assault in Edo castle)
49	1746	Takayanagi (高柳藩)	Echigo	transfer to Mikusa
50	1767	Obata (小幡藩)	Kōzuke	Meiwa incident (Yamagata Daini)
51	1786	Sagara (相良藩)	Tōtōmi	political (Tanuma fall)
52	1788	Ōmi Komuro (近江小室藩)	Ōmi	territory confiscated
53	1807	Matsumae (松前藩)	Ezo	Ezo defence (temporary seizure)
54	1866	Hamada (浜田藩)	Iwami	Chōshū War defeat
55	1868	Kaifuchi (貝淵藩)	Kazusa	Boshin War (pro-bakufu)

Table A.13: The 55 Tenryō-Conversion Events Used as Treatment in Table 1 (Part 2 of 2)

Note: Continuation of Table A.12. Same source definitions apply.

C.2. Local-Source Treatment Audit

The local *Hanshi Daijiten* and *Hanshi Sōran* volumes supply page-level dispositions for all 55 baseline rows. Twenty are retained at their recorded date, twenty-one are excluded as noncomparable, ten are ambiguous because the intervening administrator is unstated, and four have the wrong date. Three date errors are correctable: Takayanagi to 1739, Kōfu to 1724, and Sagara to 1787; Numazu is excluded because the source records no transition in 1724. The audit also fills every formerly blank reason, corrects Kaibara, and confirms Nagamine, Nariwa, and Tamanawa as conversions.

The source-stated formal causes comprise 22 succession, partition, or incapacity cases; 18 misconduct, misgovernment, or political-sanction cases; 10 administrative or strategic reorganizations; 2 wartime or regime-change cases; and 3 coding errors or non-events. No source states fiscal distress or peasant unrest as the formal cause of a baseline row. Potential links to unrest are recorded separately as a cross-cutting evidentiary screen rather than inferred from event proximity. [Table A.15](#) reports this cause-by-disposition cross-tab and retains the full 55-row audit; [Table A.18](#) reports strict, ambiguity-inclusive, and combined source-plus-unrest screens.

Grouped formal cause	All rows	Retain	Exclude	Ambig.	Wrong date
Succession, partition, or incapacity	22	6	9	6	1
Misconduct, misgovernment, or political sanction	18	9	5	3	1
Administrative or strategic reorganization	10	5	3	1	1
Wartime defeat or regime change	2	0	2	0	0
Coding error or non-event	3	0	2	0	1
Fiscal distress	0	0	0	0	0
Peasant unrest or petitioning	0	0	0	0	0
Total	55	20	21	10	4

Table A.14: Succession and Sanction Account for 40 of 55 Recorded Conversion Rows

Note: Cause category groups each source-stated formal reason independently of source disposition. Political succession is grouped with succession. Violence paired with mental derangement is grouped with sanction because violence is the recorded precipitating act. Strategic military transfer is grouped with administrative or strategic reorganization. Fiscal distress and peasant unrest or petitioning remain explicit zero-count categories because neither appears as a source-stated formal cause. The row-level audit records potentially relevant unrest separately rather than inferring its absence from a short domain history.

Domain	Year	Cause group	Formal reason	Disposition	Treatment	Local source
Itabashi	1617	Administrative/strategic	Administrative transfer	Most land transferred to another lord; no whole-domain tenryo conversion	Exclude	Hanshi Daijiten v.2 p.194 (PDF 210); Hanshi Soran p.102 (PDF 104)
Nagamine	1618	Administrative/strategic	Administrative relocation	Former Nagamine holding became tenryo after transfer to Nagaoka	Retain	Hanshi Daijiten v.3 p.105 (PDF 119)
Musashimuro	Ko-1619	Succession/household	No heir	House attained but 1,180 koku regranted to the lord's brother	Exclude	Hanshi Daijiten v.2 p.632 (PDF 648)
Omori	1632	Succession/household	Succession restriction	Heir retained 5,000 koku; only half returned temporarily to bakufu	Exclude	Hanshi Daijiten v.5 p.30 (PDF 46); Hanshi Soran pp.258-259 (PDF 260-261)
Fukozu	1639	Succession/household	Family partition and relocation	Brother received 5,000 koku including Fukozu and domain moved to Nakajima	Exclude	Hanshi Daijiten v.4 p.314 (PDF 330)
Nariwa	1642	Administrative/strategic	Administrative transfer	Former Nariwa holding became tenryo after transfer to Bitchu Matsuyama	Retain	Hanshi Daijiten v.6 p.255 (PDF 271)
Fuchu	1645	Succession/household	No heir	Domain temporarily abolished; intervening administration not identified	Ambiguous	Hanshi Daijiten v.2 p.39 (PDF 55)
Kaibara	1650	Succession/household	No heir	Full confiscation and conversion to tenryo	Retain	Hanshi Daijiten v.5 p.429 (PDF 445); Hanshi Soran p.275 (PDF 277)
Toyooka	1653	Succession/household	No heir	Former domain became bakufu land until Kyogoku entry in 1668	Retain	Hanshi Daijiten v.5 p.463 (PDF 479)
Funai	1656	Succession/household	No heir	House ended; intervening administration before 1658 entry not stated	Ambiguous	Hanshi Daijiten v.7 p.402 (PDF 418)
Marugame	1657	Succession/household	No heir	Bakufu daikan governed until Kyogoku entry in 1658	Retain	Hanshi Daijiten v.6 p.447 (PDF 463); chronology p.451 (PDF 467)
Kakogawa	1658	Succession/household	No heir	Neighboring lord held custody until 1659; bakufu-land status not stated	Ambiguous	Hanshi Daijiten v.4 p.177 (PDF 193)
Yashima	1658	Succession/household	Testamentary family partition	Two sub-10,000-koku branches became hatamoto in 1659	Exclude	Hanshi Daijiten v.1 p.389 (PDF 405)
Sakura	1660	Misconduct/sanction	Official misconduct	Lord attained; direct administration before 1661 entry not established	Ambiguous	Hanshi Daijiten v.2 p.453 (PDF 469); Aoki printed p.33 (PDF 35)
Mibu	1665	Coding/non-event	Domain-name error	No-heir abolition occurred in Minagawa, not Mibu; Mibu continued	Exclude	Hanshi Daijiten v.2 p.226 (PDF 242); Hanshi Soran p.114 (PDF 116); Mibu entry v.2 p.210 (PDF 226)
Saijo	1665	Misconduct/sanction	Official dereliction	Territory explicitly became bakufu land until 1670	Retain	Hanshi Daijiten v.6 p.469 (PDF 485); Aoki printed p.38 (PDF 40)
Miyazu	1666	Misconduct/sanction	Household conflict and misgovernment	Territory became bakufu land until 1669	Retain	Hanshi Daijiten v.5 p.443 (PDF 459); Aoki printed p.39 (PDF 41)
Shimabara	1668	Misconduct/sanction	Misgovernment	Former domain remained public land for one year and four months	Retain	Hanshi Daijiten v.7 p.212 (PDF 228); Aoki printed p.40 (PDF 42)
Harima Shingu	1670	Succession/household	Succession restriction	Brother retained the seat and 3,000 koku; domain effectively continued	Exclude	Hanshi Daijiten v.5 p.583 (PDF 599)

Table A.15: Page-Level Local-Source Audit of All Conversion Rows (Part 1 of 3)

Note: The audit reviews every one of the 55 baseline treatment rows in the local *Hanshi Daijiten* and *Hanshi Soran*. Cause group records the grouped formal reason summarized in Table A.14; it does not determine treatment status. “Retain” requires direct shogunal administration at the coded date. “Exclude” identifies a partial reduction, family partition, transfer, or post-shogunate event rather than a comparable conversion. “Ambiguous” denotes an abolition or confiscation for which the entry does not identify the intervening administrator. A year in parentheses is the source-verified correction. Nagamine, Nariwa, and Tamanawa are retained because the cited entries confirm that their former holdings became tenryō.

Domain	Year	Cause group	Formal reason	Disposition	Treatment	Local source
Ichinoseki	1671	Misconduct/sanction	Political punishment in Date dispute	Holding returned to Sendai main domain, not direct shogunal rule	Exclude	Hanshi Daijiten v.1 p.135 (PDF 151)
Aso	1676	Succession/household	No heir	Estate confiscated but regranted two months later; direct administration not stated	Ambiguous	Hanshi Daijiten v.2 p.110 (PDF 126)
Omigawa	1677	Coding/non-event	Domain-name error	Event occurred in Owa; Omigawa did not dissolve in 1677	Exclude	Hanshi Soran p.75 (PDF 77); Omigawa entry Hanshi Daijiten v.2 p.427 (PDF 443)
Yamazaki	1678	Succession/household	No heir	Shiso district explicitly became tenryo	Retain	Hanshi Daijiten v.5 p.562 (PDF 578); narrative p.557 (PDF 573)
Kururi	1679	Succession/household	Mental incapacity	Son retained 3,000 koku and most castle territory passed to Maebashi	Exclude	Hanshi Daijiten v.2 p.530 (PDF 546); chronology p.533 (PDF 549)
Niwase	1679	Succession/household	No heir	Vacancy until 1683; bakufu-land status not stated	Ambiguous	Hanshi Daijiten v.6 p.199 (PDF 215)
Numata	1681	Misconduct/sanction	Misgovernment and official failure	Territory came under daikan administration	Retain	Hanshi Daijiten v.2 p.257 (PDF 273); Aoki printed pp.47,49 (PDF 49,51)
Takada	1681	Succession/household	Household conflict	Territory became bakufu land and was surveyed in 1682	Retain	Hanshi Daijiten v.3 p.113 (PDF 129)
Tatebayashi	1683	Succession/household	Succession termination	Most land already partitioned to hatamoto; only residual domain reverted	Exclude	Hanshi Daijiten v.2 p.280 (PDF 296); chronology p.288 (PDF 304); Aoki printed p.45 (PDF 47)
Samuki	1684	Misconduct/sanction	Official misconduct	Domain abolished until 1710; intervening direct administration not stated	Ambiguous	Hanshi Daijiten v.2 p.546 (PDF 562)
Fukumoto	1687	Succession/household	Family partition	Domain status ended in 1665; 1687 was a later hatamoto split	Exclude	Hanshi Daijiten v.5 p.530 (PDF 546); Hanshi Soran p.295 (PDF 297)
Karasuyama	1687	Misconduct/sanction	Succession-rule violation	Lord removed and another daimyo entered; no whole-domain conversion	Exclude	Hanshi Daijiten v.2 p.165 (PDF 181)
Soumi	1687	Misconduct/sanction	Misconduct	All territory confiscated	Retain	Hanshi Daijiten v.3 p.54 (PDF 68); Hanshi Soran p.154 (PDF 156)
Fukami	1689	Misconduct/sanction	Official dereliction	7,800 koku confiscated but 2,200 retained	Exclude	Hanshi Daijiten v.2 p.681 (PDF 697)
Kitami	1689	Misconduct/sanction	Collective punishment after violence	Fief confiscated and lord exiled	Retain	Hanshi Daijiten v.2 p.675 (PDF 691)
Hachiman	1692	Succession/household	No heir	House continued elsewhere and another lord entered Hachiman in 1692	Exclude	Hanshi Daijiten v.4 p.33 (PDF 49)
Takayama	1692	Administrative/strategic	Administrative and political transfer	Entire Hida domain became bakufu territory under Hida daikan	Retain	Hanshi Daijiten v.4 p.3 (PDF 19)
Takahashi	1693	Succession/household	No heir	Bakufu ordered survey and castle custody; bakufu-land status not explicit	Ambiguous	Hanshi Daijiten v.6 p.243 (PDF 259)
Fukuyama	1698	Succession/household	No heir	Entire territory became bakufu land under daikan; 50,000 koku remained bakufu land after regnant	Retain	Hanshi Daijiten v.6 p.267 (PDF 283); Aoki printed p.57 (PDF 59)

Table A.16: Page-Level Local-Source Audit of All Conversion Rows (Part 2 of 3)

Note: Continuation of Table A.15. Page numbers are printed-page locators; PDF-page locators appear in parentheses in the source field.

Domain	Year	Cause group	Formal reason	Disposition	Treatment	Local source
Sano	1698	Administrative/strategic	Administrative relocation	Lord transferred to Katata; direct administration of Sano not established	Exclude	Hanshi Daijiten v.2 p.236 (PDF 252)
Yoshii	1698	Administrative/strategic	Administrative relocation	Former territory became divided holdings rather than a whole bakufu domain	Exclude	Hanshi Daijiten v.2 p.336 (PDF 352)
Ako	1701	Misconduct/sanction	Violence	Former domain became bakufu land until 1702	Retain	Hanshi Daijiten v.5 p.608 (PDF 624)
Nagashima	1702	Misconduct/sanction	Violence and mental derangement	Another lord entered in the same year; direct administration not shown	Exclude	Hanshi Daijiten v.4 p.385 (PDF 401)
Tamanawa	1703	Administrative/strategic	Administrative relocation	Former Tamanawa holding became tenryo after transfer to Otaki	Retain	Hanshi Daijiten v.2 p.683 (PDF 699); Hanshi Soran p.146 (PDF 148)
Kofu	1704	Succession/household	Political succession	Yanagisawa replaced Tokugawa house in 1704; Kofu became tenryo in 1724	Wrong date (1724)	Hanshi Daijiten v.3 p.329 (PDF 345); p.330 (PDF 346)
Yamura	1704	Administrative/strategic	Administrative relocation	Domain ended but exact former territory's bakufu-land status is not explicit	Ambiguous	Hanshi Daijiten v.3 p.322 (PDF 338); Hanshi Soran p.181 (PDF 183)
Hojo	1712	Misconduct/sanction	Misgovernment and extraction	Territory confiscated; intervening bakufu-land status not explicit	Ambiguous	Hanshi Daijiten v.2 p.569 (PDF 585); chronology p.572 (PDF 588); Aoki printed p.66 (PDF 68)
Numazu	1724	Coding/non-event	Non-event	Territory had been under daikan administration since 1632; no 1724 conversion	Wrong date	Hanshi Daijiten v.4 p.115 (PDF 131); p.116 (PDF 132)
Matsumoto	1725	Misconduct/sanction	Violence and mental derangement	Territory became bakufu land under two daikan for eight months	Retain	Hanshi Daijiten v.3 p.451 (PDF 467); Aoki printed p.70 (PDF 72)
Takayanagi	1746	Administrative/strategic	Appointment-related land exchange	Former holding reverted in 1739; 1746 records later move to Mikusa	Wrong date (1739)	Hanshi Daijiten v.3 p.122 (PDF 136); Hanshi Soran p.156 (PDF 158)
Obata	1767	Misconduct/sanction	Political punishment	Heir transferred and another lord entered Obata; no direct conversion	Exclude	Hanshi Daijiten v.2 p.361 (PDF 377); chronology p.367 (PDF 383)
Sagara	1786	Misconduct/sanction	Political fall	Reduction occurred in 1786; remaining territory became direct bakufu land in 1787	Wrong date (1787)	Hanshi Daijiten v.4 p.161 (PDF 177); chronology p.165 (PDF 181)
Omi Komuro	1788	Misconduct/sanction	Official misconduct and corruption	All territory confiscated	Retain	Hanshi Daijiten v.5 p.5 (PDF 21); Hanshi Soran pp.259-260 (PDF 261-262); Aoki printed p.146 (PDF 148)
Matsumae	1807	Administrative/strategic	Strategic and military transfer	Entire Matsumae-Ezo territory became bakufu land	Retain	Hanshi Daijiten v.1 p.5 (PDF 21); chronology p.16 (PDF 32)
Hamada	1866	Wartime collapse	Military defeat	Territory remained under Choshu occupation and custody, not shogunal rule	Exclude	Hanshi Daijiten v.6 p.119 (PDF 135); chronology p.128 (PDF 144); Aoki printed pp.291,303,305 (PDF 293,305,307)
Kaifuchi	1868	Wartime collapse	Boshin War allegiance	Meiji government confiscated Jozai territory after the shogunate ended	Exclude	Hanshi Daijiten v.2 p.526 (PDF 542); chronology p.527 (PDF 543)

Table A.17: Page-Level Local-Source Audit of All Conversion Rows (Part 3 of 3)

Note: Continuation of Table A.15. The full machine-readable audit also records the event-specific unrest screen in `analysis/output/tables/tabA_conversion_source_audit.csv`.

Specification	Outcome	Estimate	Cluster SE	Cluster p	Conley p	Treated
Baseline	Exit	0.0385	0.0147	0.0098	0.0033	55
Baseline	Confrontation	0.0076	0.0122	0.5322	0.5754	55
Initial 9-row priority screen	Exit	0.0330	0.0156	0.0361	0.0215	46
Initial 9-row priority screen	Confrontation	0.0134	0.0147	0.3624	0.3912	46
Strict, recorded dates	Exit	0.0463	0.0264	0.0817	0.0178	20
Strict, recorded dates	Confrontation	0.0480	0.0236	0.0444	0.1083	20
Strict, corrected dates	Exit	0.0385	0.0225	0.0903	0.0338	23
Strict, corrected dates	Confrontation	0.0472	0.0193	0.0158	0.0589	23
Retain + ambiguous	Exit	0.0355	0.0189	0.0625	0.0223	30
Retain + ambiguous	Confrontation	0.0231	0.0187	0.2203	0.2958	30
Retain + ambiguous, corrected	Exit	0.0310	0.0168	0.0682	0.0340	33
Retain + ambiguous, corrected	Confrontation	0.0255	0.0162	0.1171	0.1825	33
Strict, corrected; no clear unrest	Exit	0.0447	0.0233	0.0576	0.0098	21
Strict, corrected; no clear unrest	Confrontation	0.0508	0.0205	0.0144	0.0720	21
Retain + ambiguous, corrected; no clear unrest	Exit	0.0366	0.0179	0.0433	0.0140	30
Retain + ambiguous, corrected; no clear unrest	Confrontation	0.0273	0.0178	0.1272	0.2155	30

Table A.18: Sensitivity to Local-Source Treatment-Universe Corrections

Note: The page-level audit covers all 55 baseline rows. The strict screen retains only entries that establish direct shogunal administration at the stated date; the corrected version adds Takayanagi (1739), Kōfu (1724), and Sagara (1787). The ambiguity-inclusive screen also retains abolitions or confiscations whose intervening administrator is unstated. The final two specifications exclude Saijō, Shimabara, and Hōjō, the three cases clearly linked to unrest or petitioning. Each row re-estimates the 50 km two-way fixed-effects specification used in Table 1. The exit coefficient remains positive throughout. In the smallest strict samples, however, the confrontation coefficient is also positive, so those rows do not by themselves identify an exit-specific response. Cluster standard errors are two-way by domain and year; Conley p -values use a 70 km cutoff.

C.3. Potentially Unrest-Related Conversions

The available case histories are sufficient to identify a narrow set of three conversions clearly connected to peasant unrest or petitioning and a broader set that adds two mixed cases. A systematic Aoki screen adds a maximal envelope: five further conversions had an event in the same named domain or province during the preceding ten years, although the local histories do not identify those events as causes. None of the screens can certify the absence of unrest in every remaining conversion. Table A.19 therefore reports bounded exclusions rather than presenting silence in short domain chronologies as evidence that treatment timing was unrelated to local conflict.

Specification	Outcome	Estimate	Cluster SE	Cluster p	Conley p	Treated
Baseline	Exit	0.0385	0.0147	0.0098	0.0033	55
Baseline	Confrontation	0.0076	0.0122	0.5322	0.5754	55
Exclude 3 clear cases	Exit	0.0421	0.0153	0.0067	0.0014	52
Exclude 3 clear cases	Confrontation	0.0076	0.0129	0.5556	0.6020	52
Exclude 5 clear/mixed cases	Exit	0.0397	0.0158	0.0130	0.0013	50
Exclude 5 clear/mixed cases	Confrontation	0.0059	0.0136	0.6619	0.6827	50
Maximal 10-year history screen	Exit	0.0443	0.0174	0.0117	0.0010	45
Maximal 10-year history screen	Confrontation	0.0092	0.0145	0.5264	0.5643	45

Table A.19: Sensitivity to Excluding Potentially Unrest-Related Conversions

Note: The narrow set excludes Saijō (1665), Shimabara (1668), and Hōjō (1712), whose histories directly connect abolition to peasant unrest or petitioning. The broader set also excludes Numata (1681) and Ōmi Komuro (1788) as mixed cases. The maximal screen additionally removes Sakura, Miyazu, Tatebayashi, Matsumoto, and Hamada because Aoki records an event in the same named domain or province during the preceding ten years, even though the local histories do not identify those events as causes of conversion. The exit estimate remains positive and the confrontation estimate remains null under all three screens. Cluster standard errors are two-way by domain and year; Conley p -values use a 70 km cutoff. Silence in a short domain history is not treated as proof that no unrest occurred. Sources: (Aoki 1975; Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989).

C.4. Tenryo-Conversion Count Robustness

Outcome:	Nearby Exit Count			Nearby Confrontation Count		
Radius:	30km	40km	50km	30km	40km	50km
Converted $\times$ Post	0.0359	0.0592	0.0770	0.0057	0.0103	0.0398
Cluster SE	(0.0238)	(0.0323)*	(0.0396)*	(0.0162)	(0.0228)	(0.0284)
Conley SE (70km)	(0.0291)	(0.0390)	(0.0448)*	(0.0171)	(0.0256)	(0.0314)
Num. Obs.	33,877	33,877	33,877	33,877	33,877	33,877
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Tenryo-Converted/All Domains	55/294	55/294	55/294	55/294	55/294	55/294

Table A.20: Count Robustness for Domain-to-Tenryo Conversions

Note: OLS fixed-effects estimates with event counts (rather than binary indicators) as outcomes, using the same annual domain-year panels as Table 1. Tenryo-converted domains are restricted to `rel_year` in $[-5, +5]$ and matched to never-converted controls observed in the same calendar years represented by the conversion window. Two SE rows: (i) two-way clustered on domain and year; (ii) Conley spatial standard errors with a 70 km cutoff on domain capital coordinates. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

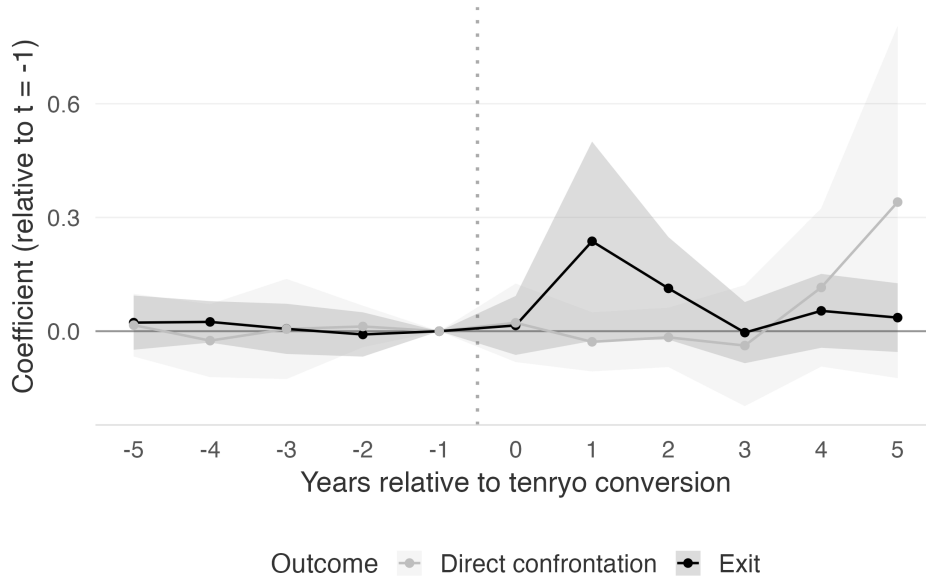


Figure A.8: OLS Count Event-Study Coefficients (50km)

Note: Year-by-year OLS coefficients from the specification $\text{count}_{j,t} = \sum_{s \neq -1} \beta_s (\mathbf{1}[t - t_j^* = s]) + \varphi_j + \omega_t + \varepsilon_{j,t}$, where the outcome is the count of nearby exit (black) or direct-confrontation (light gray) events within 50km of domain j 's capital and $s = -1$ is the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between $s = -1$ and $s = 0$. Standard errors are two-way clustered by domain and year.

C.5. Disaggregating Exit into Desertion and End-Run Appeal

Table A.21 replicates Table 1 with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals.

Outcome:	Any Nearby Desertion		
Sample Radius:	30km	40km	50km
<i>Variables</i>			
Converted $\times$ Post	0.0114	0.0176	0.0266
Cluster SE	(0.0063)*	(0.0101)*	(0.0122)**
<i>Controls</i>			
Domain & Year FE	Yes	Yes	Yes
<i>Fit Statistics</i>			
Num. Obs.	33,877	33,877	33,877
R2	0.033	0.044	0.055
Mean Dep. Var.	0.004	0.006	0.010

Table A.21: Desertion-Only DiD

Note: Replicates Table 1 with the exit outcome restricted to desertion (*chōsan*), excluding end-run appeals. Specification matches Table 1. Two-way clustered (domain & year) standard errors. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

C.6. Staggered-Robust Estimation

Table A.22 compares the main TWFE pooled estimate with the interaction-weighted estimator of Sun and Abraham (2021), which is robust to heterogeneous treatment effects under staggered adoption. The Sun and Abraham aggregate ATT for exit is 0.0413 ($p < 0.001$), virtually identical to the TWFE pooled estimate of 0.0385; the confrontation ATT remains insignificant at 0.0089 ($p = 0.262$). The near-identical TWFE and Sun-Abraham point estimates are consistent with the Goodman-Bacon decomposition in Table A.23, in which 94.5% of the TWFE estimand weight falls on clean treated-versus-never-treated 2×2 comparisons; the bad-comparison problem of heterogeneous-timing DiD is quantitatively small in this panel because never-treated domains make up the overwhelming majority of the sample.^[30]

Estimator	Nearby Exit			Nearby Confrontation		
	ATT	SE	p	ATT	SE	p
TWFE	0.0385	(0.0147)**	0.01	0.0076	(0.0122)	0.532
Sun & Abraham (2021)	0.0413	(0.0045)***	< 0.001	0.0089	(0.0079)	0.262

Table A.22: TWFE vs. Sun and Abraham Staggered-Robust Estimates

Note: The TWFE row reproduces the main 50km specification from Table 1. The Sun and Abraham row reports the interaction-weighted estimator that is robust to heterogeneous treatment effects under staggered adoption. Never-converted domains serve as the comparison group. Two-way clustered (domain & year) standard errors in parentheses. Signif. Codes: .***: 0.01, **: 0.05, *: 0.1.

^[30]The Callaway and Sant'Anna (2021) estimator, another widely used staggered-robust method, is preferred on many panels but is ill-suited here because of cell sparsity: several of the 55 distinct conversion cohorts contain only one or two treated domains, which breaks the estimator's influence-function-based inference. Sun and Abraham's interaction-weighted approach, which operates within the standard fixed-effects regression framework, accommodates sparse cohorts more naturally.

C.7. Goodman-Bacon Decomposition of the TWFE Estimator

Table A.23 reports the Goodman-Bacon (2021) decomposition of the two-way fixed-effects estimator on the balanced 50 km panel used for the tenryo DiD. The weight on clean treated-versus-never-treated 2×2 comparisons is 94.5%; only 3.8% of the weight falls on the “bad comparison” of already-treated units serving as controls for later-treated units, and 1.7% on yet-to-be-treated comparisons. The dominant never-treated weight means that the near-identical TWFE and Sun-Abraham point estimates reported in Table A.22 are a mechanical consequence of sample composition, not a test of cohort-effect homogeneity.

Comparison type	Weight	Avg. 2×2 estimate
Treated vs. never-treated	94.5%	0.0101
Treated vs. already-treated	3.8%	-0.0024
Treated vs. yet-to-be-treated	1.7%	0.0151

Table A.23: Goodman-Bacon Decomposition of the 50 km Tenryo DiD

Note: The TWFE-on-balanced-panel estimate is 0.0097. Never-treated units make up 82.6% of the sample. The dominant weight on treated-vs-never-treated 2×2 comparisons means the TWFE-vs-Sun-Abraham agreement reported in Table A.22 is mechanical, not evidence of cohort-effect homogeneity.

C.8. Cohort-Split Sun and Abraham ATT

To inspect heterogeneity directly, [Table A.24](#) re-runs the Sun and Abraham (2021) interaction-weighted estimator separately for treated domains whose tenryo conversion year falls in each of three bins: pre-1750, 1750–1799, and post-1800. The pre-1750 cohort holds 49 of the 55 conversions and delivers an aggregate ATT of 0.0443 ($p < 0.001$), essentially reproducing the pooled result. The post-1800 cohort contains only three conversions but also yields a positive ATT of 0.0367 ($p < 0.001$). The small 1750–1799 bin (three conversions) shows a marginally negative point estimate (-0.0064) that is not load-bearing for the aggregate effect. The exit response is therefore concentrated in the period when the tenryo-conversion instrument has the most statistical weight, and no cohort reverses the sign of the effect.

Cohort bin	Cohorts	Treated units	ATT	SE	p
All cohorts	44	55	0.0413	(0.0045)***	< 0.001
pre-1750	38	49	0.0443	(0.0049)***	< 0.001
1750–1799	3	3	−0.0064	(0.0019)***	0.001
post-1800	3	3	0.0367	(0.0016)***	< 0.001

Table A.24: Cohort-Split Sun and Abraham (2021) ATT on Nearby Exit (50 km)

Note: Each row re-runs the interaction-weighted estimator on the sub-sample of treated domains whose tenryo-conversion year falls in the indicated bin (plus all never-treated controls). Two-way clustered (domain and year) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.

C.9. Rambachan-Roth Relative-Magnitudes Breakdown

Table A.25 reports the full relative-magnitudes sensitivity curve of Rambachan and Roth (2023) for the average post-conversion exit ATT at 50 km. $\overline{M}$ is the ratio of the largest post-period deviation from parallel trends to the largest pre-period deviation. The 95% robust CI on the average post-conversion exit effect excludes zero for every $\overline{M} \leq 0.20$; this is the breakdown value $\overline{M}^*$. The exit pre-trend joint Wald test (the basis for the relative-magnitudes scale) is reassuring in its own right: $F = 1.26$, $p = 0.283$ on four pre-period coefficients, consistent with genuinely flat pre-trends. The sensitivity exercise therefore confirms that the main exit effect withstands post-period deviations up to one-fifth of the largest pre-period deviation, on top of a pre-trend joint test that does not reject parallel trends in the first place.

Panel A: Δ^{RM} (Relative Magnitudes)		
$\overline{M}$	95% robust CI	Conclusion
0.00	[0.0108, 0.0694]	excludes 0
0.20	[0.0022, 0.0935]	excludes 0
0.50	[−0.0355, 0.1403]	includes 0
1.00	[−0.1157, 0.2224]	includes 0
2.00	[−0.2798, 0.3081]	includes 0
Panel B: Δ^{SD} (Smoothness)		
M	95% robust CI	Conclusion
0.0000	[0.0227, 0.0773]	excludes 0
0.0010	[0.0108, 0.0855]	excludes 0
0.0016	[0.0011, 0.0929]	excludes 0
0.0020	[−0.0054, 0.0978]	includes 0
0.0030	[−0.0215, 0.1098]	includes 0
0.0050	[−0.0535, 0.1323]	includes 0
0.0100	[−0.1453, 0.1599]	includes 0

Table A.25: Rambachan-Roth (2023) Sensitivity Analysis

Note: Panel A: Δ^{RM} (relative magnitudes). $\overline{M}$ bounds the maximum post-period deviation from parallel trends as a fraction of the largest pre-period deviation. Breakdown: $\overline{M} = 0.20$. Panel B: Δ^{SD} (smoothness). M bounds the maximum second difference of the trend. Breakdown: $M = 0.0016$. Both panels report 95% robust CIs for the average post-conversion exit ATT at 50 km. Exit pre-trend joint Wald: $F = 1.26$, $p = 0.283$ (df = 4).

C.10. Rambachan-Roth Sensitivity at Canonical Benchmarks

As a companion to Table A.25, Table A.26 reports the robust 95% confidence interval for the average post-conversion exit ATT at the canonical Rambachan and Roth (2023) benchmarks $\bar{M} \in \{0.5, 1, 2\}$, so that readers can read the sensitivity at the scale RR itself recommends for comparison. The CI excludes zero at $\bar{M} \leq 0.20$; at $\bar{M} = 0.5$ the CI is $[-0.036, 0.140]$, at $\bar{M} = 1$ it is $[-0.116, 0.222]$, and at $\bar{M} = 2$ it is $[-0.280, 0.308]$. Read together with the pre-period Wald test ($F = 1.26$, $p = 0.283$, which does not reject parallel trends), the effect withstands post-period deviations up to one-fifth of the largest pre-period deviation; the $\bar{M} = 0.5, 1, 2$ columns transparently report where that envelope widens. The smoothness restriction Δ^{SD} tells the same story on a different scale: the CI excludes zero for second-difference bounds up to $M = 0.0016$, and at the empirical pre-period dispersion $\sigma_{\text{pre}} = 0.0103$ (the standard deviation of the four pre-period event-time coefficients) the CI is $[-0.148, 0.164]$. The sensitivity curve is monotone, and the breakdown values $\bar{M}^* = 0.20$ and $M^* = 0.0016$ are the sharp envelope on how much untested post-period differential trend the sign of the effect tolerates.

Panel A: Δ^{RM} (Relative Magnitudes)		
$\bar{M}$	95% robust CI	Conclusion
0.00	[0.0108, 0.0694]	excludes 0
0.20	[0.0022, 0.0935]	excludes 0
0.50	[-0.0355, 0.1403]	includes 0
1.00	[-0.1157, 0.2224]	includes 0
2.00	[-0.2798, 0.3081]	includes 0
Panel B: Δ^{SD} (Smoothness)		
M	95% robust CI	Conclusion
0.0000 (tight)	[0.0227, 0.0773]	excludes 0
0.0016 (breakdown)	[0.0011, 0.0929]	excludes 0
$\sigma_{\text{pre}} = 0.0103$	[-0.1484, 0.1635]	includes 0

Table A.26: Extended Rambachan-Roth (2023) Sensitivity Analysis

Note: Extension of Table A.25 that reports the full relative-magnitudes curve at the canonical benchmarks $\bar{M} \in \{0, 0.2, 0.5, 1, 2\}$ and the smoothness restriction Δ^{SD} at both the breakdown value ($M = 0.0016$) and the empirical pre-period dispersion ($\sigma_{\text{pre}} = 0.0103$, the SD of the four pre-period event-time coefficients). The average post-conversion exit ATT at 50 km tolerates post-period deviations up to 20% of the largest pre-period deviation under Δ^{RM} and up to a second-difference bound of $M = 0.0016$ under Δ^{SD} ; larger bounds include zero. Exit pre-trend joint Wald: $F = 1.26$, $p = 0.283$.

C.11. SUTVA and Spatial Interference

Because tenryo conversions cluster in time and space, some domain-years in the 50 km DiD panel fall within the treatment radius of multiple contemporaneous conversions, potentially violating the stable unit treatment value assumption (SUTVA). I bound this concern with three exclusion exercises. First, I map the overlap: 9.0% of domain-year observations have at least one other conversion within 50 km and ± 5 years. Second, I drop all converted domains whose capitals lie within 100 km of another conversion with a conversion date within 5 years (“contaminated” treated units), removing 30 of the 55 converted domains. Third, I exclude all never-converted control domains whose capitals fall within 50 km of any converted capital, as these controls may be indirectly treated (147 of 239 controls excluded). [Table A.27](#) reports the results. The exit coefficient retains its positive sign and comparable magnitude across all four samples. The point estimates range from 0.027 to 0.038; the reduction in statistical significance in the more restrictive samples reflects the loss of treated units rather than a change in the underlying estimate. The confrontation coefficient remains small and statistically insignificant throughout.

Sample	Nearby Exit		Nearby Confrontation	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Full sample ($N = 33,877$)	0.0385 (0.0131)***	(0.0147)***	0.0076	(0.0122) (0.0136)
Drop overlap ($N = 33,547$)	0.0302 (0.0207)	(0.0207)	0.0067	(0.0203) (0.0227)
Clean controls ($N = 13,886$)	0.0382 (0.0132)***	(0.0155)**	0.0053	(0.0125) (0.0145)
Strictest ($N = 13,556$)	0.0270 (0.0213)	(0.0216)	0.0031	(0.0214) (0.0249)
Domain FE: Yes Year FE: Yes				

Table A.27: SUTVA / Spatial Interference Robustness

Note: Each row re-estimates the main 50 km DiD specification on progressively restricted samples. “Drop overlap” removes converted domains within 100 km and 5 years of another conversion. “Clean controls” removes never-converted domains within 50 km of any converted capital. “Strictest” applies both exclusions. Two SE rows are reported in parentheses: (i) two-way clustered on domain and year and (ii) Conley spatial standard errors with a 70 km cutoff. Significance stars on each SE row reflect that estimator’s p -value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.12. Permutation Test (Full-Pool Randomization)

As a companion to the narrower permutation test reported in the main text, which randomly reassigns fake conversion events among the 239 never-converted domains and serves as a panel-structure placebo, I also run the stricter Fisher-style randomization inference in which the placebo conversion events can land on any of the 294 domains, including the actual 55. This version asks the identification-oriented question: was the actual treatment assignment unusual among all possible 55-out-of-294 assignments? I draw 55 placebo conversion events from the full 294-domain pool in each of 500 iterations, with conversion-year timing drawn from the empirical distribution of the 55 actual conversion years. Each iteration re-applies the same $[-5, +5]$ event window and matched-control-years restriction used in Table 1 and re-estimates the 50 km pooled TWFE exit coefficient. The permutation distribution is centered at zero (mean -0.0001 , SD 0.0140) and the actual estimate of 0.0385 sits far into the right tail: the two-sided p -value is 0.006 and the one-sided right-tail p -value is 0.002 . Both improve on the panel-structure version ($p = 0.012$) and confirm that the exit response is specific to the actual identity and timing of tenryo conversions rather than an artifact of the staggered-adoption panel.

Full-Pool Permutation Test (50 km, Any Nearby Exit)	
Actual $\hat{\beta}$ (Converted $\times$ Post)	0.0385
Cluster SE p -value (actual)	0.0098
Permutation iterations	500
Permutation pool	All 294 domains
Placebo events per iteration	55 (sampled without replacement from 294)
Conversion-year draw	Empirical distribution of actual 55 conversion years
Permutation mean $\hat{\beta}$	-0.0001
Permutation SD	0.0140
Permutation 2.5% quantile	0.0281
Permutation 97.5% quantile	-0.0001
Two-sided p-value ($ \text{perm} \geq \text{actual} $)	0.0060
One-sided right-tail p ($\text{perm} \geq \text{actual}$)	0.0020

Table A.28: Full-Pool Permutation Test for the Exit Coefficient

Note: Permutation test in which the 55 placebo conversion events are drawn from the full 294-domain pool (rather than only the 239 never-converted controls, which is the narrower “panel-structure” placebo reported in the main text). Conversion-year timing is drawn from the empirical distribution of the 55 actual conversion years. Each iteration assigns 55 randomly selected domains as “converted,” applies the same $[-5, +5]$ event window and matched-control-years restriction used in Table 1, and re-estimates the 50 km pooled TWFE exit coefficient. The reported two-sided p -value is the fraction of permutation estimates whose absolute value weakly exceeds the actual absolute estimate.

C.13. Hazard Model for Conversion Timing

The F-test on pre-period event-time coefficients reported in the main text (Table 1) bounds one threat to identification: a secular differential trend between treated and control domains over the five-year window leading up to conversion. A distinct concern is whether attainer decisions responded to **recent** local unrest, making conversion timing endogenous to nearby exit or confrontation. I test this with a hazard model on the panel of 294 domains over 1615–1868. The outcome indicates conversion to tenryo in year t ; domains are censored at conversion or 1868. The covariates are lagged 5-year moving averages of nearby exit and confrontation within 50 km, plus territory size, distance to the Edo–Kyoto axis, and alliance status. Neither lagged exit intensity ($p = 0.1447$ Cox, $p = 0.3608$ logit) nor confrontation intensity ($p = 0.9495$ Cox, $p = 0.5485$ logit) predicts conversion timing; the joint Wald test fails to reject at $p = 0.3439$. Fudai/kin status is the one significant predictor, consistent with the historiography of attainer politics (Ravina 1999). The hazard model, pre-trend test, and permutation analysis provide complementary checks on timing.

Panel A: Cox proportional hazards				
Variable	$\hat{\beta}$	SE	p	Hazard ratio
Nearby exit MA(5)	-5.1838	3.5539	0.1447	0.0056
Nearby confrontation MA(5)	0.1308	2.0674	0.9495	1.1398
log(area)	-0.2334	0.1615	0.1484	0.7918
log(dist to Edo-Kyoto axis)	0.1912	0.2951	0.5171	1.2106
Fudai/kin (vs. tozama)	1.3445	0.5047	0.0077	3.8364
Joint Wald (exit + confrontation = 0)	chi-square = 2.135, df = 2			$p = 0.3439$
Panel B: Discrete-time logit				
Variable	$\hat{\beta}$	SE	p	Odds ratio
Nearby exit MA(5)	-3.1981	3.4998	0.3608	0.0408
Nearby confrontation MA(5)	-1.1433	1.9052	0.5485	0.3188
log(area)	-0.1733	0.1575	0.2712	0.8409
log(dist to Edo-Kyoto axis)	0.1746	0.2901	0.5473	1.1907
Fudai/kin (vs. tozama)	1.3142	0.5070	0.0095	3.7218

Table A.29: Hazard Models for Tenryo Conversion Timing

Note: Cox proportional-hazards and discrete-time logit estimates of the tenryo conversion hazard. The panel contains 51,849 domain-years across 294 domains from 1615 to 1868, with 34 tenryo-conversion events in the estimation sample (34 of the 55 conversions fall after the first-five-year MA warm-up). Nearby-exit and nearby-confrontation variables are lagged 5-year moving averages within 50 km of the focal domain’s capital. Neither lagged exit nor confrontation intensity predicts conversion timing at conventional levels under either estimator; the joint Wald test fails to reject at $p = 0.3439$. The only significant predictor is fudai/kin alliance status, consistent with the historiography of shogunal attainer.

C.14. Conley Bandwidth Sweep (DiD)

The 70 km Conley cutoff used for the DiD main-text standard errors is approximately one first-order spatial neighbor distance, but is a researcher degree of freedom. Table A.30 sweeps the cutoff across five standard values (30, 50, 70, 100, 150 km). The point estimate is invariant to the inference method; the Conley SE varies modestly (0.0103 to 0.0131) with all p -values remaining below 0.005.

Inference	Cutoff	$\hat{\beta}$	SE	p	N
Cluster (domain+year)	–	0.0385	0.0147	0.0098	33877
Conley	30 km	0.0385	0.0123	0.0017	33877
Conley	50 km	0.0385	0.0127	0.0024	33877
Conley (paper)	70 km	0.0385	0.0131	0.0033	33877
Conley	100 km	0.0385	0.0114	0.0007	33877
Conley	150 km	0.0385	0.0103	0.0002	33877

Table A.30: Conley Bandwidth Sweep for the Tenryo DiD (50 km)

Note: Tenryo DiD exit coefficient with standard errors computed under the cluster estimator (the main-text default) and Conley spatial HAC standard errors at five cutoffs. The point estimate does not depend on the inference method; only the SE and p -value vary. The hand-picked 70 km cutoff used in Table 1 sits near the middle of the sweep.

C.15. Kelly (2020) Data-Driven Conley Bandwidth

As an alternative to the hand-picked cutoffs used in the main text (10 km at the village and grid levels, 70 km at the domain and DiD levels), I apply the Kelly (2019) data-driven rule, which selects the bandwidth from the empirical residual correlogram: the smallest distance at which the mean pairwise residual correlation falls below $\exp(-1) \approx 0.37$. Table A.31 reports this bandwidth alongside the hand-picked choice for three analyses: the village cross-section (Table 3 Model 5), the domain-level tax regression (Table 4 Model 4), and the tenryo DiD (Table 1 50 km). The data-driven choice is tighter than the hand-picked one in all three cases (6 km for villages, 10 km for tax, 10 km for DiD). Conley SEs change by less than 0.001 between the two choices in every regression. The village and DiD estimates remain significant at $p < 0.01$, while the tax estimate remains significant at $p < 0.05$.

Analysis	Bandwidth rule	Cutoff	$\hat{\beta}$	Conley SE	Conley p	N
<i>Village-level exit share (Table 3 Model 5 dyadic-FE)</i>						
	hand-picked	10km	0.0244	0.0085	0.0041	2739
	Kelly (2020) data-driven	6km	0.0244	0.0089	0.0062	2739
<i>Domain tax regression (Table 4 Model 4)</i>						
	hand-picked	70km	-0.0112	0.0055	0.0432	230
	Kelly (2020) data-driven	10km	-0.0112	0.0055	0.0449	230
<i>Tenryo DiD (Table 1 50 km)</i>						
	hand-picked	70km	0.0385	0.0131	0.0033	33877
	Kelly (2020) data-driven	10km	0.0385	0.0126	0.0022	33877

Table A.31: Kelly (2020) Data-Driven vs. Hand-Picked Conley Bandwidth

Note: The Kelly bandwidth is the smallest distance at which the empirical residual correlogram falls below $\exp(-1) \approx 0.37$. Pairwise residual products are aggregated into 40 distance bins for the village and DiD regressions and 25 bins for the tax regression. The data-driven bandwidth is tighter than the hand-picked cutoff in all three analyses, especially for the tax regression (10 km vs. 70 km). Point estimates are identical across bandwidth choices; only SEs vary. The data-driven choice moves SEs by less than 0.001 in absolute terms across all three regressions. The village and DiD estimates remain significant at $p < 0.01$; the tax estimate remains significant at $p < 0.05$.

D. VILLAGE EXIT CHOICE ANALYSIS

D.1. Full Table

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(ρ_i)	0.0458 (0.0154)***	0.0235 (0.0117)*	0.0143 (0.0089)	0.0201 (0.0071)***	0.0212 (0.0072)***	0.0244 (0.0068)***
lat					-2.417*** (0.8551)	-2.413* (1.460)
lon					-0.5462** (0.2119)	-0.5291 (0.3918)
ln(Dist. Edo/Kyoto)					0.2731** (0.1094)	0.2241 (0.1395)
ln(Rice)					-0.0048 (0.0046)	-0.0027 (0.0048)
ln(Elev.)					-0.0216 (0.0212)	-0.0291 (0.0214)
ln(Rugg.)					0.0276** (0.0150)	0.0328** (0.0152)
Lat × Lon					0.0174** (0.0064)	0.0171 (0.0112)
<i>Controls</i>						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Decade (t) FE	No	No	No	Yes	Yes	Yes
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.390	0.543	0.621	0.626	0.678

Table A.32: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Full Table)

Note: The coefficients represent OLS estimates of the relationship between each village's geographic exposure and its choice of exit conditional on recorded resistance. Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: .***: 0.01, **. 0.05, *. 0.1.

D.2. Robustness Checks

D.2.1. Alternative Covariates

Outcome:	exit _i							
	Cal. Suit.				Dist. to Edo			
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Variables</i>								
ln(ρ_i)	0.0434 (0.0150)***	0.0162 (0.0090)*	0.0209 (0.0072)***	0.0234 (0.0069)***	0.0439 (0.0149)***	0.0135 (0.0092)	0.0199 (0.0075)**	0.0240 (0.0068)***
lat	−1.3685 (0.2726)	−0.8994 (0.9761)	−2.5470 (0.8580)	−2.5127 (1.3351)	−1.7680 (0.5527)	−3.0658 (1.0682)	−3.5340 (0.9003)	−3.6532 (1.2780)
lon	−0.3040 (0.0711)	−0.0596 (0.2402)	−0.5621 (0.2012)	−0.5293 (0.3450)	−0.4533 (0.1633)	−0.7529 (0.3249)	−0.8904 (0.2710)	−0.9021 (0.3792)
ln(Dist. Edo/Ky- oto)	−0.0238 (0.0449)	0.4825 (0.2092)	0.2846 (0.1153)	0.2497 (0.1346)				
ln(Cal.)	−0.0307 (0.2051)	0.1270 (0.1825)	0.1926 (0.1566)	0.2602 (0.1590)				
ln(Elev.)	0.0132 (0.0250)	−0.0137 (0.0193)	−0.0268 (0.0188)	−0.0369 (0.0180)	0.0009 (0.0196)	−0.0204 (0.0215)	−0.0262 (0.0208)	−0.0329 (0.0212)
ln(Rugg.)	0.0324 (0.0246)	0.0210 (0.0161)	0.0274 (0.0153)	0.0319 (0.0154)	0.0445 (0.0209)	0.0316 (0.0160)	0.0331 (0.0154)	0.0368 (0.0158)
Lat × Lon	0.0096 (0.0019)	0.0057 (0.0073)	0.0183 (0.0064)	0.0178 (0.0102)	0.0127 (0.0041)	0.0223 (0.0083)	0.0260 (0.0070)	0.0266 (0.0100)
ln(Dist. Edo)					−0.1417 (0.0723)	−0.1751 (0.3853)	−0.0181 (0.2963)	−0.1370 (0.3973)
ln(Rice)					−0.0082 (0.0069)	−0.0079 (0.0064)	−0.0046 (0.0045)	−0.0027 (0.0048)
Const.	44.25*** (9.996)				64.05*** (22.24)			
<i>Controls</i>								
Home (<i>j</i>) Domain FE	No	Yes	Yes	No	No	Yes	Yes	No
Neighbor (<i>k</i>) Do- main FE	No	Yes	Yes	No	No	Yes	Yes	No
Decade (<i>t</i>) FE	No	No	Yes	Yes	No	No	Yes	Yes
Home- Neigh- bor Dyad (<i>j, k</i>) FE	No	No	No	Yes	No	No	No	Yes
<i>Fit Statistics</i>								
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.061	0.551	0.627	0.679	0.069	0.549	0.626	0.678

Table A.33: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confronta-
tion
40
(Alternative Covariates)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses.
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.2. Excluding Villages with More Than One Recorded Resistance Episode

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(ρ_i)	0.0488 (0.0153)***	0.0270 (0.0119)**	0.0228 (0.0104)**	0.0229 (0.0094)**	0.0234 (0.0094)**	0.0231 (0.0080)***
lat					-2.6613 (0.9798)	-2.8835 (1.5600)
lon					-0.6092 (0.2415)	-0.6572 (0.4214)
ln(Dist. Edo/Kyoto)					0.2075 (0.0702)	0.1153 (0.0881)
ln(Rice)					-0.0061 (0.0049)	-0.0040 (0.0051)
ln(Elev.)					-0.0267 (0.0214)	-0.0371 (0.0222)
ln(Rugg.)					0.0309* (0.0157)	0.0360** (0.0161)
Lat × Lon					0.0194 (0.0073)	0.0208 (0.0120)
Const.	0.2299*** (0.0348)					
<i>Controls</i>						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
Decade (t) FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,295	2,295	2,295	2,295	2,295	2,295
R2	0.032	0.431	0.590	0.672	0.675	0.732

Table A.34: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation
(Excluding Repeat Villages)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: .***: 0.01, **: 0.05, *: 0.1.

D.2.3. Absolute Distances

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(Dist. to Home Capital _i)	0.0399 (0.0223)*	0.0248 (0.0161)	0.0127 (0.0109)	0.0231 (0.0096)**	0.0160 (0.0104)	0.0203 (0.0083)**
ln(Dist. to Foreign Capital _i)	-0.0596 (0.0140)***	-0.0203 (0.0218)	-0.0172 (0.0199)	-0.0147 (0.0147)	-0.0306 (0.0129)**	-0.0309 (0.0135)**
lat					-2.5043 (0.8401)	-2.5427 (1.4573)
lon					-0.5607 (0.2091)	-0.5495 (0.3908)
ln(Dist. Edo/Kyoto)					0.2839 (0.1131)	0.2405 (0.1354)
ln(Rice)					-0.0047 (0.0047)	-0.0024 (0.0048)
ln(Elev.)					-0.0192 (0.0217)	-0.0252 (0.0220)
ln(Rugg.)					0.0292 (0.0157)	0.0350 (0.0155)
Lat × Lon					0.0180 (0.0063)	0.0181 (0.0112)
Const.	0.2692 (0.0683)					
<i>Controls</i>						
Home (<i>j</i>) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (<i>k</i>) Domain FE	No	No	Yes	Yes	Yes	No
Home-Neighbor Dyad FE	No	No	No	No	No	Yes
Decade (<i>t</i>) FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.030	0.390	0.543	0.621	0.627	0.678

Table A.35: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation
(Abs. Dist. Measures)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: .***: 0.01, **: 0.05, *: 0.1.

D.2.4. Water-Crossing Subsample

The main village distance-ratio regression uses straight-line great-circle distances from each village to its home-domain capital and nearest foreign-domain capital, and these lines can traverse sea straits such as Kanmon, Bungo, Naruto, or Sado. I therefore drop every resistance-event row whose village, home capital, or nearest foreign capital lies on a different main island (Honshu, Kyushu, Shikoku, Sado, or Awaji), classified from latitude and longitude. This removes 150 of 2,739 rows (5.5%). On this water-safe subsample the coefficient on $\log(\rho_i)$ remains positive across all six specifications, including the dyadic home-neighbor fixed-effects column that mirrors Table 3 Model 5 ($\hat{\beta} = 0.0253$, clustered $p < 0.001$; baseline 0.0244, $p = 0.001$).

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Distance ratio ($\ln(\rho_i)$)	0.0460	0.0232	0.0160	0.0209	0.0207	0.0253
Cluster SE	(0.0156)***	(0.0127)*	(0.0105)	(0.0082)**	(0.0080)**	(0.0063)***
Conley SE (10km)	(0.0121)***	(0.0113)**	(0.0094)*	(0.0079)***	(0.0079)***	(0.0086)***
<i>Controls</i>						
Home Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	Yes	No
Home-Nbr Dyad FE	No	No	No	No	No	Yes
Decade FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num. Obs.	2,589	2,589	2,589	2,589	2,589	2,589
R ²	0.029	0.391	0.552	0.635	0.639	0.683
Mean Dep. Var.	0.251	0.251	0.251	0.251	0.251	0.251

Table A.36: Village Distance-Ratio Regression: Water-Crossing Subsample

Note: Re-runs the six village distance-ratio specifications on the subsample that drops resistance-event rows whose straight-line village-to-home-capital or village-to-foreign-capital line crosses a sea strait (i.e., the village, the home capital, and the nearest foreign capital are not all on the same main island). Island membership (Honshu, Kyushu, Shikoku, Sado, Awaji) is assigned from lat/lon bounding boxes. 150 (5.5%) of 2,739 rows are classified as sea-crossing and dropped. The coefficient on $\log(\rho_i)$ remains positive across all six specifications and statistical significance is preserved in Model 6 (dyadic home-neighbor FE plus geography controls), which corresponds to Table 3 Model 5. SE rows: (i) three-way clustered on (home, neighbor, decade); (ii) Conley spatial SE with a 10 km cutoff on village coordinates. Stars reflect the corresponding estimator's p -value. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.5. Conley Standard Error Bandwidth Sensitivity

Table A.37 reports the bandwidth sensitivity of Conley spatial standard errors for the main village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The coefficient on $\log(\rho_i)$ is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification.

Standard error	Estimate	SE	p
Cluster (home + nbr + decade, paper)	0.0244	(0.0068)	0.001
Conley 25 km	0.0244	(0.0089)	0.006
Conley 50 km	0.0244	(0.0097)	0.012
Conley 75 km	0.0244	(0.0102)	0.017
Conley 100 km	0.0244	(0.0099)	0.014
Conley 150 km	0.0244	(0.0078)	0.002

Table A.37: Conley SE Bandwidth Sensitivity: Village Distance-Ratio Regression

Note: The coefficient on $\log(\rho_i)$ from Table 3 Model 5 (dyadic home-neighbor FE) is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification. The first row reports the three-way cluster standard error used in the main table.

D.2.6. Small- G Cluster Sensitivity of the Decade Dimension

Table A.38 reports a small- G cluster-sensitivity check for the village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The decade dimension has $G = 26$ clusters, below the conventional safe threshold for multi-way cluster-robust inference, motivating a check of whether this dimension is load-bearing for the reported significance. Dropping the decade dimension entirely and clustering only on the two higher- G dimensions (home ruler and neighbor ruler) yields $p = 0.0024$, close to the three-way number ($p = 0.0013$); clustering on decade alone yields $p = 0.0014$. The main-text significance is therefore preserved under every cluster specification considered.

Cluster specification	Estimate	SE	p
Three-way (home + nbr + decade, paper)	0.0244	(0.0068)	0.001
Two-way (home + nbr, dropping decade)	0.0244	(0.0079)	0.002
One-way on decade only ($G = 26$)	0.0244	(0.0068)	0.001

Table A.38: Small- G Cluster Sensitivity of the Decade Dimension

Note: Re-runs Table 3 Model 5 (dyadic home-neighbor FE) under three cluster specifications. The decade dimension has $G = 26$ clusters, below the conventional safe threshold for multi-way clustering. The key finding is that the paper’s three-way cluster SE is robust to dropping the decade dimension entirely — two-way clustering on home and neighbor gives nearly identical inference — indicating the decade cluster is not load-bearing for the reported p -value.

D.2.7. Hatamoto Kantō Exclusion

Morris (1990) documents that hatamoto (direct shogunal retainers holding small fiefs) were particularly dense in four Kantō provinces: Musashi, Sagami, Kazusa, and Shimōsa. Because the baseline distance-ratio measure in [Section 3.4.2](#) uses active daimyō and former-domain capitals but omits individual hatamoto fiefs, a village in these provinces may have had a hatamoto landlord closer than its nearest measured outside authority. Excluding the 40 resistance-event rows (1.5% of the sample) leaves the coefficient positive and statistically significant ([Table A.39](#)).

Sample	Num. Obs.	Estimate	SE	<i>p</i>
Full sample (paper)	2739	0.0244	(0.0068)	0.001
Excluding Musashi, Sagami, Kazusa, Shimōsa	2699	0.0245	(0.0069)	0.002

Table A.39: Village Distance-Ratio Regression: Excluding Dense-Hatamoto Kantō Provinces

Note: Re-runs [Table 3](#) Model 5 (dyadic home-neighbor FE) on the subsample that excludes resistance-event rows in the four Kantō provinces with the highest documented hatamoto density – Musashi, Sagami, Kazusa, and Shimōsa (Morris 1990) – where the distance-ratio measure may under-capture the nearest outside authority because hatamoto fiefs are not in the baseline daimyō/former-domain-capital geography. The coefficient on $\log(\rho_i)$ remains positive and statistically significant.

D.2.8. Desertion-Only Village Cross-Section

Table A.40 re-estimates the preferred specifications with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals. The preferred coefficient is positive, marginal under three-way clustering ($p = 0.051$), and significant with Conley inference ($p = 0.041$), so the narrower definition preserves the direction and most of the pooled magnitude with some inference sensitivity.

Outcome:	Desertion _{<i>i</i>}	
Models:	(4)	(5)
<i>Variables</i>		
Distance ratio ($\ln(\rho_i)$)	0.0127	0.0141
Cluster SE	(0.0060)**	(0.0068)*
Conley SE (10km)	(0.0062)**	(0.0069)**
<i>Controls</i>		
Home + Neighbor + Decade FE + Geography	Yes	No
Dyadic + Decade FE + Geography	No	Yes
<i>Fit Statistics</i>		
Num.Obs.	2,727	2,727
R2	0.598	0.674

Table A.40: Desertion-Only Village Cross-Section

Note: Replicates Table 3 Models 4–5 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Two SE rows: (i) three-way clustered; (ii) Conley spatial SE (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.9. Desertion-Only Grid-Year Panel

Table A.41 replicates Table 2 with exit restricted to desertion (*chōsan*) only. The desertion-specific coefficients track the pooled exit estimates closely in both sign and significance, across the full panel and the ever-active subsample, confirming that the within-cell exit response is driven by the prototypical desertion cases rather than by end-run appeals.

Outcome:	Full sample		Ever-active cells	
	(1)	(2)	(3)	(4)
	Desertion	Pooled exit	Desertion	Pooled exit
Distance ratio ($\tilde{\rho}_{i,t}$)	0.00024	0.00035	0.00152	0.00198
Cluster SE	(0.00014)*	(0.00017)**	(0.00087)*	(0.00109)*
Conley SE (10km)	(0.00012)**	(0.00017)**	(0.00075)**	(0.00103)*
<i>Controls</i>				
Grid & Year FE	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>				
Num.Obs.	2,225,802	2,225,802	372,364	372,364
R ²	0.005	0.006	0.006	0.008
Mean Dep. Var.	0.00009	0.00022	0.00054	0.00131

Table A.41: Desertion-Only Grid-Year Panel

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Pooled exit columns reproduce the main specification for comparison. Columns (1)–(2) use the full panel of 8,763 cells; columns (3)–(4) restrict to the 1,466 ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.2.10. Grid-Panel Count Models: Poisson and Negative Binomial

Table A.42 replicates the main-text Table 2 using two count-data estimators with hexagon and year fixed effects: Poisson (columns 1–3) and the negative binomial (columns 4–6). Because both estimators use a log link ($\log \mathbb{E}[y_{i,t} | X_{i,t}] = \beta X_{i,t} + \alpha_i + \tau_t$), the raw coefficient $\hat{\beta}$ sits on the log-count scale. The *incidence rate ratio*, $\text{IRR} \equiv \exp(\hat{\beta})$, back-transforms this to the count scale: a one-unit increase in the regressor multiplies the expected count by IRR. Both estimators agree on the exit-versus-confrontation asymmetry. The exit coefficient is positive and corresponds to a multiplicative association of roughly 4–5 $\times$ per unit distance ratio (Poisson IRR = 4.020; NB IRR = 5.357). The confrontation coefficient is small and statistically indistinguishable from zero under both estimators (Poisson $p = 0.5676$; NB $p = 0.4891$), and the strict-confrontation coefficient turns negative (IRRs below one). The NB columns drop all-zero cells via the conditional likelihood, so sample size varies by outcome; the Poisson columns use the full 8,763-cell panel.

Outcome:	Poisson			Neg. Binom.		
	(1)	(2)	(3)	(4)	(5)	(6)
	Exit count	Confronta- tion count	Confronta- tion count (strict)	Exit count	Confronta- tion count	Confronta- tion count (strict)
Distance ratio ($\tilde{\rho}_{i,t}$)	1.3912	0.3240	−0.4231	1.6784	0.4453	−0.5370
IRR	4.020	1.383	0.655	5.357	1.561	0.584
Cluster SE	(0.9599)	(0.5668)	(0.5299)	(0.8860)*	(0.6437)	(0.7985)
Conley SE (10km)	(0.9043)	(0.5257)	(0.6052)	(0.8601)*	(0.6620)	(0.7938)
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	62,900	202,130	40,810	62,900	202,130	40,810
Pseudo R ²	0.135	0.130	0.127	0.093	0.092	0.087
Mean Dep. Var.	0.0003	0.0009	0.0004	0.0003	0.0009	0.0004

Table A.42: Grid-Panel Count Models: Poisson and Negative Binomial with FE

Note: Columns (1)–(3) estimate a Poisson model and columns (4)–(6) a negative binomial, both with hexagon and year fixed effects on the 5 km grid-year panel (1615–1868). Coefficients are on the log-count scale; $\text{IRR} = \exp(\hat{\beta})$ gives the multiplicative effect on the expected count. Num. Obs. in the NB columns reflects the ever-event subsample retained by the conditional likelihood; the Poisson columns use the full 8,763-cell panel. SEs: (i) two-way clustered on cell and year; (ii) Conley (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.11. Ever-Active Cells Subsample

Table A.43 replicates Table 2 on the 1,466 “ever-active” hexagons — cells with at least one exit or confrontation event between 1615 and 1868. This restriction filters out the 83% of cells that recorded neither form of resistance and sharpens within-cell variation among active locales while using the same binary-outcome LPM and fixed-effects structure. Columns (1)–(3) use the distance ratio $\tilde{\rho}_{i,t}$; columns (4)–(6) use the closest-capital distance $d_{i,t}^1$ (km).

The distance-ratio coefficient on exit rises to 0.00198 ($p = 0.054$ Conley, $p = 0.070$ clustered), while the confrontation and strict-confrontation coefficients remain small and statistically indistinguishable from zero. The closest-capital regressor remains a null predictor of exit on this subsample ($p = 0.179$ clustered, $p = 0.259$ Conley). It carries a marginal negative signal for strict confrontation (Conley $p = 0.028$), plausibly because uprisings and castle attacks require a nearby state target. Remoteness therefore contains some information about where direct confrontation is mechanically feasible but not where exit is chosen; relative geographic exposure is associated with exit.

Outcome:	Distance ratio ($\tilde{\rho}_{i,t}$)			Closest-capital dist. ($d_{i,t}^1$, km)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any con- frontation	Any con- frontation (strict)	Any exit	Any con- frontation	Any con- frontation (strict)
Coefficient	0.00198	0.00096	−0.00097	0.00004	−0.00004	−0.00004
Cluster SE	(0.00109)*	(0.00105)	(0.00067)	(0.00003)	(0.00003)	(0.00002)*
Conley SE (10km)	(0.00103)*	(0.00105)	(0.00062)	(0.00004)	(0.00002)*	(0.00002)**
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	372,364	372,364	372,364	372,364	372,364	372,364
R ²	0.008	0.011	0.011	0.008	0.011	0.011
Mean Dep. Var.	0.00131	0.00384	0.00155	0.00131	0.00384	0.00155

Table A.43: Grid-Year Panel on Ever-Active Cells: Distance Ratio vs. Closest-Capital

Distance

Note: Replicates Table 2 on the 1,466 “ever-active” hexagons — cells with at least one exit or confrontation event between 1615 and 1868 — which sharpens within-variation among active locales while filtering out cells with no recorded resistance. Columns (1)–(3) use the time-varying distance ratio $\tilde{\rho}_{i,t} = \frac{d_{i,t}^1}{d_{i,t}^2}$. Columns (4)–(6) replace it with the raw closest-capital distance $d_{i,t}^1$ (km). Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.2.12. Least-Cost-Path Distance Ratio

Table A.44 replicates Table 2 with the great-circle distance ratio $\tilde{\rho}_{i,t} = \frac{d_{i,t}^1}{d_{i,t}^2}$ replaced by a terrain-weighted least-cost-path variant $\tilde{\rho}_{i,t}^{\text{LCP}}$, computed from a $(1 + \text{slope})$ cost surface on the SRTM elevation raster. For each of the 8,763 5 km hexagon centroids and each unique capital location, I run a single-source Dijkstra over the cost surface and record the cumulative terrain-weighted travel cost; $\tilde{\rho}_{i,t}^{\text{LCP}}$ is the ratio of the nearest-to-second-nearest such cost among capitals active in year t . The two regressors capture overlapping but distinct information: their Pearson correlation is 0.44, and the LCP ratio has mean 0.82 and standard deviation 0.24 (vs. 0.63 and 0.26 for the great-circle ratio), reflecting the common terrain component that travel costs to multiple capitals share when paths cross the same ridges and valleys.

The exit coefficient preserves its sign and standardized magnitude: a one-standard-deviation increase in $\tilde{\rho}_{i,t}^{\text{LCP}}$ corresponds to $0.00045 \times 0.24 \approx 0.011\%$ -points, close to the 0.009%-points implied by Table 2 column (1). The exit-versus-confrontation asymmetry also survives: confrontation coefficients remain small and statistically indistinguishable from zero across the full panel and ever-active subsample. Statistical significance on exit falls below conventional thresholds ($p = 0.278$ clustered; $p = 0.203$ Conley). Terrain friction adds a shared component to the cost of reaching any capital and compresses the contrast between the nearest two, making the LCP ratio a noisier proxy for relative geographic exposure. The alternative measure therefore preserves sign and standardized magnitude but yields less precise inference.

Outcome:	<i>Full sample</i>			<i>Ever-active cells</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any con- frontation	Any con- frontation (strict)	Any exit	Any con- frontation	Any con- frontation (strict)
Distance ratio (LCP, $\tilde{\rho}_{i,t}^{\text{LCP}}$)	0.00045	0.00053	−0.00008	0.00173	0.00177	−0.00051
Cluster SE	(0.00041)	(0.00060)	(0.00031)	(0.00169)	(0.00221)	(0.00122)
Conley SE (10km)	(0.00035)	(0.00068)	(0.00026)	(0.00141)	(0.00192)	(0.00074)
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,225,802	2,225,802	2,225,802	372,364	372,364	372,364
R ²	0.006	0.007	0.006	0.008	0.011	0.011
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00131	0.00384	0.00155

Table A.44: Grid-Year Panel with Least-Cost-Path Distance Ratio

Note: LPM on binary any-event-of-type indicators. Replicates [Table 2](#) with the haversine distance ratio replaced by a terrain-weighted least-cost-path (LCP) variant: $\tilde{\rho}_{i,t}^{\text{LCP}}$ equals the ratio of nearest-to-second-nearest LCP cost from each hexagon centroid to any active domain capital in year t , where cost is cumulative across pixels of a $(1 + \text{slope})$ cost surface derived from the SRTM elevation raster. Columns (1)–(3) use the full panel; (4)–(6) restrict to ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.13. Daikansho-Inclusive Distance Ratio (Village)

Table A.45 replicates Table 3 with d_i^{FC} redefined as the distance to the nearest non-home administrative center of any type, including shogunal intendant offices (*daikansho*). The preferred dyadic-FE specification (Model 5) is virtually unchanged ($\hat{\beta} = 0.0268$ vs. 0.0244 in Table 3).

Outcome:	Exit _i				
Models:	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
Distance ratio ($\ln(\rho_i)$)	0.0422	0.0496	0.0201	0.0202	0.0268
Cluster SE	(0.0148)***	(0.0137)***	(0.0074)**	(0.0075)**	(0.0067)***
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0077)***	(0.0079)**	(0.0085)***
<i>Controls</i>					
Decade FE	No	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No
Dyadic FE	No	No	No	No	Yes
Geography	No	No	No	Yes	Yes
<i>Fit Statistics</i>					
Num. Obs.	2,739	2,739	2,739	2,739	2,739
R ²	0.024	0.222	0.633	0.638	0.686
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246

Table A.45: Village Exit Choice with Daikansho-Inclusive Distance Ratio

Note: Replicates Table 3 with d_i^{FC} redefined to include shogunal intendant offices (*daikansho*). Two SE rows: (i) clustered on home, nearest neighboring domain, and decade; (ii) Conley spatial standard errors (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.14. Daikansho-Inclusive Distance Ratio (Grid)

Table A.46 replicates Table 2 with the daikansho-inclusive distance ratio. Adding daikansho to the capital set compresses the ratio toward unity and reduces within-cell variation; all grid-level coefficients attenuate and lose statistical significance. The village-level and tax results are unaffected by this inclusion (Supporting Information D.2.13; Supporting Information D.3.1).

Outcome:	Full sample				Ever-active cells		
	(1)	(2)	(3)	(4)	(5)	(6)	
	Any exit	Any confrontation	Any confrontation (strict)	Any exit	Any confrontation	Any confrontation (strict)	
Distance ratio ($\hat{\rho}_{i,t}$)	0.00019	0.00023	−0.00006	0.00095	−0.00012	−0.00096	
Cluster SE	(0.00019)	(0.00021)	(0.00012)	(0.00111)	(0.00111)	(0.00068)	
Conley SE (10km)	(0.00018)	(0.00021)	(0.00012)	(0.00107)	(0.00097)	(0.00060)	
<i>Controls</i>							
Grid & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	
<i>Fit Statistics</i>							
Num.Obs.	2,225,802	2,225,802	2,225,802	372,364	372,364	372,364	
R ²	0.006	0.007	0.006	0.008	0.011	0.011	
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00131	0.00384	0.00155	

Table A.46: Grid-Year Panel with Daikansho-Inclusive Distance Ratio

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with the daikansho-inclusive distance ratio. Columns (1)–(3) use the full panel of 8,763 cells; columns (4)–(6) restrict to the 1,466 ever-active hexagons. See Table 2 notes for details. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.3. Additional Analyses

D.3.1. Daikansho-Inclusive Distance Ratio (Tax)

Table A.47 replicates Table 4 with the daikansho-inclusive distance ratio. Models 1–2 remain significant at $p < 0.01$; models 3–4 show modest p -value drift (from $p < 0.05$ to $p \approx 0.07$) with stable point estimates, consistent with daikansho offices compressing the distance ratio’s variance. The spatial lag models (5–6) preserve the negative sign but are not statistically significant, mirroring the pattern in the main table where the fully controlled spatial lag is only marginally significant.

Outcome:	Tax Rate (τ_j)					
	OLS				Spatial Lag	
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Distance ratio ($\ln(\rho_j)$)	−0.0233	−0.0192	−0.0098	−0.0096	−0.0130	−0.0080
HC SE	(0.0054)***	(0.0058)***	(0.0054)*	(0.0054)*	(0.0047)***	(0.0050)
<i>Controls</i>						
Domain Characteristics	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbor Characteristics	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	232	230	230	230	230	230
R2 / Pseudo R2	0.071	0.138	0.368	0.386	0.309	0.437
Mean Dep. Var.	0.391	0.391	0.391	0.391	0.394	0.394

Table A.47: Domain Tax Rates with Daikansho-Inclusive Distance Ratio

Note: Replicates Table 4 with the daikansho-inclusive distance ratio. Models 1–4: OLS with heteroskedasticity-robust standard errors. Models 5–6: spatial lag (SAR) with neighborhood-weighted tax rates as spatial regressor; standard errors from the spatial likelihood. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.3.2. Controlling for Distance to the Border

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(ρ_i)	0.0487 (0.0161)***	0.0225 (0.0124)*	0.0152 (0.0098)	0.0206 (0.0068)***	0.0206 (0.0065)***	0.0264 (0.0075)***
ln(Shortest Distance to Border) _{t}	0.0060 (0.0103)	0.0073 (0.0085)	0.0010 (0.0106)	0.0057 (0.0089)	0.0058 (0.0086)	0.0037 (0.0086)
<i>Controls</i>						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Decade (t) FE	No	No	No	Yes	Yes	Yes
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,391	2,391	2,391	2,391	2,391	2,391
R2	0.037	0.377	0.535	0.632	0.638	0.676

Table A.48: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Shortest Distance to Border)

Note: The coefficients relate the distance ratio (home-capital distance divided by distance to the nearest active non-home capital in the baseline daimyō/former-domain-capital geography) to exit choice conditional on recorded resistance, controlling for shortest distance to another domain's border. Border data match 2,391 of the 2,739 preferred-panel rows. On this covered sample, the preferred dyadic estimate is 0.0265 without and 0.0264 with the border control; the border coefficient is 0.0037 ($p = 0.668$). Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.3.3. Sequential Decomposition of the Village Choice

The main analysis in [Section 3.4.3](#) conditions on recorded resistance and so cannot, on its own, rule out an association between geographic exposure and resistance incidence. I therefore combine the resisting-village data with the comprehensive end-period village ledger, adding compliant villages to the analysis. For each resisting village, I identify its counterpart in the full ledger, using the geographically nearest match when names do not resolve the record, and remove matched ledger rows to avoid duplication. A village-event remains a separate observation when the home or neighboring domain differs or when the event records a distinct tactic. The combined data contain 46,322 observations; actions by resisting villages are classified as exit or direct confrontation, while unmatched villages are labeled compliance.

On this combined sample, I decompose the village's three-way choice into two binary steps: Step 1 estimates recorded resistance versus compliance over the full village sample, and Step 2 estimates exit versus direct confrontation conditional on resistance. This is the two-step structure implied by a nested logit with resistance nest = {exit, confrontation}, and it can be estimated as two internally consistent binary models without the IIA restriction imposed by a flat multinomial logit.

Formally, let i index villages and let the choice set be $\mathcal{J} = \{C, E, F\}$ (compliance, exit, direct confrontation) with nest structure $B_0 = \{C\}$ and $B_1 = \{E, F\}$. Random utility is $U_{ij} = V_{ij} + \varepsilon_{ij}$, with the GEV distribution

$$F(\varepsilon_i) = \exp\left\{-e^{-\varepsilon_{iC}} - \left(e^{-\frac{\varepsilon_{iE}}{\lambda}} + e^{-\frac{\varepsilon_{iF}}{\lambda}}\right)^\lambda\right\},$$

where $\lambda \in (0, 1]$ is the dissimilarity parameter for the resistance nest ($\lambda = 1$ collapses to multinomial logit). With covariates X_i governing resistance incidence and Z_i governing tactic composition, the linear indices are $V_{iC} = 0$ (normalization), $V_{iE} = X_{i'}\gamma + Z_{i'}\delta_E$, and $V_{iF} = X_{i'}\gamma + Z_{i'}\delta_F$. Defining the inclusive value $I_i = \log\left[\exp\left(\frac{V_{iE}}{\lambda}\right) + \exp\left(\frac{V_{iF}}{\lambda}\right)\right]$, the closed-form choice probabilities are

$$\mathbb{P}(R_i = 1 \mid X_i, Z_i) = \frac{\exp(\lambda I_i)}{1 + \exp(\lambda I_i)}, \quad \mathbb{P}(E_i = 1 \mid R_i = 1, Z_i) = \Lambda\left((\delta_E - \delta_F)' \frac{Z_i}{\lambda}\right),$$

where $\Lambda(\cdot)$ is the logistic CDF. Letting $r_i = \mathbb{1}\{y_i \in \{E, F\}\}$ and $e_i = \mathbb{1}\{y_i = E\}$, the nested-logit log-likelihood factors as

$$\begin{aligned} \log \mathcal{L} = & \underbrace{\sum_i [r_i \log \mathbb{P}(R_i) + (1 - r_i) \log(1 - \mathbb{P}(R_i))]}_{\log \mathcal{L}_1(\psi_1)} \\ & + \underbrace{\sum_{i:r_i=1} [e_i \log \mathbb{P}(E_i | R_i) + (1 - e_i) \log(1 - \mathbb{P}(E_i | R_i))]}_{\log \mathcal{L}_2(\psi_2)}. \end{aligned}$$

Because $\log \mathcal{L}_1$ depends only on the top-nest parameters ψ_1 and $\log \mathcal{L}_2$ only on the within-nest contrast $\psi_2 = \frac{\delta_E - \delta_F}{\lambda}$, joint MLE coincides with the concatenation of two independent maximizations. Reparameterizing the singleton-nest top level into a single binary index $X_{i'}\beta_1$, the two pieces actually estimated are

$$\text{Step 1: } \mathbb{P}(R_i = 1 | X_i) = \Lambda(X_{i'}\beta_1), \quad i = 1, \dots, N,$$

$$\text{Step 2: } \mathbb{P}(E_i = 1 | R_i = 1, Z_i) = \Lambda(Z_{i'}\beta_2), \quad i \in \{i : r_i = 1\}.$$

β_1 identifies correlates of resistance incidence; β_2 identifies the within-resistance contrast $\frac{\delta_E - \delta_F}{\lambda}$, so its sign and significance are interpretable directly while its magnitude is identified jointly with the dissimilarity parameter λ . Estimating Step 1 and Step 2 separately writes the nested logit in sequential form, provided X_i and Z_i enter the linear indices as a clean partition and V_{iC} is normalized to zero.

[Table A.49](#) reports three complementary specifications side by side. Columns (1)–(2) report a flat multinomial logit (no fixed effects) as an IIA-restricted anchor. Columns (3)–(4) and (5)–(6) estimate the sequential decomposition as binary logit and linear probability model respectively. Step 1 (Columns (3) and (5)) runs on one row per unique village and uses no fixed effects because compliant villages drawn from the full ledger have no observed neighboring ruler or event date. Step 2 (Columns (4) and (6)) runs on the full resistance-event panel and matches [Table 3](#) Model 5: home $\times$ neighbor dyadic fixed effects, decade fixed effects, the same geographic controls, and three-way clustering on home ruler, neighbor, and decade.

The sequential decomposition addresses selection into recorded resistance directly. Step 1 uses one row per unique village (46,296 villages, of which 2,482 appear in the resisting set). The binary logit gives $\beta = +0.028$ ($p = 0.58$), while the LPM gives $\beta = -0.00078$ ($p = 0.12$ heteroskedasticity-robust; $p = 0.42$ Conley); both are small and statistically indistinguishable from zero. Step 2 is positive in both estimators: the binary logit gives $\beta = 0.276$ (cluster $p = 0.009$; Conley $p = 0.017$) and the LPM gives $\beta = 0.0244$ (cluster $p = 0.001$; Conley $p =$

0.004). The Step 2 LPM reproduces Table 3 Model 5. Geographic exposure is therefore associated with tactic composition (exit versus direct confrontation) without a detectable positive association with resistance incidence in this specification.

	<i>Multinomial logit</i>		<i>Seq. binary logit</i>		<i>Seq. LPM</i>	
Outcome:	exit	direct confrontation	non-compliance	exit non-compliance	non-compliance	exit non-compliance
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Distance ratio ($\ln(\rho_i)$)	0.2823	−0.0055	0.0281	0.2762	−0.00078	0.0244
Robust / 3-way cluster SE	(0.0149)***	(0.0167)	(0.0514)	(0.1054)***	(0.00050)	(0.0068)***
Conley SE (10km)	—	—	(0.0771)	(0.1159)**	(0.00096)	(0.0085)***
<i>Controls</i>						
Geo controls	Yes	Yes	Yes	Yes	Yes	Yes
Home × Neighbor FE	No	No	No	Yes	No	Yes
Decade FE	No	No	No	Yes	No	Yes
<i>Fit Statistics</i>						
Num. Obs.	46,387	46,387	46,296	1,310	46,296	2,739
Mean Dep. Var.	0.014	0.042	0.054	0.246	0.054	0.246

Table A.49: Sequential Decomposition of the Village Choice: Logit and LPM

Note: Columns (1)–(2) report a flat multinomial logit (no fixed effects) as an IIA-restricted anchor; both coefficients come from a single regression with compliance as the baseline category. Columns (3)–(4) and (5)–(6) estimate the sequential decomposition as binary logit and linear probability model respectively. Step 1 (Columns (3) and (5)) runs on one row per unique village (non-compliance vs. compliance); it omits dyadic and decade fixed effects because compliant villages drawn from the full ledger have no observed neighboring ruler or event date. Step 2 (Columns (4) and (6)) runs on the full non-compliance-event panel (exit vs. direct confrontation conditional on non-compliance) and reproduces Table 3 Model 5: home × neighbor dyadic fixed effects, decade fixed effects, and the same geographic controls. Two SE rows are reported in parentheses. The first is heteroskedasticity-robust for the multinomial logit, Step-1 logit, and Step-1 LPM, and three-way clustered on home ruler, neighbor, and decade for Step 2. The second is Conley spatial inference with a 10 km cutoff, omitted for Columns (1)–(2) because no standard implementation exists for the multinomial likelihood. Stars reflect the corresponding estimator’s p -value. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.3.4. Sequential Decomposition Under Time-Averaged Distance Ratio

The distance ratio used in the main Step-1 regression (Table A.49) differs across outcomes: resisting villages receive the event-time ratio, while compliant villages from the full ledger receive the end-period (1868) ratio. Table A.50 removes this asymmetry by applying to both groups each village’s mean annual home-capital distance divided by nearest-foreign-capital distance over 1615–1868, with active capitals reconstructed each year. Under this symmetric construction Step 1 is small and negative ($\hat{\beta}_{\text{logit}} = -0.0303$, $p = 0.0402$; LPM $\hat{\beta} = -0.00155$, $p = 0.0387$). Step 2, exit versus direct confrontation conditional on resistance, retains its positive sign ($\hat{\beta} = 0.01436$, cluster $p = 0.0089$, Conley $p = 0.0524$). The symmetric measure therefore preserves the composition result without evidence that exposure positively selects villages into recorded resistance.

Inference	Step 1: non-compliance vs. compliance			Step 2: exit vs. direct confrontation		
	Logit	LPM (robust)	LPM (Conley 10 km)	LPM (3-way cluster)	LPM (Conley 10 km)	
$\hat{\beta}$ on $\ln(\bar{p}_i^{\text{avg}})$	-0.0303	-0.00155	-0.00155	0.01436	0.01436	
SE	0.0148	0.00075	0.00072	0.00506	0.00740	
p	0.0402	0.0387	0.0311	0.0089	0.0524	
Geo controls	Yes	Yes	Yes	Yes	Yes	
Home $\times$ Neighbor FE	No	No	No	Yes	Yes	
Decade FE	No	No	No	Yes	Yes	
Num. Obs.	46024	46024	46024	2740	2740	
Mean Dep. Var.	0.0543	0.0543	0.0543	0.2464	0.2464	

Table A.50: Sequential Decomposition Under Time-Averaged Distance Ratio

Note: Replicates the sequential decomposition in Table A.49 with a time-averaged distance ratio applied symmetrically to resisting and compliant villages. The time-averaged ratio is the mean of the annual distance-to-home-domain capital / distance-to-nearest-foreign-domain capital across 1615–1868, with the active-domain panel recomputed each year using the augmented tenryo spells. This symmetric construction removes the measurement asymmetry in the existing Step-1 table, where the distance ratio for resisting villages used event-time capitals while that for compliant villages used end-period positions. Step 1 (non-compliance vs. compliance) is estimated on all 46,024 villages (2,497 non-compliant); Step 2 (exit vs. direct confrontation conditional on non-compliance) uses its own 2,740-event time-averaged analytic sample. Under time-averaged capitals Step 1 is a small negative association ($\hat{\beta}_{\text{logit}} = -0.030$, $p = 0.040$; LPM $\hat{\beta} = -0.00155$, $p = 0.039$), and Step 2 retains its positive sign ($\hat{\beta} = 0.014$, $p = 0.009$ cluster, $p = 0.052$ Conley). The symmetric specification therefore provides no evidence that exposure positively selects villages into recorded non-compliance while preserving the positive composition estimate conditional on non-compliance.

D.3.5. Linear Hypothesis Testing

In the village-level analysis for validating the village mobility measure, I use the log-ratio of home-capital distance to nearby-capital distance:

$$\ln\left(\frac{\text{Dist. to Home Cap.}}{\text{Dist. to Foreign Cap.}}\right) = \ln(\text{Dist. to Home Cap.}) - \ln(\text{Dist. to Foreign Cap.})$$

Implicit in this choice is the restriction: $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$

To check whether the data are consistent with this restriction, I estimate the unrestricted model including both $\ln(\text{Dist. to Home Cap.})$ and $\ln(\text{Dist. to Foreign Cap.})$ separately, then conduct a Wald test of the linear hypothesis $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$.

Corresponding Model	Outcome	Restriction	Df	χ^2	p-value
Table 3 (5)	exit_i	$\beta_{\ln(\text{DHC})} + \beta_{\ln(\text{DFC})} = 0$	1	0.009	0.9244

Table A.51: Wald Test of the Log-Ratio Restriction

The Wald tests indicates that the log-ratio specification is consistent with the data; the high p-values (0.92 and 0.69, respectively) indicate no significant evidence against the restriction.

E. DOMAIN TAXATION ANALYSIS

E.1. Full Tables

E.1.1. Main Model

Outcome:	τ_j											
Models:	(1)		(2)		(3)		(4)		(5: SPL)		(6: SPL)	
<i>Variables</i>												
$\ln(\rho_j)$	−0.0241	(0.0053)***	−0.0203	(0.0058)***	−0.0110	(0.0055)**	−0.0112	(0.0055)**	−0.0140	(0.0047)***	−0.0094	(0.0051)*
$\ln(\text{Rice})$			−0.0010	(0.0158)	0.0076	(0.0157)	0.0116	(0.0163)			0.0004	(0.0135)
$\ln(\text{Area})$			−0.0260	(0.0112)**	−0.0318	(0.0144)**	−0.0340	(0.0151)**			−0.0227	(0.0130)*
$\ln(\text{Pop.})$			0.0392	(0.0167)**	0.0263	(0.0144)*	0.0232	(0.0150)			0.0222	(0.0147)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)			0.0004	(0.0002)
Shog. Ally			−0.0316	(0.0177)*	0.0059	(0.0161)	0.0011	(0.0164)			0.0025	(0.0143)
Goshi			0.0217	(0.0256)	0.0196	(0.0261)	0.0274	(0.0271)			0.0242	(0.0215)
lat					−0.5279	(0.2934)*	−0.4747	(0.3077)			−0.3535	(0.2855)
lon					−0.1758	(0.0754)**	−0.1639	(0.0796)**			−0.1252	(0.0743)*
$\ln(\text{Elev.})$					−0.0118	(0.0160)	−0.0137	(0.0165)			−0.0100	(0.0134)
$\ln(\text{Rugg.})$					0.0398	(0.0160)**	0.0373	(0.0158)**			0.0238	(0.0152)
$\ln(\text{Dist. Edo/Kyoto})$					−0.0370	(0.0158)**	−0.0441	(0.0171)**			−0.0335	(0.0167)**
Rainy					−1.6629	(0.8969)*	−1.5887	(0.9078)*			−1.0143	(0.7369)
Snowy					0.7322	(1.8580)	0.6920	(1.8597)			−0.2419	(1.4193)
Lat × Lon					0.0042	(0.0021)*	0.0038	(0.0023)*			0.0029	(0.0021)
Kin Nbs.							0.0396	(0.0296)			0.0188	(0.0257)
Nb. FL							0.0002	(0.0001)			0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							−0.0244	(0.0189)			−0.0221	(0.0172)
$\ln(\text{Nb. Area})$							0.0180	(0.0176)			0.0116	(0.0159)
$\ln(\text{Nb. Pop.})$							−0.0047	(0.0207)			−0.0015	(0.0196)
Const.	0.4152	(0.0091)***	0.1760	(0.1247)	23.1744	(10.2659)**	21.8425	(10.7990)**	0.1761	(0.0274)***	16.4558	(10.0938)
<i>Fit Statistics</i>												
Num.Obs.	232		230		230		230		230		230	
R2	0.074		0.141		0.370		0.389		0.312 ^[a]		0.439 ^[a]	

Table A.52: Geographic Exposure and Reported Domain Tax Rates

Note: OLS estimates (maximum likelihood in Models 5–6) of reported effective tax rates on the log domain-average distance ratio. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Models 5–6. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. ^[a] Nagelkerke pseudo-R2

E.1.2. Interaction Model

Outcome:	τ_j			
Models:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Distance ratio $(\ln(\rho_j) \times \hat{y}_{-j})$: High	−0.0350 (0.0084)***	−0.0286 (0.0089)***	−0.0143 (0.0070)**	−0.0158 (0.0069)**
Distance ratio $(\ln(\rho_j) \times \hat{y}_{-j})$: Low	−0.0161 (0.0053)***	−0.0151 (0.0058)***	−0.0092 (0.0063)	−0.0085 (0.0064)
<i>Controls</i>				
Main	No	Yes	Yes	Yes
Geography	No	No	Yes	Yes
Neighbors	No	No	No	Yes
<i>Fit Statistics</i>				
Num.Obs.	232	230	230	230
R2	0.091	0.150	0.371	0.391
Mean Dep. Var.	0.392	0.394	0.394	0.394

Table A.53: Geographic Exposure and Tax Rates, Interacted with Outside-Option Quality

Note: The coefficients represent OLS estimates of the relationship between the average distance ratio of each domain interacted with $\hat{y}_{-j}$ (indicator of whether the weighted average labor-to-potential-yield ratio of neighboring domains is above its median) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.1.3. Interaction Model (Full Table)

Outcome:	τ_j							
Models:	(1)		(2)		(3)		(4)	
<i>Variables</i>								
$\ln(\rho_j) \times \text{High}$	-0.0350	(0.0084)***	-0.0286	(0.0089)***	-0.0143	(0.0070)**	-0.0158	(0.0069)**
$\ln(\rho_j) \times \text{Low}$	-0.0161	(0.0053)***	-0.0151	(0.0058)***	-0.0092	(0.0063)	-0.0085	(0.0064)
$\ln(\text{Rice})$			0.0014	(0.0156)	0.0083	(0.0153)	0.0122	(0.0159)
$\ln(\text{Area})$			-0.0256	(0.0111)**	-0.0314	(0.0144)**	-0.0336	(0.0151)**
$\ln(\text{Pop.})$			0.0360	(0.0171)**	0.0253	(0.0145)*	0.0221	(0.0151)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)
Shog. Ally			-0.0300	(0.0180)*	0.0064	(0.0163)	0.0016	(0.0166)
Goshi			0.0221	(0.0257)	0.0197	(0.0262)	0.0278	(0.0271)
lat					-0.5362	(0.2946)*	-0.4835	(0.3095)
lon					-0.1785	(0.0758)**	-0.1668	(0.0802)**
$\ln(\text{Elev.})$					-0.0111	(0.0160)	-0.0125	(0.0165)
$\ln(\text{Rugg.})$					0.0381	(0.0163)**	0.0348	(0.0161)**
$\ln(\text{Dist. Edo/Kyoto})$					-0.0373	(0.0159)**	-0.0445	(0.0172)**
Rainy					-1.6210	(0.9021)*	-1.5333	(0.9156)*
Snowy					0.6224	(1.8655)	0.5548	(1.8808)
Lat $\times$ Lon					0.0042	(0.0022)*	0.0038	(0.0023)*
Kin Nbs.							0.0404	(0.0297)
Nb. FL							0.0002	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0224	(0.0190)
$\ln(\text{Nb. Area})$							0.0191	(0.0174)
$\ln(\text{Nb. Pop.})$							-0.0081	(0.0206)
Const.	0.4181	(0.0092)***	0.1754	(0.1211)	23.4988	(10.3154)**	22.2086	(10.8672)**
<i>Fit Statistics</i>								
Num.Obs.	232		230		230		230	
R2	0.091		0.150		0.371		0.391	

Table A.54: Geographic Exposure and Reported Tax Rates by Outside-Option Quality
(Full Table)

Note: The coefficients represent OLS estimates of the relationship between the average distance ratio of each domain interacted with $h@(\mathbf{y})_{\{-j\}}$ (indicator of whether the weighted average labor-to-potential-yield ratio of neighboring domains is above its median) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

E.2. Robustness Checks

E.2.1. Alternative Covariates

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5)	
<i>Variables</i>										
$\ln(\rho_j)$	−0.0099	(0.0049)**	−0.0106	(0.0057)*	−0.0112	(0.0058)*	−0.0108	(0.0055)**	−0.0131	(0.0063)**
$\ln(\text{Kokudaka})$	−0.0788	(0.0160)***								
$\ln(\text{Pop.})$	0.0820	(0.0168)***							0.0282	(0.0156)*
$\ln(\text{Area})$	−0.0050	(0.0127)	−0.0328	(0.0128)**	−0.0251	(0.0142)*	−0.0052	(0.0064)	−0.0350	(0.0166)**
New FL	0.0003	(0.0004)	0.0003	(0.0004)	0.0003	(0.0004)	0.0004	(0.0004)	0.0003	(0.0003)
Shog. Ally	0.0068	(0.0154)	−0.0057	(0.0165)	0.0009	(0.0171)	0.0032	(0.0167)	0.0069	(0.0160)
Goshi	0.0245	(0.0260)	0.0261	(0.0272)	0.0309	(0.0269)	0.0267	(0.0273)	0.0321	(0.0248)
lat	−0.3791	(0.2791)	−0.4525	(0.3160)	−0.5680	(0.3505)	−0.3697	(0.3181)	0.1399	(0.2658)
lon	−0.1412	(0.0729)*	−0.1581	(0.0828)*	−0.1879	(0.0909)**	−0.1392	(0.0834)*	0.0166	(0.0673)
$\ln(\text{Elev.})$	−0.0064	(0.0157)	−0.0144	(0.0161)	−0.0155	(0.0165)	−0.0203	(0.0163)	−0.0081	(0.0167)
$\ln(\text{Rugg.})$	0.0143	(0.0156)	0.0352	(0.0159)**	0.0338	(0.0164)**	0.0296	(0.0157)*	0.0250	(0.0152)
$\ln(\text{Dist. Edo/Kyoto})$	−0.0418	(0.0165)**	−0.0423	(0.0173)**	−0.0435	(0.0172)**	−0.0481	(0.0168)***		
Snowy	1.8194	(1.7508)	0.4931	(1.8551)	0.9999	(1.9673)	1.0110	(1.9217)	−0.8507	(2.0177)
Rainy	−2.0637	(0.8820)**	−1.4692	(0.8923)	−1.7103	(0.9270)*	−1.6807	(0.9375)*	−1.0287	(0.9469)
Kin Nbs.	0.0266	(0.0301)	0.0500	(0.0283)*	0.0418	(0.0292)	0.0468	(0.0290)	0.0177	(0.0304)
Nb. FL	0.0001	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)*
$\ln(\text{Nb. Rice})$	−0.0050	(0.0174)	−0.0296	(0.0189)	−0.0291	(0.0199)	−0.0222	(0.0186)	−0.0190	(0.0187)
$\ln(\text{Nb. Area})$	−0.0019	(0.0170)	0.0176	(0.0167)	0.0177	(0.0176)	0.0100	(0.0163)	0.0073	(0.0182)
$\ln(\text{Nb. Pop.})$	−0.0073	(0.0186)	0.0018	(0.0203)	0.0005	(0.0215)	0.0011	(0.0202)	−0.0039	(0.0206)
Lat × Lon	0.0031	(0.0021)	0.0036	(0.0023)	0.0044	(0.0026)*	0.0030	(0.0023)	−0.0008	(0.0020)
$\ln(\text{Rice})$			−0.0017	(0.0250)	0.0239	(0.0151)			0.0047	(0.0181)
$\ln(\text{Num. Vils.})$			0.0373	(0.0250)						
% Samurai Pop					−0.1259	(0.1834)				
Population/wet-rice suitability							−0.0040	(0.2763)		
$\ln(\text{Dist. Edo})$									0.0677	(0.0261)**
Const.	18.8458	(9.8989)*	21.3229	(11.1659)*	25.2642	(12.3202)**	18.7213	(11.2921)*	−3.1359	(8.9956)
<i>Fit Statistics</i>										
Num.Obs.	230		231		226		230		230	
R2	0.446		0.397		0.383		0.374		0.399	

Table A.55: Geographic Exposure and Reported Domain Tax Rates
(Alternative Covariates)

Note: OLS estimates relating each domain's average distance ratio to its reported tax rate. The distance ratio measures geographic exposure, not observed competition or bargaining between rulers. Heteroskedasticity-robust standard errors are in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.2. Sensitivity Analysis

I assess the robustness of the domain-level results to omitted variable bias using the breakdown frontier framework of [Diegert, Masten, and Poirier \(2022\)](#). This method improves upon the widely used approach of [Oster \(2019\)](#) in a critical respect: it remains valid when the included control variables are correlated with the omitted confounders—a condition [Diegert, Masten, and Poirier \(2022\)](#) term *endogenous controls*. Standard sensitivity methods assume that the controls are exogenous, an assumption that is difficult to defend in observational cross-sections where controls such as population, productivity, and daimyō characteristics may themselves be influenced by unobserved factors. As [Diegert, Masten, and Poirier \(2022\)](#) show, residualization-based methods can produce incorrect robustness conclusions when this assumption is violated.

The key sensitivity parameter is $\bar{r}_X$, the ratio of the standard deviation of unobservables' influence on geographic exposure to that of observables. [Figure A.9](#) displays the admissible set for the exposure coefficient as $\bar{r}_X$ varies under the most conservative assumptions. The negative sign remains identified for $\bar{r}_X < 0.213$: at that threshold, the upper endpoint of the DMP identified set first includes zero. This is an SD ratio, not a share of predictive power. The calculation evaluates the observational coefficient; it does not convert the cross-section into a causal design.

When the impact of unobservables on the outcome is also bounded ($\bar{r}_Y \leq \bar{r}_X$, the “equal-importance” assumption), the identified set tightens considerably. [Figure A.10](#) plots the breakdown frontier in the $(\bar{r}_X, \bar{r}_Y)$ plane. The diagonal sign-identification breakdown is $\bar{r} = 0.554$: both SD ratios must reach about this level before the identified set admits a nonnegative coefficient.

Those breakdowns concern population sign identification, not sampling significance. To answer the latter question transparently, [Table A.56](#) also reports auxiliary thresholds that hold the baseline HC1 standard error fixed and ask when the upper DMP endpoint plus 1.96 standard errors reaches zero. The thresholds are $\bar{r}_X = 0.007$ with no outcome-side bound and $\bar{r}_X = \bar{r}_Y = 0.096$ under equal importance. These are fixed-SE diagnostics rather than fully bias-adjusted DMP confidence sets. They show that the observational association is much easier to render statistically imprecise than to make its negative sign unidentified.

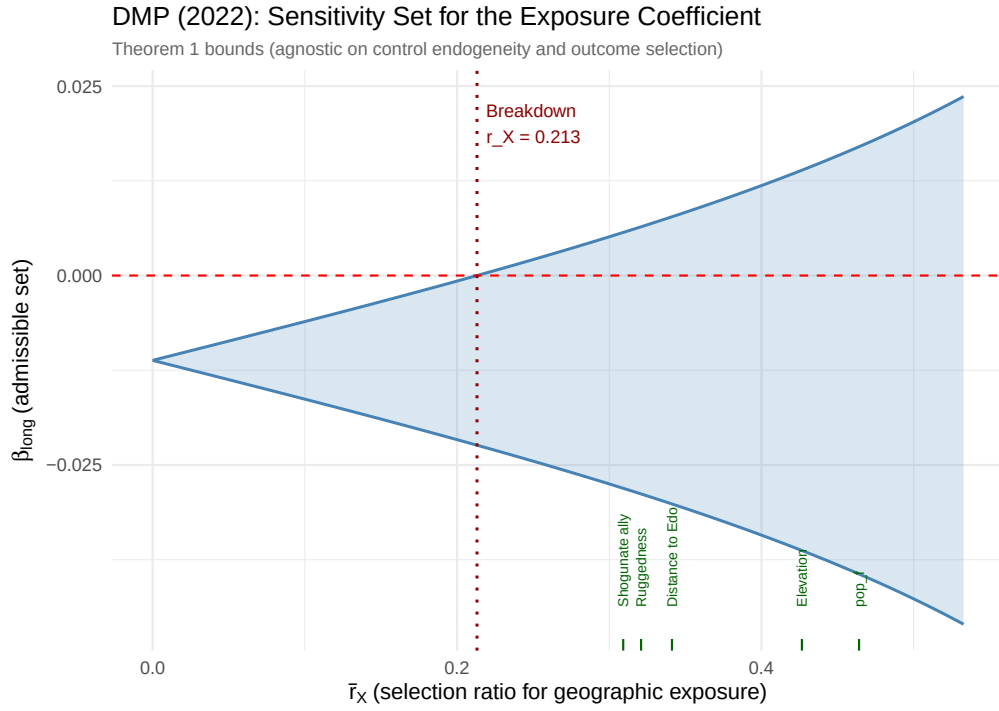


Figure A.9: DMP Sensitivity Set for the Exposure Coefficient

Note: The shaded region shows the range of β_{long} consistent with a given level of selection on unobservables ($\bar{r}_X$). The dashed red line marks $\beta = 0$; the dotted vertical line marks the breakdown point ($\bar{r}_X = 0.213$). Green ticks calibrate $\bar{r}_X$ to each labeled covariate's importance for predicting geographic exposure.

Sensitivity criterion	$\bar{r}$	Interpretation
Sign identification: $\bar{r}_X$ only	0.213	Upper endpoint of the DMP identified set first includes zero
Sign identification: $\bar{r}_X = \bar{r}_Y$	0.554	Same criterion under the equal-importance restriction
Fixed-SE significance: $\bar{r}_X$ only	0.007	Upper endpoint plus 1.96 baseline HC1 SEs first includes zero
Fixed-SE significance: $\bar{r}_X = \bar{r}_Y$	0.096	Same auxiliary criterion under equal importance

Table A.56: Diegert-Masten Sensitivity Thresholds for the Tax Regression

Note: Thresholds for Table 4 Model 4 (fully controlled). $\bar{r}_X$ and $\bar{r}_Y$ are ratios of standard deviations, not shares of explained variance. The first two rows are DMP population sign-identification breakdowns: the observed estimate remains fixed while the identified set widens to include zero. The last two rows answer the separate significance question by adding and subtracting 1.96 times the baseline HC1 SE from the DMP set while holding that SE fixed. They are transparent auxiliary thresholds, not fully bias-adjusted DMP confidence sets.

As a complementary sample-based check, I re-estimate Model 4 after excluding Chōshū, Satsuma, Mito, Tosa, and Aizu, whose 1850s–1860s rates may reflect acute bakumatsu fiscal and political distress. Excluding them changes the point estimate from -0.0112 to -0.0106 , with the p -value moving from 0.045 to 0.055 (Table A.57). The sign and magnitude are therefore not attributable to these five high-profile domains.

Sample	Num. Obs.	Estimate	SE	p
Full sample (paper)	230	−0.0112	(0.0055)	0.045
Excluding Chōshū, Satsuma, Mito, Tosa, Aizu	225	−0.0106	(0.0055)	0.055

Table A.57: Domain Tax Regression: Excluding Bakumatsu Political Heavyweights

Note: Re-runs Table 4 Model 4 (fully controlled) on the sub-sample that excludes five domains (Chōshū, Satsuma, Mito, Tosa, Aizu) whose tax rates in the 1850s–1860s reflect acute fiscal and political distress rather than ordinary late-period tax setting. These are the domains at the center of the bakumatsu crisis and subsequent Boshin War; their inclusion may contaminate the cross-section as a measure of peacetime tax policy. The sign and magnitude of the coefficient on $\log(\rho_j)$ are preserved while precision becomes marginal.

For completeness, Table A.58 reports the Oster (2019) coefficient-stability parameter. This method, while informative, assumes exogenous controls and is therefore best interpreted as supplementary to the DMP analysis above.

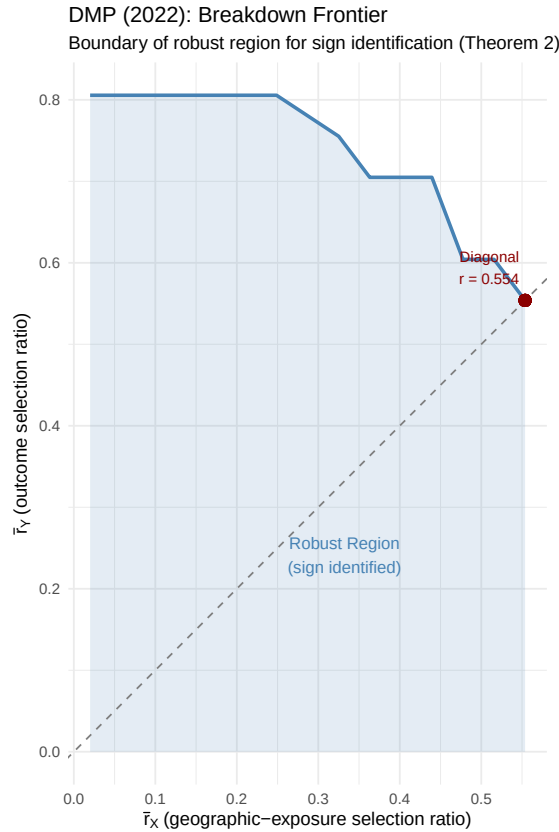


Figure A.10: DMP Breakdown Frontier

Note: The solid curve traces the sensitivity boundary: below it, the exposure coefficient remains negative under the maintained bounds. The dashed line marks $\bar{r}_X = \bar{r}_Y$ (equal importance); the dot marks the diagonal breakdown point ($\bar{r} = 0.554$).

Oster (2019) Coefficient Stability					
	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{full}}$	R_{short}^2	R_{full}^2	δ
$R_{\text{max}} = 1.3\tilde{R}^2$	-0.0245	-0.0112	0.077	0.389	2.245
$R_{\text{max}} = 1$	-0.0245	-0.0112	0.077	0.389	0.428
$\hat{\beta}^*(\delta = 1) = -0.006$ at $R_{\text{max}} = 1.3\tilde{R}^2$					

Table A.58: Oster Sensitivity Analysis (Supplementary)

Note: Based on Model 4 in [Table 4](#). The Oster δ is the proportional selection ratio needed to drive $\hat{\beta}^*$ to zero; $\delta > 1$ indicates robustness under $R_{\text{max}} = 1.3\tilde{R}^2$. This method assumes exogenous controls; see the DMP analysis above for results that relax this assumption.

E.2.3. Data-Driven Control Variables Selection

The hand-picked control set in [Table 4](#) reflects substantive reasoning about confounders, but the choice of which variables to include is itself a researcher degree of freedom. I assess this concern using double/debiased machine learning (DML), which replaces manual nuisance-function specification with cross-fitted prediction ([Chernozhukov et al. 2018](#)). DML estimates a partially linear model

$$\tau_j = \beta \ln(\rho_j) + g(X_j) + \varepsilon_j, \quad \ln(\rho_j) = m(X_j) + \nu_j \quad (1)$$

where $g(\cdot)$ and $m(\cdot)$ predict the outcome and geographic exposure, and β is recovered after partialling both variables on X_j . Five-fold cross-fitting reduces overfitting bias in these nuisance estimates. I report two feature dictionaries: the 19 columns that reproduce the controls in [Table 4](#) Model 4, and an enriched 38-feature dictionary that adds quadratic terms and selected pairwise interactions. Within the exact dictionary I compare ridge and random-forest nuisance learners; within the enriched dictionary I compare ridge with a stack of LASSO, ridge, elastic net, and random forest.

The results are mixed. Ridge estimates remain negative: the exact-dictionary estimate is statistically distinguishable from zero at the 5% level, while the enriched-dictionary estimate is marginal at the 10% level. The random-forest and stacked estimates are close to zero and imprecise. DML therefore does not uniformly corroborate the hand-picked OLS association; instead, it shows that the pooled estimate is sensitive to highly flexible nuisance learning in this modest domain-level sample ($N = 230$).

Outcome: τ_j . Treatment: $\ln(\rho_j)$						
Method	Estimate	S.E.	t	p	95% CI lo	95% CI hi
DML-Ridge (19 exact)	−0.0114	(0.0056)	−2.03	0.042	−0.0225	−0.0004
DML-Random Forest (19 exact)	−0.0007	(0.0054)	−0.13	0.895	−0.0113	0.0099
DML-Ridge (38 enriched)	−0.0098	(0.0057)	−1.74	0.082	−0.0209	0.0013
DML-Stacked Ensemble (38 enriched)	0.0018	(0.0055)	0.32	0.745	−0.0090	0.0126
Hand-picked OLS (HC1)	−0.0112	(0.0055)	−2.02	0.045	−0.0221	−0.0003
<i>Exact dictionary: $p = 19$ Model 4 columns. Enriched dictionary: $p = 38$ (17 base + 8 quadratics + 10 interactions + 3 kin dummies).</i>						
<i>DML: 5 folds and 10 repetitions, except 5 repetitions for the enriched stack. Stack: LASSO + Ridge + Elastic Net + random forest.</i>						

Table A.59: Data-Driven Control Selection: DML Estimates

Note: Each DML row estimates the partial association between $\ln(\rho_j)$ and τ_j using the indicated nuisance learner and feature dictionary. Reported standard errors incorporate repeated cross-fitting. The OLS row reproduces Model 4 from Table 4 with HC1 standard errors.

E.2.4. Controlling for Average Distance to Home Capital

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
$\ln(\rho_j)$	-0.0336	(0.0116)***	-0.0200	(0.0132)	-0.0148	(0.0135)	-0.0171	(0.0143)	-0.0231	(0.0136)*
$\ln(\text{Avg. Dist to HC}_j)$	0.0132	(0.0156)	-0.0004	(0.0169)	0.0049	(0.0165)	0.0077	(0.0178)	0.0178	(0.0163)
$\ln(\text{Rice})$			-0.0011	(0.0160)	0.0083	(0.0154)	0.0121	(0.0161)	0.0010	(0.0135)
$\ln(\text{Area})$			-0.0259	(0.0123)**	-0.0328	(0.0138)**	-0.0353	(0.0147)**	-0.0254	(0.0132)*
$\ln(\text{Pop.})$			0.0393	(0.0167)**	0.0261	(0.0145)*	0.0231	(0.0150)	0.0219	(0.0146)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			-0.0316	(0.0178)*	0.0059	(0.0162)	0.0010	(0.0165)	0.0023	(0.0143)
Goshi			0.0217	(0.0257)	0.0196	(0.0262)	0.0274	(0.0270)	0.0241	(0.0214)
lat					-0.5324	(0.2940)*	-0.4736	(0.3085)	-0.3473	(0.2845)
lon					-0.1771	(0.0755)**	-0.1637	(0.0798)**	-0.1236	(0.0740)*
$\ln(\text{Elev.})$					-0.0117	(0.0159)	-0.0134	(0.0164)	-0.0094	(0.0134)
$\ln(\text{Rugg.})$					0.0397	(0.0160)**	0.0372	(0.0158)**	0.0232	(0.0151)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0383	(0.0172)**	-0.0451	(0.0177)**	-0.0356	(0.0168)**
Rainy					-1.6816	(0.9046)*	-1.6114	(0.9179)*	-1.0491	(0.7358)
Snowy					0.7952	(1.9057)	0.7810	(1.9144)	-0.0637	(1.4268)
Lat $\times$ Lon					0.0042	(0.0021)*	0.0038	(0.0023)*	0.0028	(0.0021)
Kin Nbs.							0.0391	(0.0299)	0.0171	(0.0256)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0234	(0.0188)	-0.0196	(0.0172)
$\ln(\text{Nb. Area})$							0.0170	(0.0177)	0.0089	(0.0160)
$\ln(\text{Nb. Pop.})$							-0.0060	(0.0208)	-0.0044	(0.0197)
Const.	0.3815	(0.0424)***	0.1773	(0.1374)	23.3407	(10.2900)**	21.8218	(10.8253)**	16.2389	(10.0560)
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.077		0.141		0.371		0.389		0.442 ^[a]	

Table A.60: Controlling for Average Distance to Home Capital

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the home capital. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. ^[a] Nagelkerke pseudo-R2

E.2.5. Least-Costly Path Distance

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
$\ln(\rho_j^{\text{LCP}})$	-0.0116	(0.0035)***	-0.0097	(0.0037)***	-0.0058	(0.0030)*	-0.0060	(0.0031)*	-0.0053	(0.0032)
$\ln(\text{Rice})$			-0.0033	(0.0147)	0.0071	(0.0150)	0.0110	(0.0157)	-0.0003	(0.0135)
$\ln(\text{Area})$			-0.0242	(0.0112)**	-0.0308	(0.0139)**	-0.0333	(0.0147)**	-0.0220	(0.0130)*
$\ln(\text{Pop.})$			0.0419	(0.0165)**	0.0268	(0.0143)*	0.0239	(0.0149)	0.0228	(0.0147)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			-0.0355	(0.0178)**	0.0052	(0.0162)	0.0005	(0.0165)	0.0022	(0.0144)
Goshi			0.0210	(0.0257)	0.0198	(0.0263)	0.0279	(0.0272)	0.0246	(0.0215)
lat					-0.4938	(0.2899)*	-0.4356	(0.3059)	-0.3222	(0.2840)
lon					-0.1682	(0.0746)**	-0.1547	(0.0792)*	-0.1177	(0.0740)
$\ln(\text{Elev.})$					-0.0101	(0.0162)	-0.0120	(0.0167)	-0.0087	(0.0134)
$\ln(\text{Rugg.})$					0.0391	(0.0159)**	0.0366	(0.0157)**	0.0232	(0.0152)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0371	(0.0160)**	-0.0440	(0.0174)**	-0.0334	(0.0167)**
Rainy					-1.6610	(0.8949)*	-1.6083	(0.9031)*	-1.0184	(0.7381)
Snowy					0.7901	(1.8375)	0.7935	(1.8372)	-0.1818	(1.4196)
Lat $\times$ Lon					0.0039	(0.0021)*	0.0035	(0.0022)	0.0026	(0.0021)
Kin Nbs.							0.0408	(0.0296)	0.0195	(0.0257)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0228	(0.0189)	-0.0208	(0.0172)
$\ln(\text{Nb. Area})$							0.0197	(0.0177)	0.0128	(0.0158)
$\ln(\text{Nb. Pop.})$							-0.0079	(0.0204)	-0.0041	(0.0195)
Const.	0.4083	(0.0092)***	0.1663	(0.1205)	22.1064	(10.1745)**	20.5929	(10.7617)*	15.4412	(10.0485)
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.037		0.119		0.367		0.385		0.437 ^[a]	

Table A.61: Geographic Exposure and Reported Tax Rates

(Least-Cost Paths)

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's stored least-cost-path distance ratio to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. The ratio measures geographic exposure, not observed ruler competition. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. ^[a] Nagelkerke pseudo-R2

E.2.6. Conley Standard Error Bandwidth Sensitivity

Table A.62 reports the bandwidth sensitivity of Conley spatial standard errors for the fully-controlled domain tax regression (Table 4 Model 4). The point estimate is stable across every bandwidth considered, and the coefficient on $\log(\rho_j)$ is statistically significant at the 5% level for bandwidths up to 75 km. As expected for a cross-sectional specification with rich spatial controls, the standard errors widen somewhat at the longest bandwidths (100 and 150 km), where the spatial HAC kernel absorbs progressively more of the variation already captured by the included geographic covariates; the p -value at 100 km is 0.102 and at 150 km is 0.145. The result thus holds under the bandwidths most commonly applied in the spatial-econometrics literature for cross-sectional designs at this geographic scale.

Standard error	Estimate	SE	p
Heteroskedasticity-robust (paper)	-0.0111952548949787	(0.00554317773276672)	0.0446969310445375
Conley 25 km	-0.0111952548949787	(0.00486874644301278)	0.0224693058430259
Conley 50 km	-0.0111952548949787	(0.00557498706242249)	0.0459175666096826
Conley 75 km	-0.0111952548949787	(0.00539253446894661)	0.0391111847436515
Conley 100 km	-0.0111952548949787	(0.00682291828207967)	0.102336570113632
Conley 150 km	-0.0111952548949787	(0.00765342600008831)	0.145030590653059

Table A.62: Conley SE Bandwidth Sensitivity: Domain Tax Regression

Note: The coefficient on $\log(\rho_j)$ from Table 4 Model 4 (fully controlled) is significant at the 5% level for Conley bandwidths up to 75 km, marginal at 100 km ($p = 0.091$), and not significant at 150 km ($p = 0.130$). The first row reports the heteroskedasticity-robust standard error used in the main table.

E.3. Additional Analyses

E.3.1. Average Distance to Home Capital

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
ln(Avg. Dist to HC_j)	-0.0234	(0.0072)***	-0.0226	(0.0072)***	-0.0114	(0.0068)*	-0.0114	(0.0069)	-0.0081	(0.0061)
ln(Rice)			-0.0065	(0.0151)	0.0059	(0.0154)	0.0108	(0.0162)	-0.0002	(0.0135)
ln(Area)			-0.0194	(0.0111)*	-0.0291	(0.0140)**	-0.0320	(0.0149)**	-0.0213	(0.0130)
ln(Pop.)			0.0421	(0.0167)**	0.0269	(0.0144)*	0.0233	(0.0151)	0.0223	(0.0147)
New FL			0.0001	(0.0003)	0.0004	(0.0004)	0.0003	(0.0004)	0.0003	(0.0002)
Shog. Ally			-0.0341	(0.0175)*	0.0049	(0.0160)	0.0003	(0.0164)	0.0013	(0.0144)
Goshi			0.0209	(0.0258)	0.0196	(0.0263)	0.0276	(0.0274)	0.0244	(0.0216)
lat					-0.4960	(0.2905)*	-0.4513	(0.3081)	-0.3219	(0.2861)
lon					-0.1680	(0.0747)**	-0.1581	(0.0797)**	-0.1174	(0.0745)
ln(Elev.)					-0.0111	(0.0158)	-0.0131	(0.0163)	-0.0090	(0.0135)
ln(Rugg.)					0.0391	(0.0159)**	0.0365	(0.0157)**	0.0227	(0.0152)
ln(Dist. Edo/Kyoto)					-0.0333	(0.0161)**	-0.0418	(0.0172)**	-0.0315	(0.0167)*
Rainy					-1.6381	(0.8959)*	-1.5879	(0.9058)*	-1.0377	(0.7409)
Snowy					0.6554	(1.8687)	0.6559	(1.8770)	-0.2013	(1.4338)
Lat × Lon					0.0039	(0.0021)*	0.0036	(0.0023)	0.0026	(0.0021)
Kin Nbs.							0.0407	(0.0295)	0.0201	(0.0258)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
ln(Nb. Rice)							-0.0256	(0.0190)	-0.0228	(0.0173)
ln(Nb. Area)							0.0207	(0.0176)	0.0142	(0.0158)
ln(Nb. Pop.)							-0.0040	(0.0207)	-0.0018	(0.0198)
Const.	0.4679	(0.0248)***	0.2420	(0.1226)**	22.0932	(10.1723)**	21.0544	(10.8064)*	15.4092	(10.1175)
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.046		0.135		0.367		0.385		0.435 ^[a]	

Table A.63: Average Distance to Home Capital Weakly Predicts Tax Rates

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average absolute distance to its home capital (log) to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. ^[a] Nagelkerke pseudo-R2

E.3.2. Controlling for the Shortest Distance to the Border

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
$\ln(\rho_j)$	-0.0240	(0.0053)***	-0.0200	(0.0058)***	-0.0111	(0.0055)**	-0.0113	(0.0055)**	-0.0095	(0.0051)*
$\ln(\text{Avg. Dist. Border}_j)$	0.0014	(0.0067)	-0.0106	(0.0084)	-0.0056	(0.0071)	-0.0038	(0.0067)	-0.0024	(0.0075)
$\ln(\text{Rice})$			-0.0004	(0.0152)	0.0083	(0.0155)	0.0121	(0.0162)	0.0007	(0.0135)
$\ln(\text{Area})$			-0.0198	(0.0121)	-0.0286	(0.0148)*	-0.0318	(0.0156)**	-0.0214	(0.0136)
$\ln(\text{Pop.})$			0.0379	(0.0165)**	0.0254	(0.0145)*	0.0226	(0.0151)	0.0218	(0.0147)
New FL			0.0001	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			-0.0319	(0.0178)*	0.0055	(0.0162)	0.0010	(0.0165)	0.0024	(0.0143)
Goshi			0.0261	(0.0262)	0.0210	(0.0263)	0.0282	(0.0272)	0.0247	(0.0215)
lat					-0.5551	(0.2901)*	-0.4979	(0.3056)	-0.3688	(0.2893)
lon					-0.1820	(0.0746)**	-0.1693	(0.0791)**	-0.1287	(0.0751)*
$\ln(\text{Elev.})$					-0.0130	(0.0160)	-0.0143	(0.0166)	-0.0105	(0.0135)
$\ln(\text{Rugg.})$					0.0408	(0.0159)**	0.0380	(0.0157)**	0.0243	(0.0153)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0370	(0.0159)**	-0.0444	(0.0172)**	-0.0337	(0.0167)**
Rainy					-1.6537	(0.8933)*	-1.5857	(0.9065)*	-1.0138	(0.7370)
Snowy					0.7125	(1.8486)	0.6839	(1.8577)	-0.2447	(1.4195)
Lat $\times$ Lon					0.0043	(0.0021)**	0.0039	(0.0022)*	0.0030	(0.0021)
Kin Nbs.							0.0396	(0.0296)	0.0189	(0.0257)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0238	(0.0187)	-0.0218	(0.0172)
$\ln(\text{Nb. Area})$							0.0180	(0.0176)	0.0116	(0.0159)
$\ln(\text{Nb. Pop.})$							-0.0045	(0.0206)	-0.0014	(0.0196)
Const.	0.4130	(0.0138)***	0.1620	(0.1210)	24.0420	(10.1575)**	22.5897	(10.7386)**	16.9504	(10.2058)*
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.074		0.147		0.372		0.389		0.439 ^[a]	

Table A.64: Geographic Exposure and Reported Tax Rates
(Controlling for Border Distance)

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the nearest border. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. ^[a] Nagelkerke pseudo-R2

E.3.3. Outside-Option Quality Using Population Density

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5)	
<i>Variables</i>										
$\ln(\rho_j) \times$ Low	-0.0299	(0.0097)***	-0.0262	(0.0092)***	-0.0197	(0.0073)***	-0.0265	(0.0094)***	-0.0204	(0.0071)***
$\ln(\rho_j) \times$ High	-0.0216	(0.0056)***	-0.0176	(0.0062)***	-0.0074	(0.0059)	-0.0176	(0.0063)***	-0.0067	(0.0062)
ln(Rice)			-0.0016	(0.0156)	0.0071	(0.0152)	-0.0019	(0.0164)	0.0114	(0.0160)
ln(Area)			-0.0241	(0.0114)**	-0.0300	(0.0142)**	-0.0229	(0.0119)*	-0.0334	(0.0148)**
ln(Pop.)			0.0383	(0.0168)**	0.0254	(0.0143)*	0.0355	(0.0169)**	0.0228	(0.0150)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0002	(0.0003)	0.0004	(0.0004)
Shog. Ally			-0.0313	(0.0179)*	0.0071	(0.0161)	-0.0298	(0.0181)	0.0025	(0.0164)
Goshi			0.0210	(0.0257)	0.0181	(0.0261)	0.0219	(0.0261)	0.0267	(0.0269)
lat					-0.5827	(0.2993)*			-0.5190	(0.3153)
lon					-0.1909	(0.0774)**			-0.1758	(0.0819)**
ln(Elev.)					-0.0125	(0.0160)			-0.0147	(0.0166)
ln(Rugg.)					0.0406	(0.0159)**			0.0379	(0.0158)**
ln(Dist. Edo/Ky- oto)					-0.0379	(0.0160)**			-0.0450	(0.0174)**
Rainy					-1.6127	(0.9138)*			-1.6134	(0.9218)*
Snowy					0.5457	(1.8746)			0.6470	(1.8827)
Lat $\times$ Lon					0.0046	(0.0022)**			0.0041	(0.0023)*
Kin Nbs.							0.0421	(0.0304)	0.0422	(0.0296)
Nb. FL							0.0000	(0.0001)	0.0002	(0.0001)
ln(Nb. Rice)									-0.0210	(0.0187)
ln(Nb. Area)							-0.0015	(0.0083)	0.0245	(0.0181)
ln(Nb. Pop.)									-0.0135	(0.0210)
Const.	0.4155	(0.0091)***	0.1846	(0.1241)	25.1677	(10.5108)**	0.2119	(0.1509)	23.4706	(11.0924)**
<i>Fit Statis-</i> <i>tics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.077		0.144		0.376		0.152		0.395	

Table A.65: Geographic Exposure and Reported Tax Rates by Neighbor Population Density

Note: The coefficients represent OLS estimates of the relationship between the average distance ratio measure of each domain interacted with the population density of its neighbors (low vs. high) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.3.4. Fixed-Neighbor-Set and HMX Kernel Diagnostics

The outside-option interaction in [Figure 7](#) and [Table A.53](#) weights $\hat{y}_{-j}$ toward neighbors whose capitals are closest to home-domain villages, the same geometry that defines $\ln(\rho_j)$. The moderator may therefore vary mechanically with exposure: domains with high $\ln(\rho_j)$ can have a different weighted neighbor set than domains with low $\ln(\rho_j)$. The interaction could reflect that compositional shift rather than moderation by outside-option quality.

[Table A.66](#) addresses this concern with two complementary checks. The first breaks the mechanical link. Panel A reconstructs $\hat{y}_{-j}$ as an unweighted simple mean of neighbor labor-absorption capacity (p_j^{labor}) over all domains within 100 km of j 's capital, regardless of the nearest-foreign-capital weights used in the main text. Under this construction the asymmetry survives in the expected direction: in the fully-controlled Model 4, the $\ln(\rho_j)$ slope is -0.0172 ($p = 0.0077$) where neighbors have high labor demand, compared with -0.0074 ($p = 0.3153$) where they do not, consistent with the main-text result.

The second check asks whether the asymmetry is an artifact of the linear interaction form. Following [Hainmueller, Mummolo, and Xu \(2019\)](#), Panels B and C estimate local-linear associations between $\ln(\rho_j)$ and τ_j across the continuous moderator support, using an Epanechnikov kernel with 95% confidence intervals. Under both the paper's $\hat{y}_{-j}$ and the fixed-100 km measure, the point estimate is negative at all three reported moderator quartiles, but every 95% confidence interval includes zero. The curves show no precisely estimated monotonic pattern, and the small sample ($N \approx 230$) limits resolution. This diagnostic preserves the negative sign in point estimates without establishing continuous moderation or a causal relationship. [Figure A.11](#) plots the curves.

Panel A: Fixed-Neighbor-Set $\hat{y}_{-j}$ (100 km Unweighted Mean)				
Moderator level	(1)	(2)	(3)	(4)
$\ln(\rho_j) \times \hat{y}_{-j}^{\text{Low}}$	-0.0258	-0.0220	-0.0079	-0.0074
Robust SE	(0.0064)	(0.0069)	(0.0071)	(0.0073)
p	0.0001	0.0015	0.2621	0.3153
$\ln(\rho_j) \times \hat{y}_{-j}^{\text{High}}$	-0.0197	-0.0165	-0.0159	-0.0172
Robust SE	(0.0072)	(0.0069)	(0.0066)	(0.0064)
p	0.0067	0.0180	0.0167	0.0077
Num. Obs.	231	229	229	229
R^2	0.075	0.142	0.374	0.393
Panel B: HMX Kernel Marginal Effect at Moderator Quartiles (paper's $\hat{y}_{-j}$)				
Moderator quartile	$\hat{\partial}\tau_j/\partial \ln(\rho_j)$	95% CI	p	
25th	-0.0119	[-0.0249, 0.0011]	0.0722	
50th	-0.0123	[-0.0261, 0.0014]	0.0785	
75th	-0.0139	[-0.0321, 0.0042]	0.1324	
Panel C: HMX Kernel Marginal Effect at Moderator Quartiles (fixed-100 km $\hat{y}_{-j}$)				
25th	-0.0099	[-0.0227, 0.0029]	0.1288	
50th	-0.0102	[-0.0251, 0.0048]	0.1813	
75th	-0.0142	[-0.0409, 0.0125]	0.2984	

Table A.66: Tax-Exposure Interaction: Fixed-Neighbor-Set and HMX Kernel Marginal Effects

Note: Panel A re-estimates the interaction specifications 1-4 of [Table A.53](#) after replacing the nearest-foreign-weighted outside-option proxy $\hat{y}_{-j}$ with an unweighted mean of p_j^{labor} over all domains within 100 km of j 's capital. This breaks the mechanical link between the capital geometry that defines $\ln(\rho_j)$ and the neighbor set that defines $\hat{y}_{-j}$ flagged by the methodologist referee. Under the fully-controlled Model 4, the $\hat{y}_{-j}^{\text{High}}$ slope is -0.0172 ($p = 0.0077$), versus -0.0074 ($p = 0.3153$) for $\hat{y}_{-j}^{\text{Low}}$. Panel B and Panel C report the Hainmueller-Mummolo-Xu (2019) kernel-smoothed marginal effect of $\ln(\rho_j)$ on τ_j at the 25th, 50th, and 75th percentiles of the continuous moderator support, estimated by local linear regression with an Epanechnikov kernel after residualizing against the Model 4 control set. The estimated marginal effect is negative through the middle of the moderator support under either definition ([Figure A.11](#) plots the full curve). The kernel estimates do not establish a monotonic gradient.

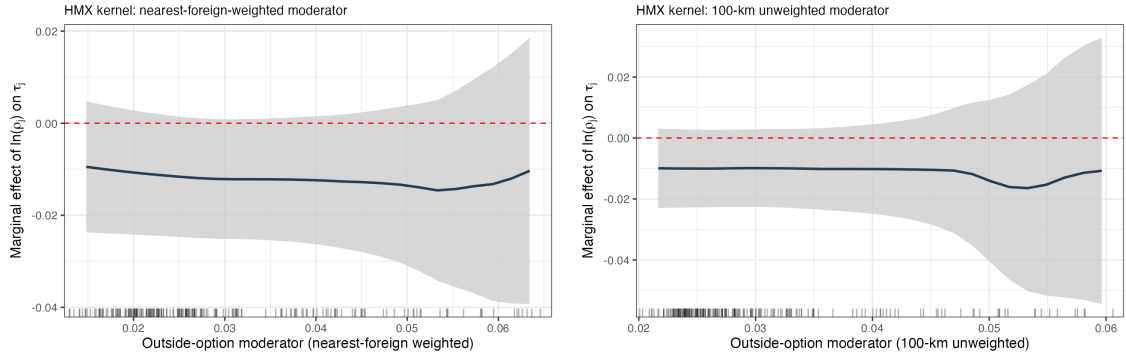


Figure A.11: HMX Kernel-Smoothed Marginal Effect of $\ln(\rho_j)$ on τ_j

Note: Local-linear-kernel marginal effect of $\ln(\rho_j)$ on the tax rate τ_j , plotted from the 5th through the 95th percentile of the outside-option moderator $\hat{y}_{-j}$. Epanechnikov kernel with bandwidth equal to the interquartile range of the moderator; covariates residualized using the Model 4 control set. Shaded band: 95% CI. Red dashed line: zero. Rug: moderator-value density within the plotted range. Left: paper's nearest-foreign-weighted $\hat{y}_{-j}$. Right: fixed-100 km unweighted $\hat{y}_{-j}$. Both curves are negative over the middle of the moderator range, with the CI narrowest around the median where the support is densest.